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# ST7102

## Datasheet

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深圳市双禹盛泰科技有限公司
联系电话：0755-2772 1006

**Sitronix Technology Corporation**

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# 1 GENERAL DESCRIPTION

The ST7102 is a System-on-Chip (SoC) driver LSI designed for TFT LCD controller with a build-in touch panel controller and suitable for small to medium size portable devices such as mobile phone or tablet. ST7102 can support up to 480RGBx1280 dots panel and support 16,777,216-color. There is no internal RAM in ST7102. The 480-channel source driver has true 8-bit resolution, which generates 256 Gamma-corrected values by an internal D/A converter.

The ST7102 incorporates with several charge pumps to generate various voltage levels that form an on-chip power management system for gate driver and source driver. The built-in timing controller in ST7102 can support MIPI interface ( included 4-lane/1-port) and RGB interface display serial interface with low EMI noise and touch protocol via standard integrated circuit bus($I^2C$) or serial peripheral interface(SPI). The ST7102 also supports a standby mode for power control consideration. For further power control, the dynamic backlight control function basing on displaying image content is also supported.

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# 2 FEATURES

- Single-chip FWVGA Amorphous TFT Controller/Driver.
- Display Resolution
  - 480RGB(H) x 864(V) (WVGA)
  - 480RGB(H) x 960(V) (FWVGA)
  - 480RGB(H) x 1280(V)
- Display Modes (Color Mode)
  - Full Color: 16M, RGB=(888) max. Idle Mode Off
  - Color Reduce: 8-color, RGB=(111), Idle Mode On
- System Interfaces
  - MIPI DSI: MIPI DSI (DSI v1.01.00, D-PHY v1.00.00 and DCS v1.01)
  - 16/18/24 RGB interface
- Display Features
  - Outputs 256γ-corrected values using an internal true 8-bit resolution D/A converter to achieve 16,777,216 colors
  - Supports CGOUTR[1:16]/CGOUTL[1:16] GIP control signal
  - Individual gamma correction setting for RGB dots(1 analog/ 3 digital gamma)
  - Supports column/1-dot/2-dot/4-dot inversion
  - Power saving mode (standby mode)
  - Supports low frame rate mode
- Built-in Color Image Processing Functions
  - Support WB function
  - Color enhance (CE 2.0)
  - Content adaptive brightness control (CABC 2.0)
- On Chip Function
  - Support VCOM ground level driving scheme
  - Internal oscillator for display clock generation
  - Timing controller
  - Built–in NVM to store VCOM/GVDD calibration and ID1~ID3
  - Built–in NVM to store analog gamma, digital gamma and color enhancement.
  - Built–in NVM to store panel timing, analog power setting, and etc.
- Supply Power Range
  - Logic power supply voltage (IOVCC): 1.65 ~ 3.3V
  - Positive analog power supply voltage (AVDD): 4.5 ~ 6.3V
  - Negative analog power supply voltage (AVEE): -4.5 ~ -6.3V
- Touch Feature
  - 16-bits MCU optimized for capacitive sensing and other human interactions

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- 64 analog front ends (AFEs) supports up to 512 receiver pads
- Supports I2C and SPI protocol for communication with the host
- 64 analog front ends (AFEs) supports up to 512 receiver pads
- 10 fingers support
- Support Long-V sensing mode
- Support noise detection and automatic frequency hopping
- Support passive stylus
- Support proximity
- Hopping frequency range from 50kHz to 120kHz to minimize noise interference
- High signal- to noise ratio (>50dB SNR) touch AFE enables
- Wake-up gesture
- Touch FW host download

– Optimized Layout For COG Assembly

– Operating Temperature Range: −30ºC to +75 ºC

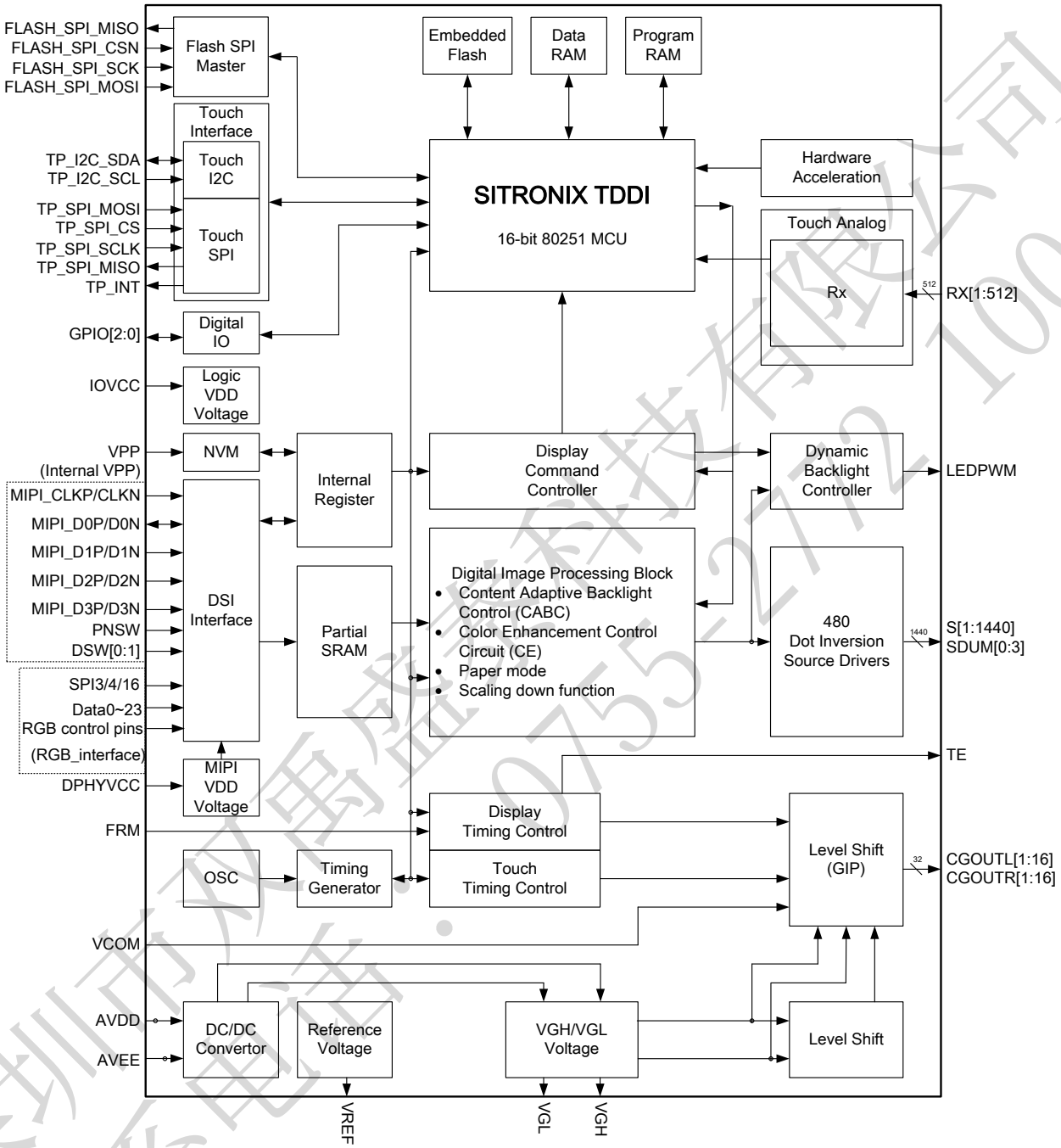
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# 3 BLOCK DIAGRAM

## 3.1 Function Block




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# 4 PIN INFOMATION

## 4.1 Pad Arrangement

- **Chip Information**

| | |
|---|---|
| Chip size | 22500um x 918um |
| Chip height & width tolerance | 20um |
| Chip thickness | 250um |
| Pad Location | Pad center |
| Coordinate Origin | Chip center |

*Note:*

*Chip size do not include scribe line*

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## 4.2 Pin Definition

### 4.2.1 Voltage Pins

| Name | I/O | PAD Type (Voltage Level) | Description |
|---|---|---|---|
| IOVCC | I | Power Supply | External power supply for internal logic circuit |
| AVDD | I | Power Supply | External positive power supply for analog circuit |
| AVEE | I | Power Supply | External negative power supply for analog circuit |
| AGND | - | GND | Analog ground, need to connect to GND from FPC |
| DGND | - | GND | Digital ground, need to connect to GND from FPC |
| DPHYGND | - | GND | Ground for MIPI DPHY circuit, need to connect to GND from FPC |
| VDD | O | Analog | Regulator output for logic, all pins need to connect together from FPC |
| VDDM | O | Analog | Regulator output for MIPI DSI |
| DPHYVCC | I | Analog | External power supply for MIPI PHY circuit<br>This pin need connect to IOVCC from FPC |
| TVH | O | Analog | Touch output high voltage level, all pins need to connect together from FPC. |
| TVL | O | Analog | Touch output low voltage level, all pins need to connect together from FPC. |
| VAG | O | Analog | Active guard signal, all pins need to connect together from FPC.<br>For IGZO panel, VGH/VGHO1/VGL need have capacitor to VAG |
| VGH | O | Analog | Step-up output voltage for panel, all pins need to connect together from FPC. |
| VGHO1 | O | Analog | Step-up output voltage, all pins need to connect together from FPC. |
| VGL | O | Analog | Step-up output voltage for panel, all pins need to connect together from FPC. |
| GVDDP | O | Analog | Positive LDO output for gamma circuit.<br>If not used, please keep it open. |
| GVDDN | O | Analog | Negative LDO output for gamma circuit.<br>If not used, please keep it open. |

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### 4.2.2 Control Pins

<table>
<tr><th>Name</th><th>I/O</th><th>PAD Type<br>(Voltage Level)</th><th>Description</th></tr>
<tr><td>RESX</td><td>I</td><td>Digital(IOVCC)</td><td>Global reset signal. Low active.</td></tr>
<tr><td>IM[2:0]</td><td>I</td><td>Digital(IOVCC)</td><td>Interface mode select pins
<table>
<tr><th>IM2</th><th>IM1</th><th>IM0</th><th>Interface Selection</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>MIPI</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>Reserved</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>Reserved</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>RGB(video)+SPI9 control</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>RGB(video)+SPI8 control</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>RGB(video)+SPI16 (rise) control</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>RGB(video) +SPI16 (fall) control</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>LANSEL[1:0]</td><td>I</td><td>Digital(IOVCC)</td><td>Input pin to select 1 data lane to 4 data lanes in MIPI interface.
<table>
<tr><th>Lanesel 1</th><th>Lanesel 0</th><th>Interface Selection</th></tr>
<tr><td>0</td><td>0</td><td>1 Lane</td></tr>
<tr><td>0</td><td>1</td><td>2 Lanes</td></tr>
<tr><td>1</td><td>0</td><td>3 Lanes</td></tr>
<tr><td>1</td><td>1</td><td>4 Lanes</td></tr>
</table></td></tr>
<tr><td>TE</td><td>O</td><td>Digital(IOVCC)</td><td>Tearing effect output pin to synchronies MCU to frame writing, activated by S/W command. When this pin is not activated (TE function OFF), this pin is GND level.</td></tr>
<tr><td>LEDPWM</td><td>O</td><td>Digital(IOVCC)</td><td>LCD backlight control PWM output pin</td></tr>
<tr><td>FRM</td><td>I</td><td>--</td><td>Test pin, please connect to ground or floating.</td></tr>
<tr><td>PSWAP</td><td>I</td><td>Digital(IOVCC)</td><td>MIPI Lane polarity swap pin.</td></tr>
</table>

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<table>
<tr><td>DSWAP[1:0]</td><td>I</td><td>Digital(IOVCC)</td><td>MIPI data lane swap and polarity swap table.
<table>
<tr><th>PNSWAP</th><th>DSWAP[1:0]</th><th>D2P</th><th>D2N</th><th>D1P</th><th>D1N</th><th>CLKP</th><th>CLKN</th><th>D0P</th><th>D0N</th><th>D3P</th><th>D3N</th></tr>
<tr><td rowspan="4">0</td><td>00</td><td>D3N</td><td>D3P</td><td>D2N</td><td>D2P</td><td>CLKN</td><td>CLKP</td><td>D1N</td><td>D1P</td><td>D0N</td><td>D0P</td></tr>
<tr><td>01</td><td>D3N</td><td>D3P</td><td>D0N</td><td>D0P</td><td>CLKN</td><td>CLKP</td><td>D1N</td><td>D1P</td><td>D2N</td><td>D2P</td></tr>
<tr><td>10</td><td>D0N</td><td>D0P</td><td>D1N</td><td>D1P</td><td>CLKN</td><td>CLKP</td><td>D2N</td><td>D2P</td><td>D3N</td><td>D3P</td></tr>
<tr><td>11</td><td>D2N</td><td>D2P</td><td>D1N</td><td>D1P</td><td>CLKN</td><td>CLKP</td><td>D0N</td><td>D0P</td><td>D3N</td><td>D3P</td></tr>
<tr><td rowspan="4">1</td><td>00</td><td>D3P</td><td>D3N</td><td>D2P</td><td>D2N</td><td>CLKP</td><td>CLKN</td><td>D1P</td><td>D1N</td><td>D0P</td><td>D0N</td></tr>
<tr><td>01</td><td>D3P</td><td>D3N</td><td>D0P</td><td>D0N</td><td>CLKP</td><td>CLKN</td><td>D1P</td><td>D1N</td><td>D2P</td><td>D2N</td></tr>
<tr><td>10</td><td>D0P</td><td>D0N</td><td>D1P</td><td>D1N</td><td>CLKP</td><td>CLKN</td><td>D2P</td><td>D2N</td><td>D3P</td><td>D3N</td></tr>
<tr><td>11</td><td>D2P</td><td>D2N</td><td>D1P</td><td>D1N</td><td>CLKP</td><td>CLKN</td><td>D0P</td><td>D0N</td><td>D3P</td><td>D3N</td></tr>
</table></td></tr>
<tr><td>SDO</td><td>O</td><td>IOVCC</td><td>RGB/MIPI SPI9, RGB/MIPI SPI8, RGB/MIPI SP16 output data Pin.<br>-If not used, please floating</td></tr>
<tr><td>SDA</td><td>I</td><td>IOVCC</td><td>RGB/MIPI SPI9, RGB/MIPI SPI8, RGB/MIPI SP16 input data Pin.<br>-If not used, please fix this pin at GND.</td></tr>
<tr><td>CSX</td><td>I</td><td>IOVCC</td><td>RGB/MIPI SPI9, RGB/MIPI SPI8, RGB/MIPI SP16 Chip select pin.<br>CSX=’0’ : Low enable.<br>CSX=’1’ : High disable<br>-If not used, please fix this pin at IOVCC.</td></tr>
<tr><td>SCL</td><td>I</td><td>IOVCC</td><td>RGB/MIPI SPI9, RGB/MIPI SPI8, RGB/MIPI SP16 CLK Pin.<br>-If not used, please fix this pin at IOVCC.</td></tr>
</table>

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### 4.2.3 MIPI Interface Pins

| Name | I/O | PAD Type (Voltage Level) | Description |
|---|---|---|---|
| CLKP | I | MIPI | MIPI-DSI clock lane positive-end input pin |
| CLKN | I | MIPI | MIPI-DSI clock lane negative-end input pin |
| DATA0P | I/O | MIPI | MIPI-DSI data lane 0 positive-end input/output pin<br>* Please connected to GND if not used |
| DATA0N | I/O | MIPI | MIPI-DSI data lane 0 negative-end input/output pin<br>* Please connected to GND if not used |
| DATA1P | I | MIPI | MIPI-DSI data lane 1 positive-end input pin<br>* Please connected to GND if not used |
| DATA1N | I | MIPI | MIPI-DSI data lane 1 negative-end input pin<br>* Please connected to GND if not used. |
| DATA 2P | I | MIPI | MIPI-DSI data lane 2 positive-end input pin<br>* Please connected to GND if not used |
| DATA 2N | I | MIPI | MIPI-DSI data lane 2 negative-end input pin<br>* Please connected to GND if not used |
| DATA 3P | I | MIPI | MIPI-DSI data lane 3 positive-end input pin<br>* Please connected to GND if not used |
| DATA 3N | I | MIPI | MIPI-DSI data lane 3 negative-end input pin<br>* Please connected to GND if not used |

### 4.2.4 RGB Interface Pins

| Name | I/O | PAD Type (Voltage Level) | Description |
|---|---|---|---|
| D0~D23 | I | IOVCC | RGB interface data bus. -If not used, please fix this pin at GND. |
| ENABLE | I | IOVCC | Data enable signal in RGB interface. -If not used, please fix this pin at IOVCC. |
| PCLK | I | IOVCC | Dot clock signal in RGB interface. (DOTCLK)<br>-If not used, please fix this pin at IOVCC |
| HSYNC | I | IOVCC | -Horizontal (Line) synchronizing input signal in RGB interface.<br>-If not used, please fix to the IOVCC. |
| VSYNC | I | IOVCC | Vertical (Frame) synchronizing input signal in RGB interface.<br>-If not used, please fix to the IOVCC. |

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### 4.2.5 Driver Panel Related Pins

| Name | I/O | PAD Type (Voltage Level) | Description |
|---|---|---|---|
| CGOUTL[16:1] | O | Analog | Panel control signal output pads for left side GIP. |
| CGOUTR[16:1] | O | Analog | Panel control signal output pads for right side GIP. |
| S1:S1440 | O | Analog | Output source driver signals. The D/A converted 256-gray-scale analog voltage is output. |
| VCOM | O | Analog | VCOM voltage output |
| COGTEST12<br>COGTEST34 | O | No level | For ITO resistance trace |
| VCOM_PASS_R<br>VCOM_PASS_L | - | No level | Pass line for VCOM_OPT |
| VCOM_OPT_R<br>VCOM_OPT_L | O | Analog | VCOM optional buffer output. |

### 4.2.6 Touch Related Pins

| Name | I/O | PAD Type (Voltage Level) | Description |
|---|---|---|---|
| TP_RESX | I | Digital(IOVCC) | External reset for TP |
| TP_UART_TX | I/O | Digital(IOVCC) | UART TX pad |
| TP_INT | O | Digital(IOVCC) | Touch screen interrupt |
| TP_I2C_SDA | I/O | Digital(IOVCC) | I2C interface data pin |
| TP_I2C_SCL | I | Digital(IOVCC) | I2C interface clock pin |
| TP_SPI_CS | I | Digital(IOVCC) | Slave chip select pin in SPI interface<br>-If not used, please fix this pin at IOVCC |
| TP_SPI_MISO | O | Digital(IOVCC) | Slave input data pin in SPI interface |
| TP_SPI_MOSI | I | Digital(IOVCC) | Slave output data pin in SPI interface |
| TP_SPI_SCL | I | Digital(IOVCC) | Slave clock pin in SPI interface |
| FLASH_HOLD | O | Digital(IOVCC) | Hold signal to flash. |
| FLASH_WP | O | Digital(IOVCC) | Write protect signal to flash. |
| FLASH_CS | O | Digital(IOVCC) | Master chip select in SPI interface |
| FLASH_MISO | I | Digital(IOVCC) | Master input data in SPI interface |
| FLASH_MOSI | I/O | Digital(IOVCC) | Master output data in SPI interface |
| FLASH_SCL | I | Digital(IOVCC) | Master clock signal in SPI interface |

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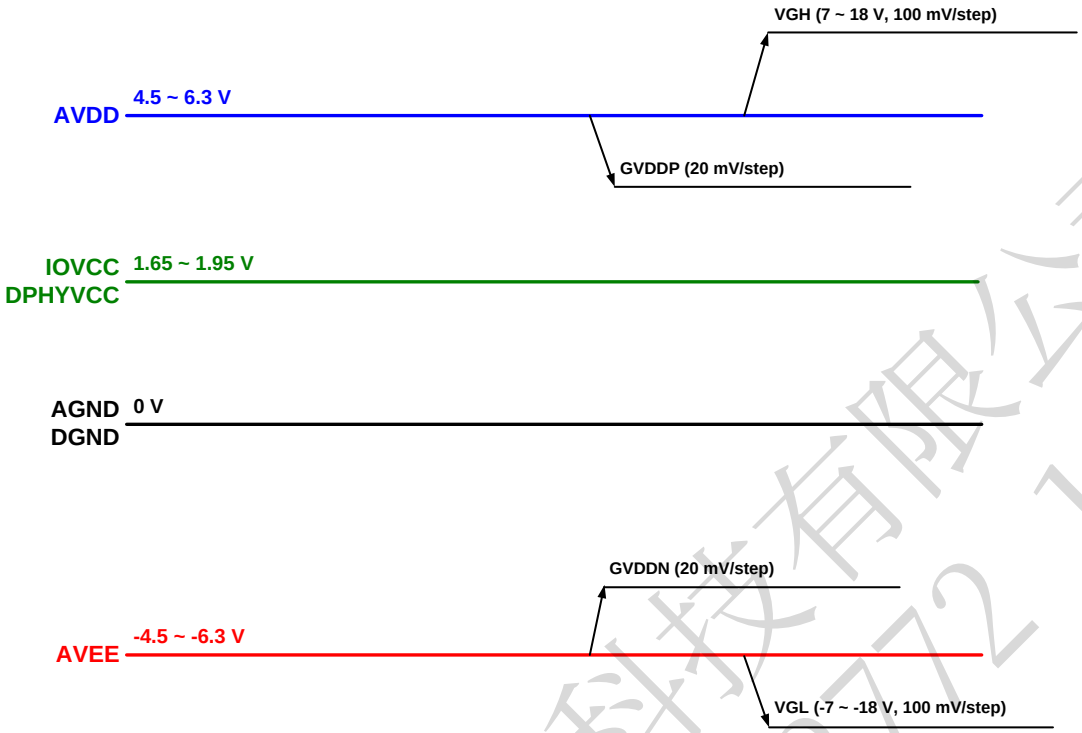
| | | | |
|---|---|---|---|
| RX[512:1] | O | Analog | Output source driver signals. The D/A converted 256-gray-scale analog voltage is output. |
| TP_OPT_1 | I | Digital(IOVCC) | If flash downloading firmware need to be used, this pin need connect to IOVCC. |

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### 4.3 Power Supply Configuration




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# 5 FUNCTION DESCRIPTION

## 5.1 Frame Tearing Effect Interface

The Tearing Effect output can be show the synchronize status with MPU (Host). This signal can be enabled or disabled by the TE off & on commands. The mode of the Tearing Effect signal is defined by the parameter of the Tearing Effect Line On command.

Tearing Effect Line Modes

Mode 1, the Tearing Effect Output signal consists of V-Blanking Information only (Figure 1):

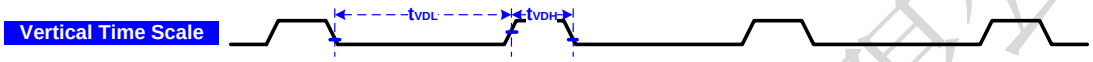



**Figure 1 Mode1**

tVDH= The LCD display is not updated video signal.

tVDL= The LCD display is updated video signal

Mode 2, the Tearing Effect Output signal consists of V-Blanking and H-Blanking Information, there is one V-sync and 800 H-sync pulses per field. (Figure 2)

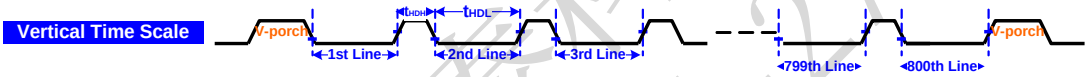



**Figure 2 Mode2**

tHDH= The LCD display is not updated video signal.

tHDL= The LCD display is updated f video signal.

Mode3, in this mode, the tearing effect output when the display reaches line N. The output signal length of the high level is one line period. In below figure, it shows the TE pulse that can be select from 1st line to 800th line by CMD44h.P1 and CMD44.P2 (Figure 3)

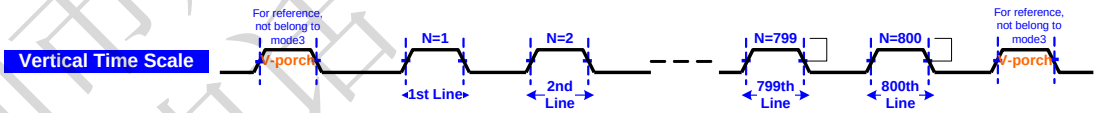



**Figure 3 Mode3**

| CMD 35h | CMD 44h | TE output |
|---|---|---|
| TEM | TESN | |
| 0 | 0 | TE high in V-porch region (mode1) |
| 1 | 0 | TE high in all V-porch and H-porch region (mode2) |
| 0 | ≠0 | TE high at N-th line (mode3) |
| 1 | ≠0 | Same as mode2 |



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Where mode1, mode2 and mode3 timing chart is shown in below (Figure 4):

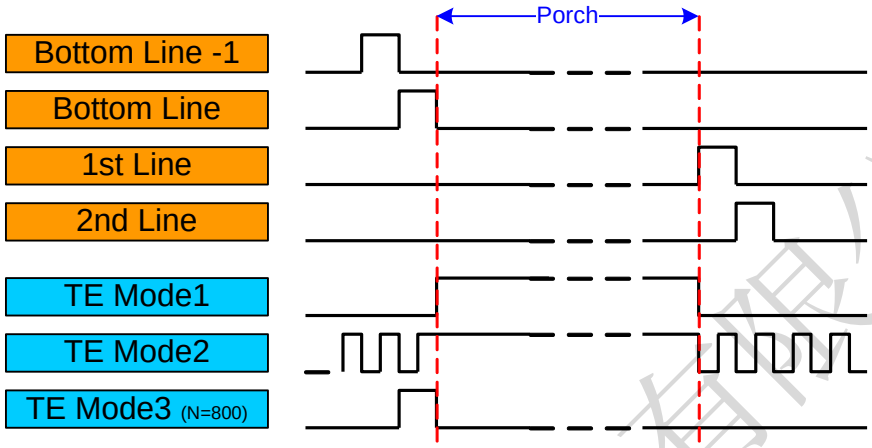



**Figure 4 TE Mode**

Note: during sleep in mode, the Tearing output pin is active low.I

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## 5.2 Content Adaptive Backlight Control (CABC2.0)

### 5.2.1 Definition of CABC

A Content Adaptive Brightness Control function can be used to reduce the power consumption of the luminance source. Content adaptation means that content gray level scale can be increased while simultaneously lowering brightness of the backlight to achieve same perceived brightness. The adjusted gray level scale and thus the power consumption reduction

Definition of Modes and target power reduction ratio:

Off mode: Content Adaptive Brightness Control functionality is totally off.

UI [User Interface] image mode: Optimized for UI image. It is kept image quality as much as possible. Target power consumption reduction ratio: 10% or less.

Still picture mode: Optimized for still picture. Some image quality degradation would be acceptable. Target power consumption reduction ratio: more than 30%.

Moving image mode: Optimized for moving image. It is focused on the biggest power reduction with image quality degradation. Target power consumption reduction ratio: more than 30%.

| | Off ode | UI image mode | Still picture mode | Moving image mode |
|---|---|---|---|---|
| Power Reduction | 0% | 10% or Less | More than 30% | More than 30% |
| Image Quality | Best | Approaching Best | Some image degradation | Some image degradation |

*Note 1: Updating partial area of the image data should be supported by CABC functionality.*
*Note 2: Processing power consumption of CABC should be minimized.*
*Note 3: Customer need program NVM GAMMA when using CABC.*

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The transition time for dimming function is illustrated below.

- Content Adaptive Brightness Control

Display brightness is changed, according to the image contents. The following graph mentions the case of displaying three different images.

- Image A: -20% brightness reduction
- Image B: -30% brightness reduction
- Image C: -10% brightness reduction

Transition time from the previous image to the current displayed image is “transition time A”.

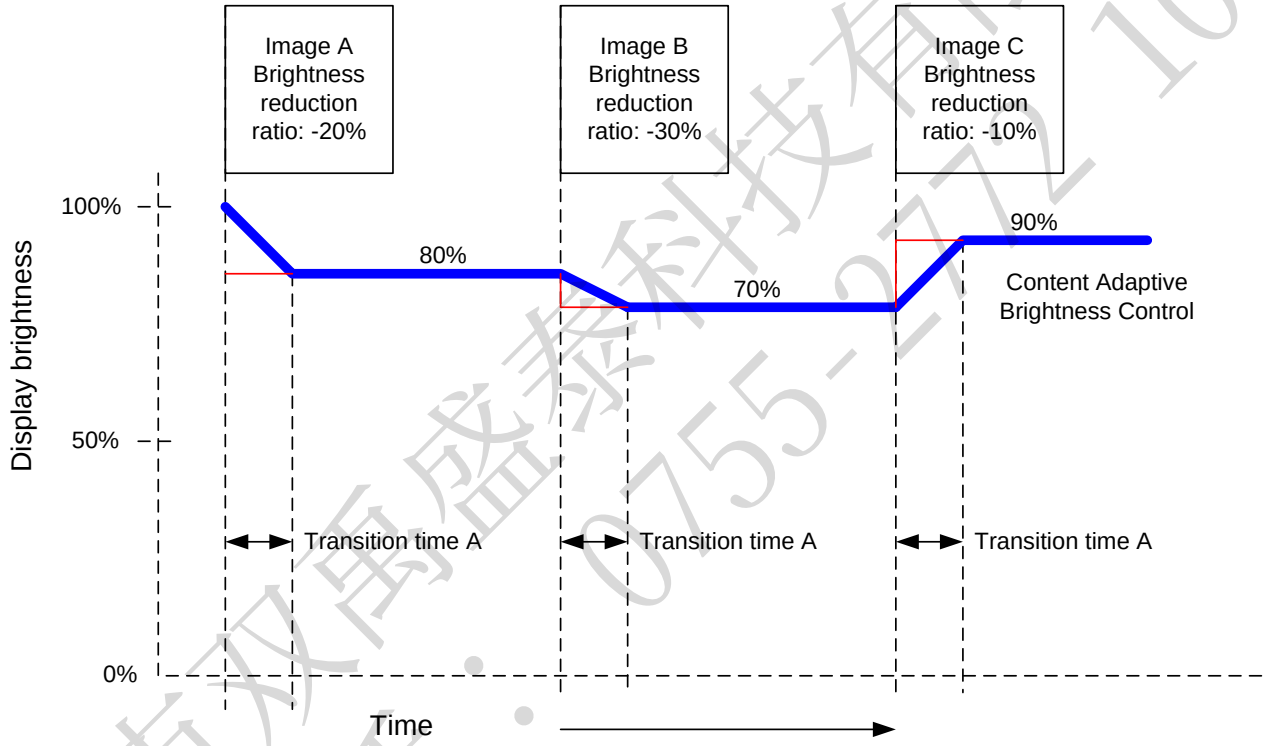



**Figure 5 Transition time A**

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- Manual brightness setting and Dimming function

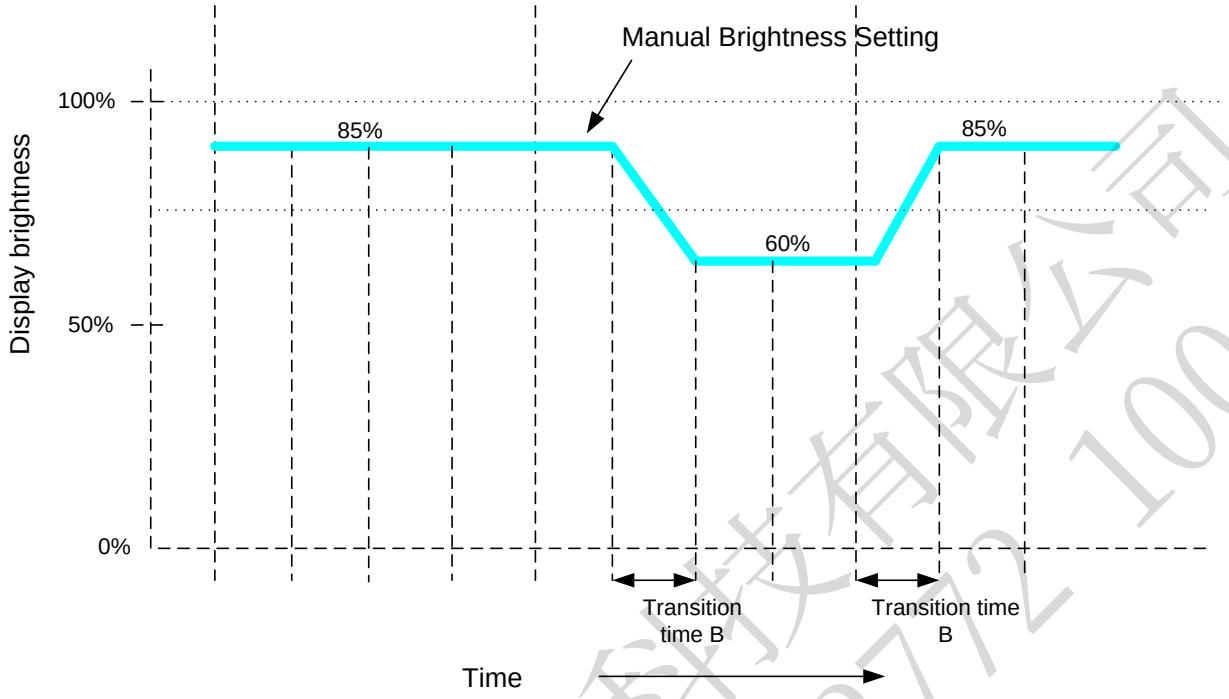



**Figure 6 Manual brightness setting and Dimming function**

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- Combine Display brightness

Green line in the following graph is for the output brightness of display. It is combined with both display brightness, which are defined in the above graphs.

Maximum transition time is transition time A+B.

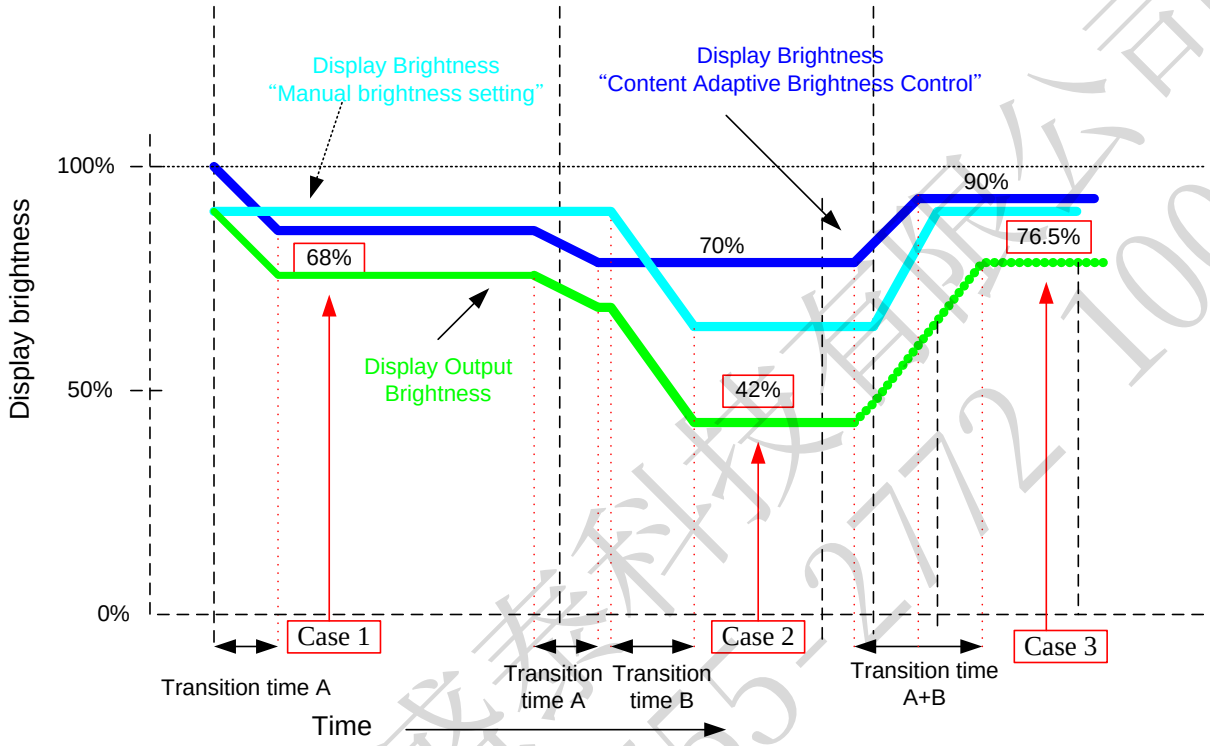



**Figure 7 Maximum transition time is transition time A+B**

Brightness level calculates with the following formula.

Display Output brightness = Manual Brightness setting * CABC brightness ratio

| | Manual Brightness setting | Brightness ratio [CABC] | Display Output brightness |
|---|---|---|---|
| Case 1 | 85% | 80% | 68% |
| Case 2 | 60% | 70% | 42% |
| Case 3 | 85% | 90% | 76.5% |

Transition time from the current brightness to target brightness is A+B in the worst case.

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### 5.2.2 Minimum Brightness Setting of CABC Function

CABC function is automatically reduced backlight brightness based on image contents. In the case of the combination with the LABC or manual brightness setting, display brightness is too dark. It must affect to image quality degradation. CABC minimum brightness setting is to avoid too much brightness reduction. When CABC is active, CABC cannot reduce the display brightness to less than CABC minimum brightness setting. If CABC algorithm works without any abnormal visual effect, image processing function can operate even when the brightness cannot be changed.

This function does not affect to the other function, manual brightness setting. Manual brightness can be set the display brightness to less than CABC minimum brightness. Smooth transition and dimming function can be worked as normal.

When display brightness is turned off (BCTRL=0 of “Write CTRL Display (53h)”), CABC minimum brightness setting is ignored. “Read CABC minimum brightness (5Fh)” always read the setting value of “Write CABC minimum brightness (5Eh)”.

| | WRCABC (55h) | Function | RDCABCMB (5Fh) | Image |
|---|---|---|---|---|
| Sleep-in | | NA | WRCABCMB (5Eh) | |
| CABC off | 00b | Disable | WRCABCMB (5Eh) | Original |
| CABC on | 01b/10b/11b | Enable | WRCABCMB (5Eh) | CABC modified |

Brightness level calculates with the following formula.

Display Output Brightness = Manual brightness setting * CABC brightness ratio

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Below drawing is for the explanation of the CABC minimum brightness setting.

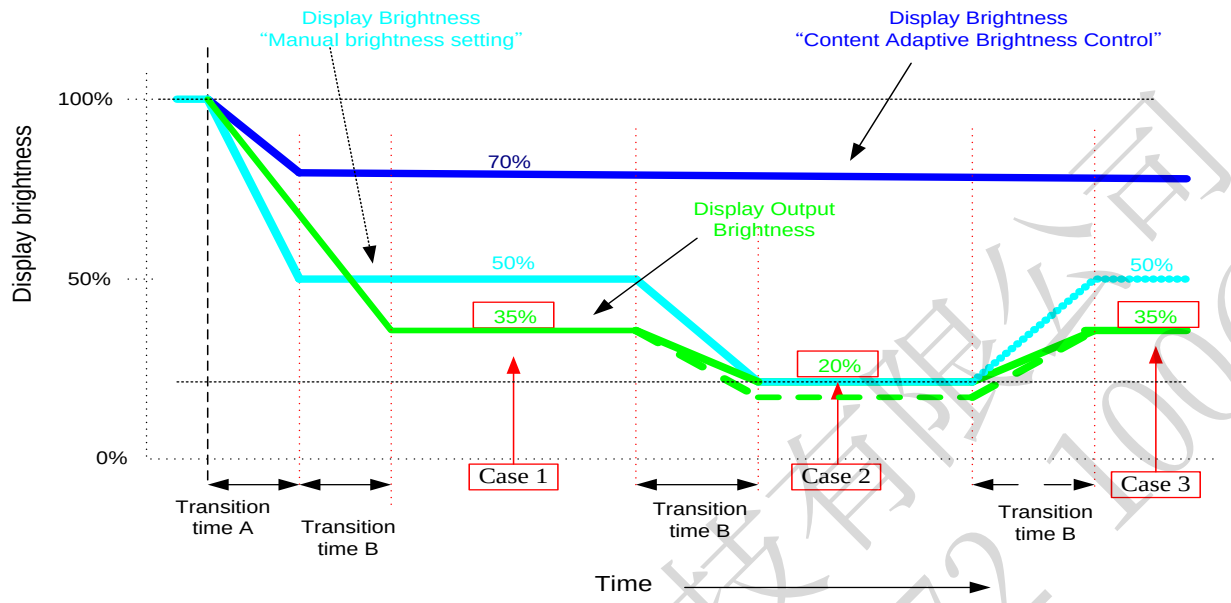



**Figure 8 CABC minimum brightness setting**

CABC minimum brightness value = 51 (33h: 20% display brightness)

| | Display Brightness [manual setting] | Brightness ratio [CABC] | Calculation result of the display brightness formula | Display Output Brightness | Image |
|---|---|---|---|---|---|
| Case 1 | 50% | 70% | 35% | 35% | CABC modified |
| Case 2 | 20% | 70% | 14% | 20% | CABC modified |
| Case 3 | 50% | 70% | 35% | 35% | CABC modified |

At the case 2, the calculation result of the display brightness is 14%. CABC minimum brightness value is set to 20% brightness. Actual display brightness is 20% as the CABC minimum brightness setting.

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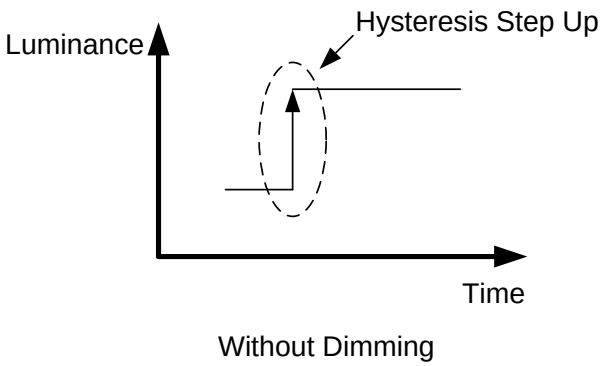


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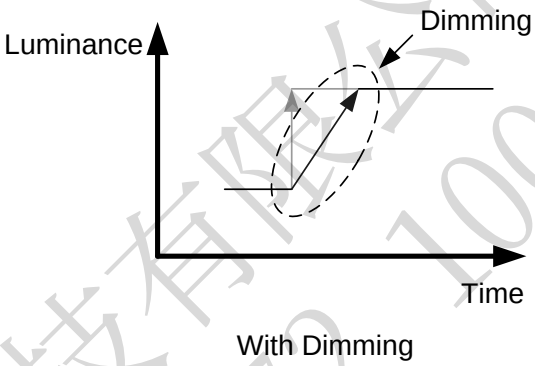
### 5.2.3 Display Dimming

#### 5.2.3.1 General Description

A dimming function (how fast to change the brightness from old to new level and what are brightness levels during the change) is used when changing from one brightness level to another. This dimming function curve is the same in increment and decrement. The basic idea is described below.




Without Dimming




With Dimming

Dimming function can be enable and disable. See “Write CTRL Display (53h)” (bit DD) for more information.

#### 5.2.3.2 Dimming Requirement

Dimming function in the display module should be implemented so that 400-600ms is used for the transition between the original brightness value and the target brightness value. The transferring time steps between these two brightness values are equal making the transition linear.

The dimming function is working similarly in both upward and downward directions.

An upward example is illustrating below

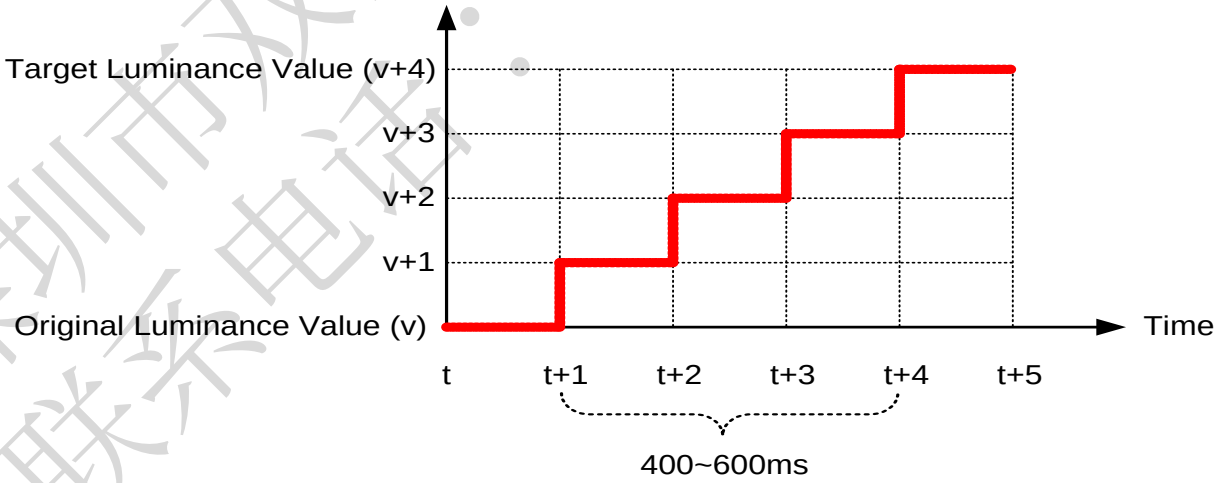



**Figure 9 Dimming Requirement**

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### 5.2.4 Definition of Brightness Transition Time

- Shorter transition time than 500ms.

There is some stable time between transitions. Below drawing is for transition time: 400ms.

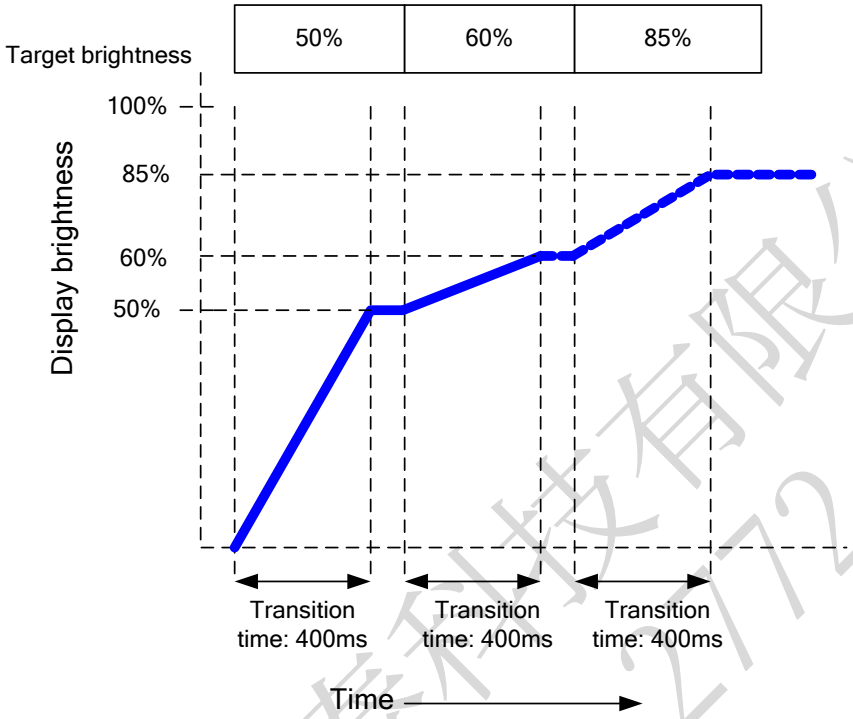



**Figure 10 Shorter Transition time than 500ms**

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- Longer transition time than 500ms

There is no any stable time between transitions. Below drawing is for transition time: 600ms.

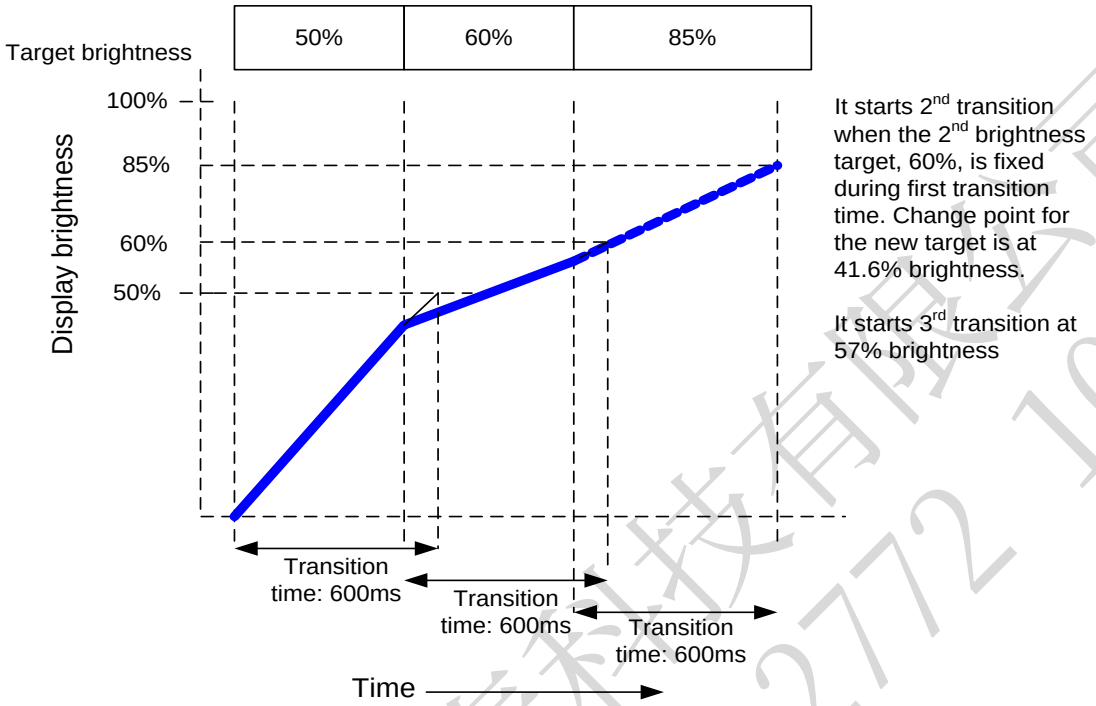



**Figure 11 Longer Transition time than 500ms**

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### 5.3 Color Enhancement (CE2.0)

Color enhancement function enhances the color saturation by gamut expansion.

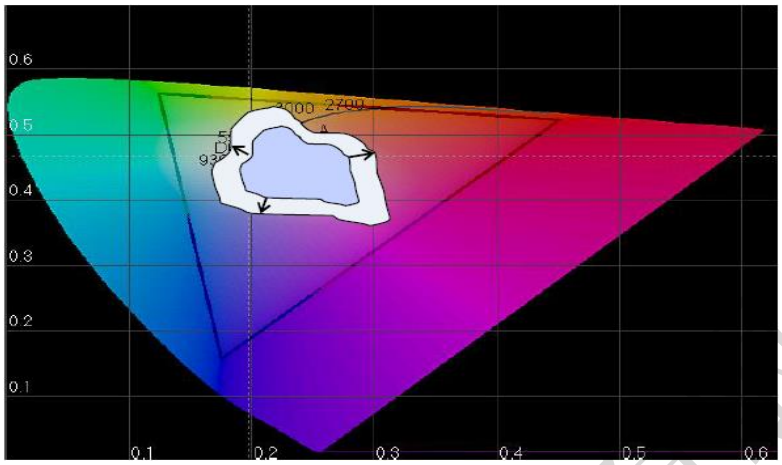
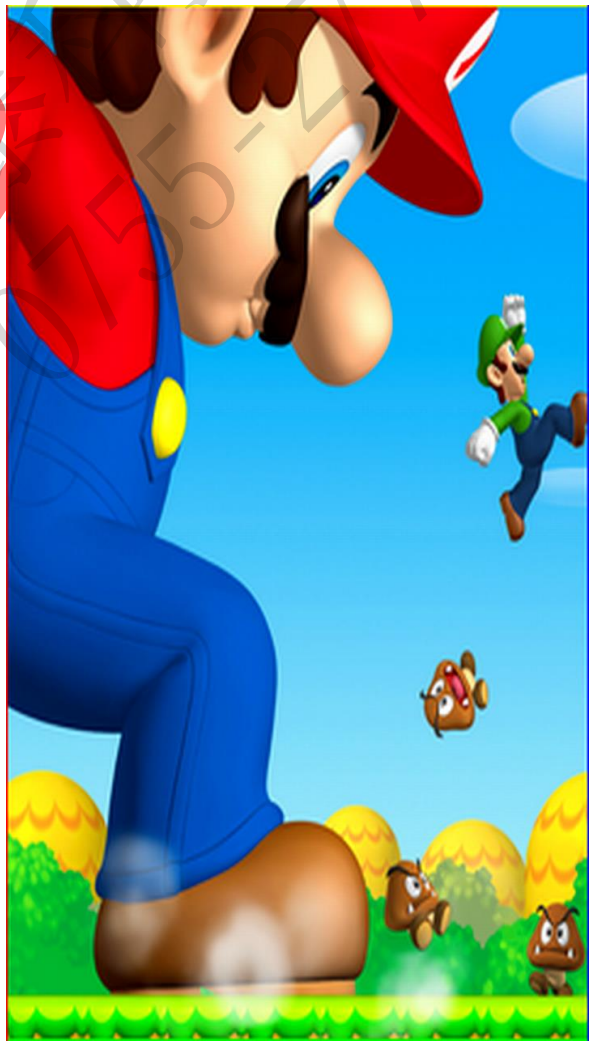



**Figure 12 Color Gamut Expansion**

An example of the color enhancement function is illustrated below:

| Color Enhancement Off | Color Enhancement On |
|---|---|
|  |  |



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### 5.4 MIPI-DSI Interface

The Display Serial Interface standard defines protocols between a host processor and peripheral devices that adhere to MIPI Alliance standards for mobile device interfaces. The DSI standard builds on existing standards by adopting pixel formats and command set defined in MIPI Alliance standards.

DSI-compliant peripherals support either of two basic modes of operation: Command Mode and Video Mode. Which mode is used depends on the architecture and capabilities of the peripheral. The mode definitions reflect the primary intended use of DSI for display interconnect, but are not intended to restrict DSI from operating in other applications.

Typically, a peripheral is capable of Command Mode operation or Video Mode operation. Some Video Mode display modules also include a simplified form of Command Mode operation in which the display module may refresh its screen from a reduced-size, or partial, frame buffer, and the interface (DSI) to the host processor may be shut down to reduce power consumption.

Command Mode refers to operation in which transactions primarily take the form of sending commands to a peripheral, such as a display module, that incorporates a display controller. The display controller may include local registers and a frame buffer. Systems using Command Mode write to, and read from, the registers. The host processor indirectly controls activity at the peripheral by sending commands, parameters to the display controller. The host processor can also read display module status information. Command Mode operation requires a bidirectional interface.

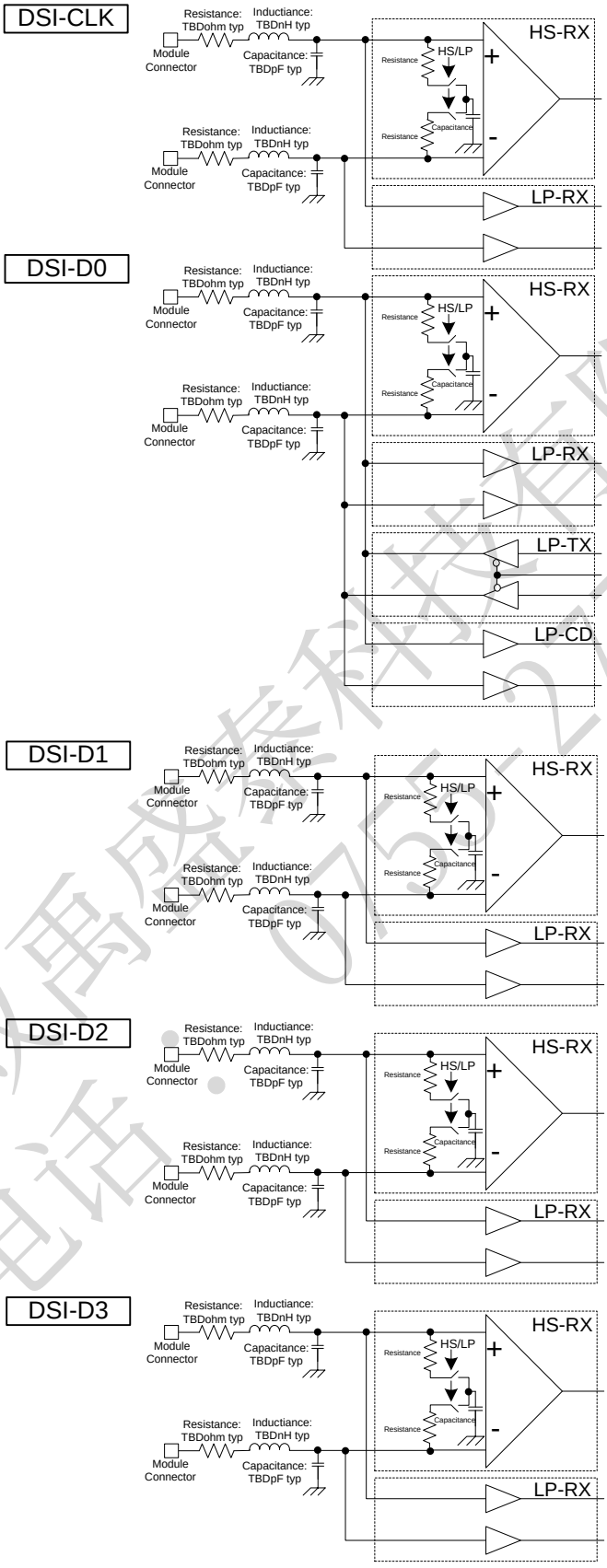
Video Mode refers to operation in which transfers from the host processor to the peripheral take the form of a real-time pixel stream. In normal operation, the display module relies on the host processor to provide image data at sufficient bandwidth to avoid flicker or other visible artifacts in the displayed image. Video information should only be transmitted using High-Speed Mode. Some Video Mode architectures may include a simple timing controller and partial frame buffer, used to maintain a partial-screen or lower-resolution image in standby or Low-Power Mode. This permits the interface to be shut down to reduce power consumption. To reduce complexity and cost, systems that only operate in Video Mode may use a unidirectional data path.

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### 5.4.1 Display Module Pin Configuration for DSI




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### 5.4.2 Display Serial Interface (DSI)

#### 5.4.2.1 General description

The communication can be separated 2 different levels between the MCU and the display module:

- Interface level : Low level communication

- Packet level : High level communication

#### 5.4.2.2 Interface level communication

##### 5.4.2.2.1 General

The display module uses data and clock lane differential pairs for DSI. Both clock lane and data lane0 can be driven Low-Power (LP) or High-Speed (HS) mode. Data lane1, data lane2 and data lane3 can be driven High-Speed mode only.

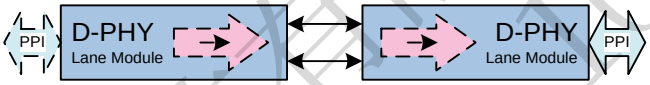
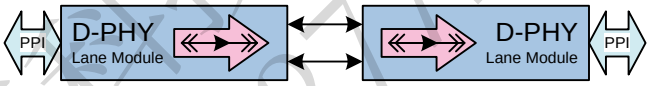
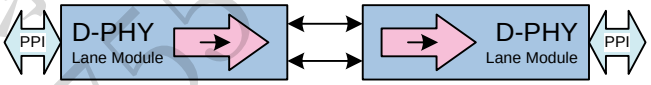
| | Lane support mode | |
|---|---|---|
| Clock Lane | Unidirectional lane<br>High-Speed Clock only<br>Simplified Escape Mode (ULPS Only) |  |
| Data Lane0 | Bi-directional lane<br>Forward high-speed only<br>Bi-directional Escape Mode<br>Bi-direction LPDT |  |
| Data Lane1/2/3 | Unidirectional lane<br>Forward high-speed only<br>Simplified Escape Mode(ULPS Only) |  |

**Figure 13 The Interface Color Lane Types and Support Mode**

Low-Power mode means that each line of the differential pair is used in single end mode and a differential receiver is disable (A termination resistor of the receiver is disable) and it can be driven into a Low-Power mode.

High-Speed mode means that differential pairs (The termination resistor of the receiver is enable) are not used in the single end mode.

There are used different modes and protocols in each mode when there are wanted to transfer information from the MCU to the display module and vice versa.

The State Codes of the High-Speed (HS) and Low-Power (LP) lane pair are defined below.

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| Lane Pair State Code | Line Voltage Levels | | High-Speed(HS) | Low-Power(LP) | |
|---|---|---|---|---|---|
| | Dn+ Line | Dn- Line | Burst Mode | Control Mode | Escape Mode |
| HS-0 | HS Low | HS High | Differential-0 | N/A, Note 1 | N/A, Note 1 |
| HS-1 | HS High | HS Low | Differential-1 | N/A, Note 1 | N/A, Note 1 |
| LP-00 | LP Low | LP Low | N/A | Bridge | Space |
| LP-01 | LP Low | LP High | N/A | HS-Request | Mark-0 |
| LP-10 | LP High | LP Low | N/A | LP-Request | Mark-1 |
| LP-11 | LP High | LP High | N/A | Stop | N/A, Note 2 |

**High-Speed and Low-Power Lane Pair State Descriptions**

*Notes:*

*1. During High-Speed transmission the Low-Power observe LP-00 on the Lines.*

*2. If LP-11 occurs during Escape mode the Lane returns to Stop state.*

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#### 5.4.2.2.2 DSI-CLOCK Lanes

DSI-CLK+/- lanes can be driven into three different power modes: Low-Power Mode (LPM LP-11), Ultra-Low Power Mode (ULPM) or High-Speed Clock Mode (HSCM).

Clock lanes are in a single end mode (LP = Low-Power) when there is entering or leaving Low-Power Mode (LPM) or Ultra-Low Power Mode (ULPM).

Clock lanes are in the single end mode (LP = Low-Power) when there is entering in or leaving out High-Speed Clock Mode (HSCM).

These entering and leaving protocols are using clock lanes in the single end mode to generate an entering or leaving sequences.

The principal flow chart of the different clock lanes power modes is illustrated below.

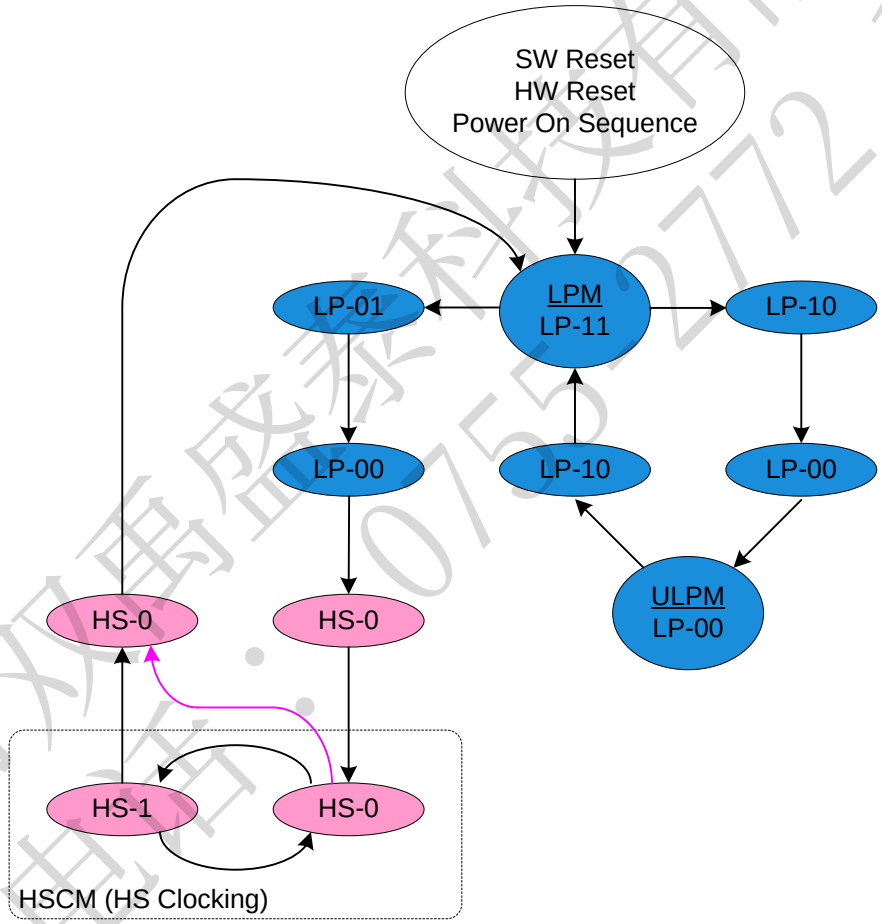



**Figure 14 Clock Lanes Power Modes**

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#### 5.4.2.2.1 Low-Power Mode (LPM)

DSI-CLK+/- lanes can be driven to the Low-Power Mode (LPM),when DSI-CLK lanes are entering LP-11 State Code , in three different ways:

After SW Reset, HW Reset or Power On Sequence=>LP-11

After DSI-CLK+/- lanes are leaving Ultra Low Power Mode (ULPM, LP-00 State Code) =>LP10=>LP-11(LPM).

This sequence is illustrated below.

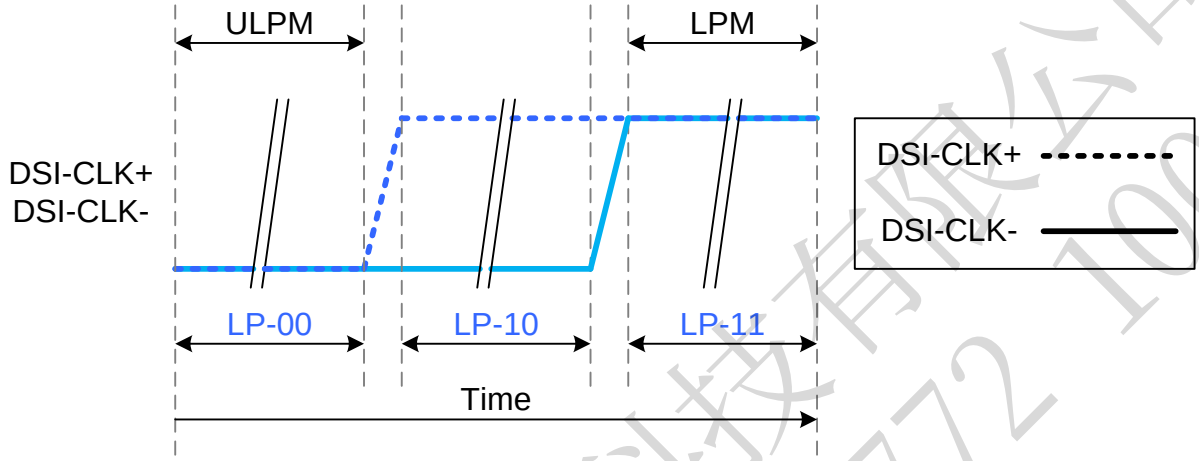



**Figure 15 From ULPM to LPM**

After DSI-CLK+/- lanes are leaving High-Speed Clock Mode (HSCM, HS-0 or HS-1 State Code) =>HS-0 =>LP-11 (LPM).

This sequence is illustrated below.

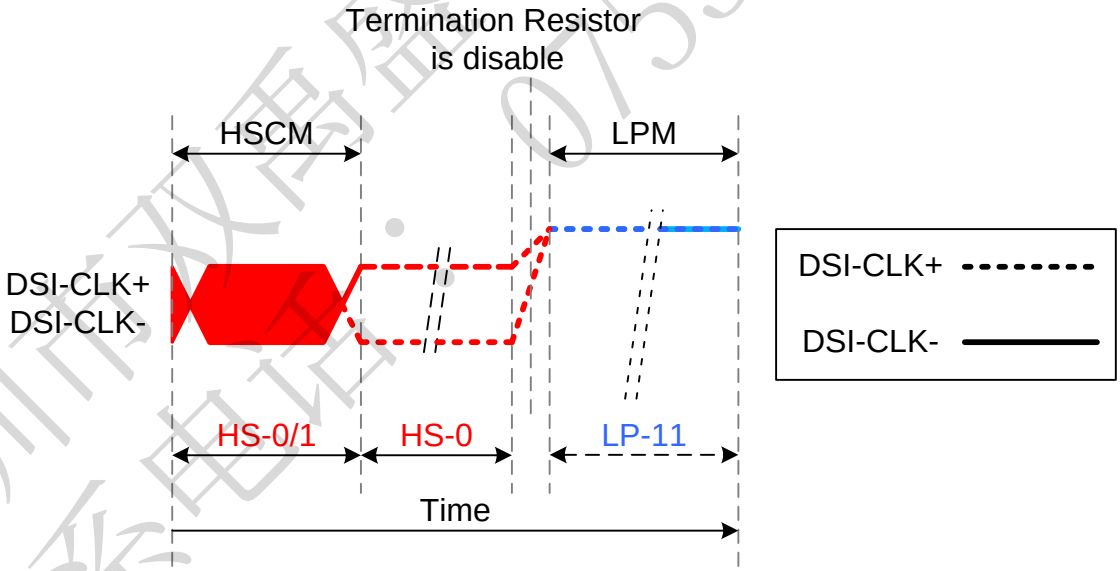



**Figure 16 From HSCM to LPM**

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All three mode changes are illustrated a flow chart below.

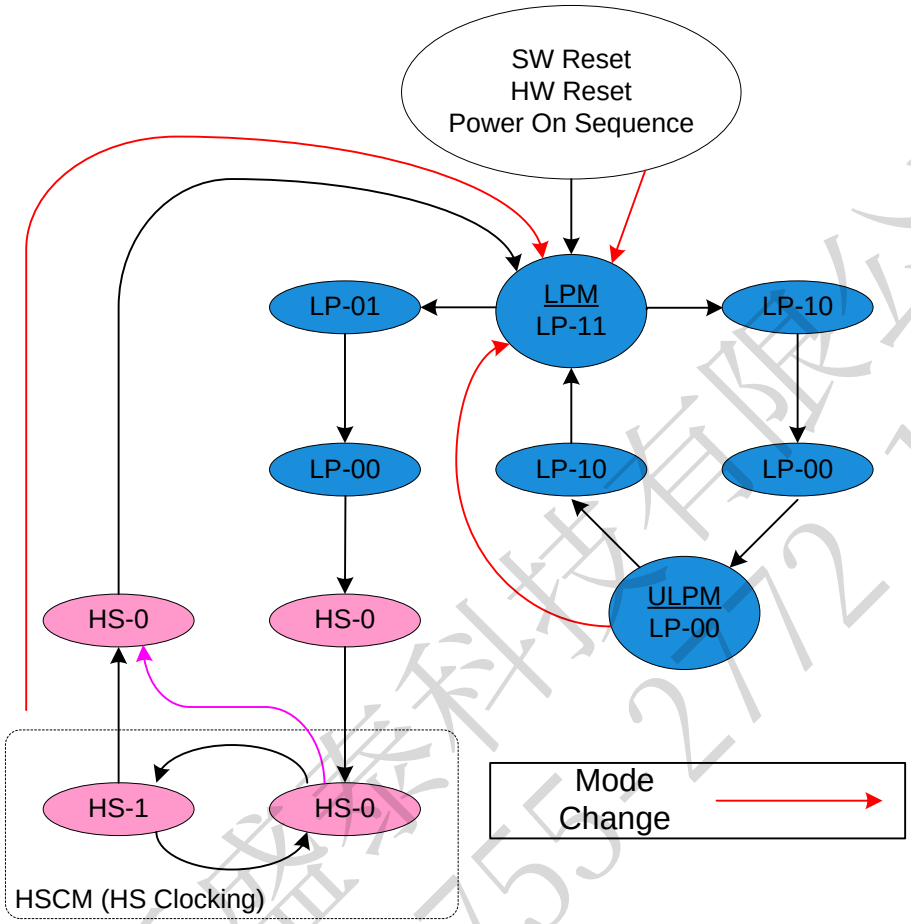



**Figure 17 All three mode changes to LPM**

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#### 5.4.2.2.2.2 Ultra-Low Power Mode (ULPM)

DSI-CLK+/- lanes can be driven to the Ultra-Low Power Mode (ULPM), when DSI-CLK lanes are entering LP-00 State Code. The only entering possibility if from the Low-Power Mode (LPM, LP-11 State Code) => LP-10 => LP-00 (ULPM).

This sequence is illustrated below:

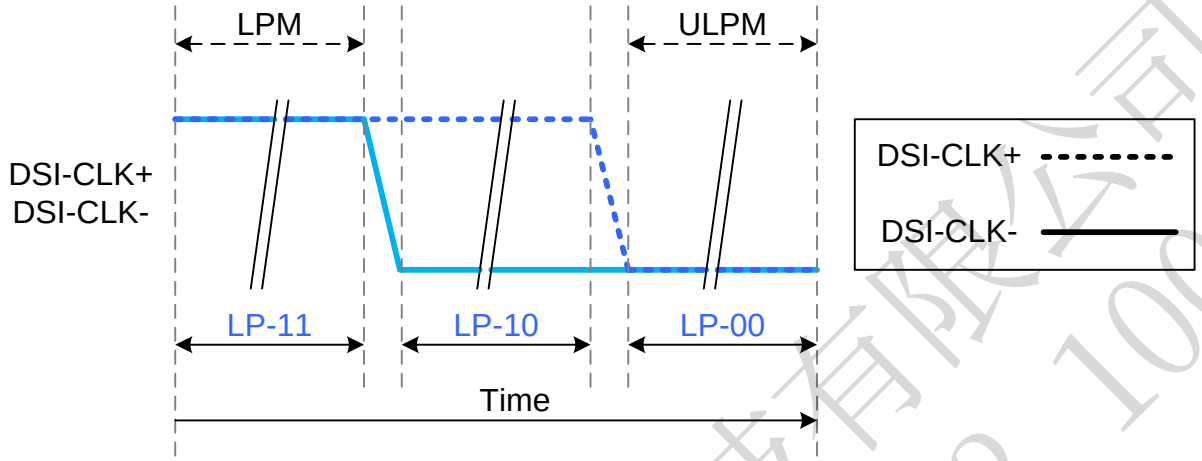



**Figure 18 From LPM to UPLM**

The mode change is also illustrated below:

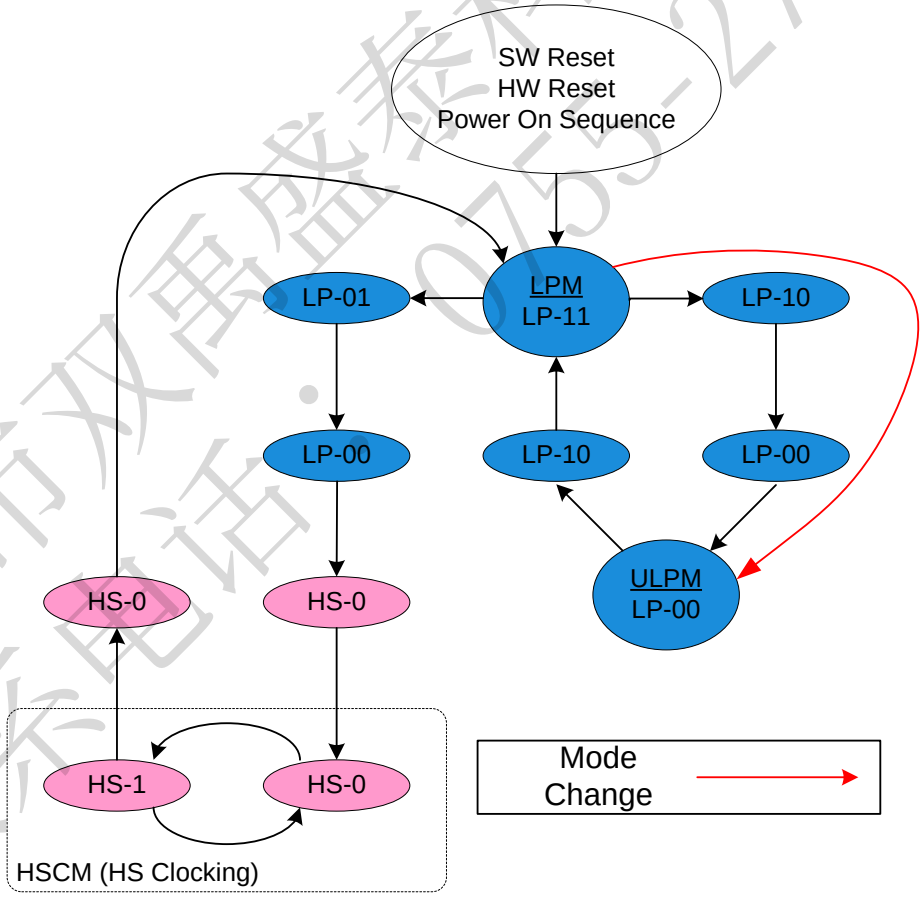



**Figure 19 The mode change from LPM to UPLM**



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#### 5.4.2.2.2.3 High-Speed Clock Mode (HSCM)

DSI-CLK+/- lanes can be driven to the High-Speed Clock Mode (HSCM), when DSI-CLK lanes are starting to work between HS-0 and HS-1 State Codes. The only entering possibility is from the Low-Power Mode (LPM, LP-11 State Code) =>LP-01 =>LP-00 =>HS-0 =>HS-0/1 (HSCM). This sequence is illustrated below.

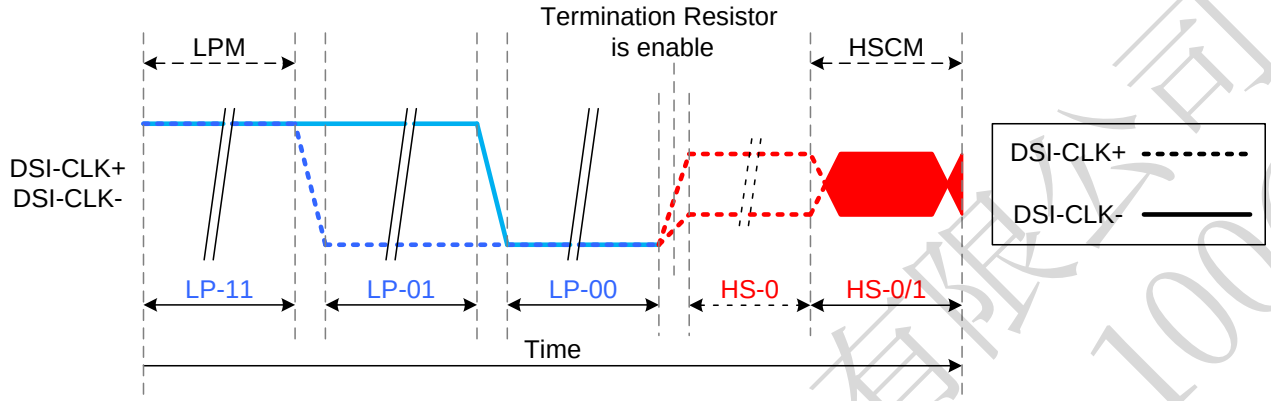



**Figure 20 From LPM to HSCM**

The mode change is also illustrated below:

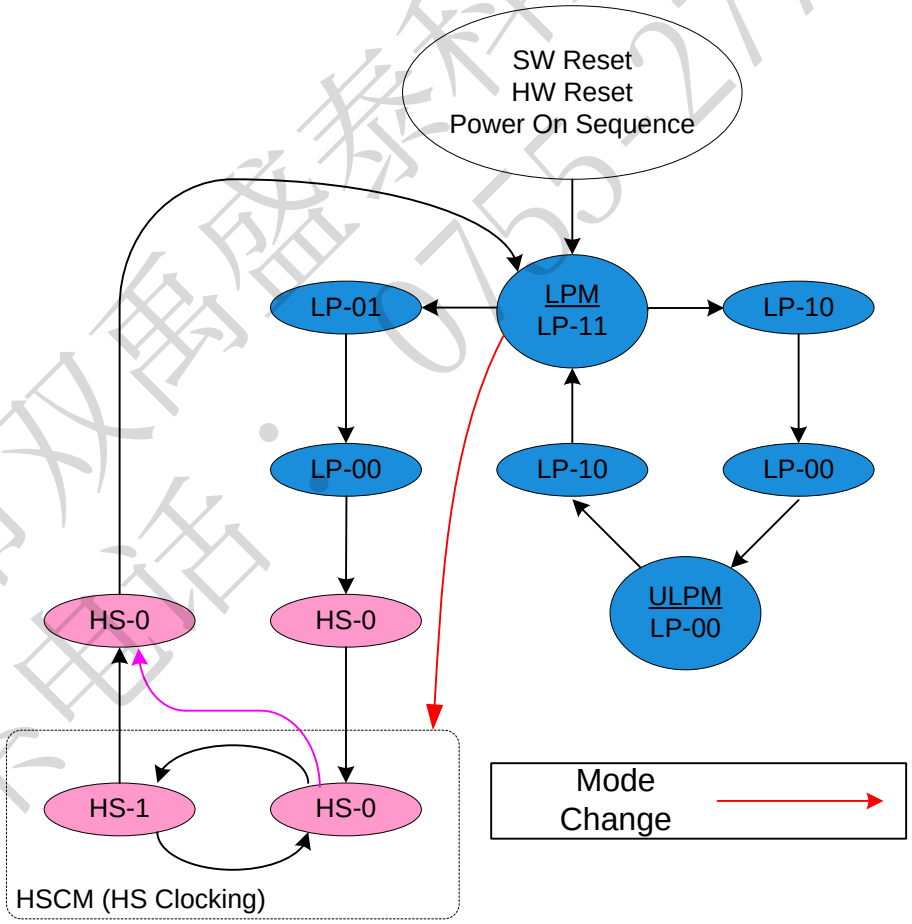



**Figure 21 Mode Change from LPM to HSCM on the Flow Chart**



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The High-Speed clock (DSI-CLK+/-) is started before High-Speed data is sent via DSI-Dn+/- lanes. The High-Speed clock continues clocking after the High-Speed data sending has been stopped.

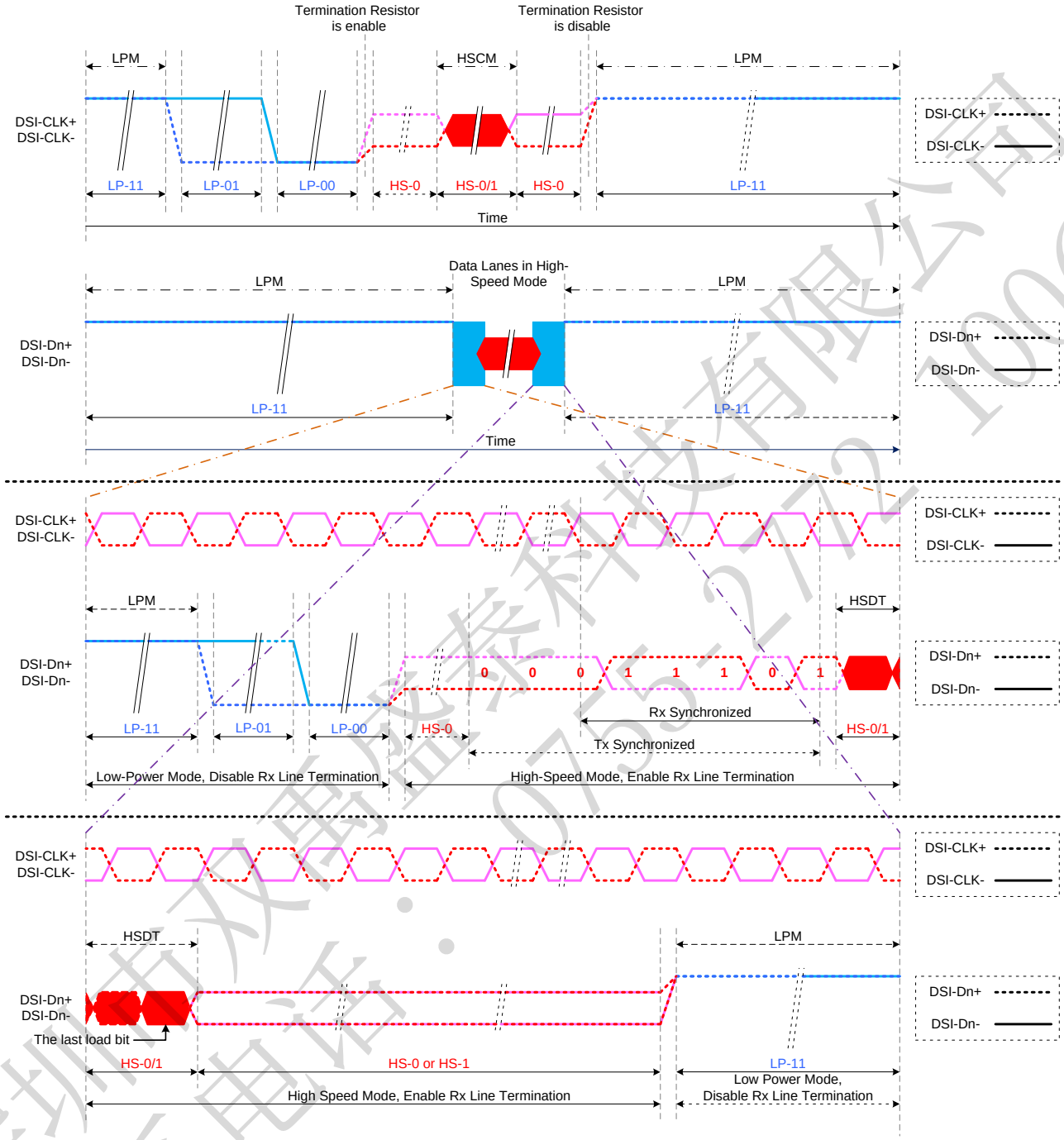



**Figure 22 High-Speed Clock Burst**

*Note:*

*If the last load bits is HS-0, the transmitter changes form HS-0 to HS-1.*

*If the last load bits is HS-1, the transmitter changes form HS-1 to HS-0.*



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#### 5.4.2.2.3 DSI-DATA Lanes

##### 5.4.2.2.3.1 General

DSI-D0+/- data lanes can be driven in different modes which are:

• Escape Mode (Only DSI-D0+/- data lane is used)

• High-Speed Data Transmission (DSI-D0+/-, DSI-D1+/-, DSI-D2+/- and DSI-D3+/- data lanes are used)

• Bus Turnaround Request (Only DSI-D0+/- data lane is used)

These modes and their entering codes are defined on the following table.

| Mode | Entering Mode Sequence | Leaving Mode Sequence |
|---|---|---|
| Escape Mode | LP-11=>LP-10=>LP-00=>LP-01=>LP-00 | LP-00=>LP-10=>LP11(Mark1) |
| High-Speed Data Transmission | LP-11=>LP-01=>LP-00=>HS-0 | (HS-0 or HS-1) =>LP-11 |
| Bus Turnaround Request | LP-11=>LP-10=>LP-00=>LP-10=>LP-00 | High-Z |

**Entering and leaving sequence**

##### 5.4.2.2.3.2 ESCAPE MODE

Data lane0 (DSI-D0+/-) can be used in different Escape Modes when data lanes are in Low-Power (LP) mode.

These Escape Modes are used to:

• Send “Low-Power Data Transmission” (LPDT) e.g. from the MCU to the display module

• Drive data lanes to “Ultra-Low Power State” (ULPS)

• Indicate “Remote Application Reset” (RAR), which is reset the display module

• Indicate “Tearing Effect” (TEE), which is used for a TE trigger event from the display module to the MCU

• Indicate “Acknowledge” (ACK), which is used for a non-error event from the display module to the MCU

The basic sequence of the Escape Mode is as follow

• Start: LP-11

• Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00

• Escape Command (EC), which is coded, when one of the data lanes is changing from low-to-high-to-low then this changed data lane is presenting a value of the current data bit (DSI-D0+ = 1, DSI-D0- = 0) e.g. when DSI-D0- is changing from low-to-high-to-low, the receiver is latching a data bit, which value is logical 0. The receiver is using this low-to-high-to-low transition for its internal clock.

• A load if it is needed

• Exit Escape (Mark-1) LP-00 =>LP-10 =>LP-11

• End: LP-11

This basic construction is illustrated below:

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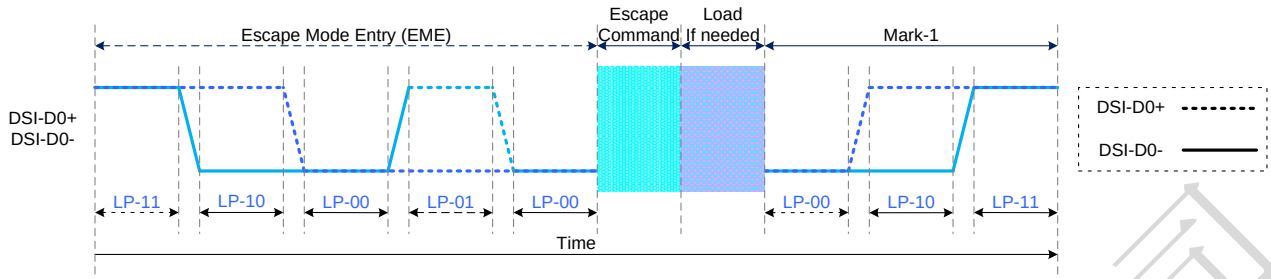



**Figure 23 General Escape Mode Sequence**

The number of the different Escape Commands (EC) is eight. These eight different escape commands (EC) can be divided 2 different groups: Mode or Trigger. The MCU is informing to the display module that it is controlling data lanes (DSI-D0+/-) with the mode e.g. The MCU can inform to the display module that it can put data lanes in the Low-Power mode. The MCU is waiting from the display module event information, which has been set by the MCU, with the trigger e.g. when the display module reaches a new V-synch, the display module sent to the MCU a TE trigger (TEE), if the MCU has been requested it.

Escape commands are defined on the next table.

This basic construction is illustrated below:

| Escape Command | Command Type Mode/Trigger | Entry Command Pattern (First Bit➔Last Bit Transmitted) | Dn | D0 |
|---|---|---|---|---|
| Low-Power Data Transmission | Mode | 1110 0001 bin | - | ○ |
| Ultra-Low Power Mode | Mode | 0001 1110 bin | ○ | ○ |
| Underfined-1, Note 1 | Mode | 1001 1111 bin | - | - |
| Underfined-2, Note 1 | Mode | 1101 1110 bin | - | - |
| Remote Application Reset | Trigger | 0110 0010 bin | - | ○ |
| Tearing Effect | Trigger | 0101 1101 bin | - | - |
| Acknowledge | Trigger | 0010 0001 bin | - | ○ |
| Unknow-5, Note 1 | Trigger | 1010 0000 bin | - | - |

*Notes:*

*1. This Escape command support has not been implemented on the display module.*

*2. n=1.*

*3. "○"=Supported*

*4. "-"=Not Supported*

*5. Tearing Effect Trigger cannot be used in MIPI Video mode.*

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**Low-Power Data Transmission (LPDT)**

The MCU can send data to the display module in Low-Power Data Transmission (LPDT) mode when data lanes are entering in Escape Mode and Low-Power Data Transmission (LPDT) command has been sent to the display module. The display module is also using the same sequence when it is sending data to the MCU.

The Low-Power Data Transmission (LPDT) is using a following sequence:

• Start: LP-11

• Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00

• Low-Power Data Transmission (LPDT) command in Escape Mode: 1110 0001 (First to Last bit)

• Load (Data):

One or more bytes (8 bits)

Data lanes are in pause mode when data lanes are stopped (Both lanes are low ) between bytes

• Mark-1: LP-00 =>LP-10 =>LP-11

• End: LP-11

This sequence is illustrated for reference purposes below:

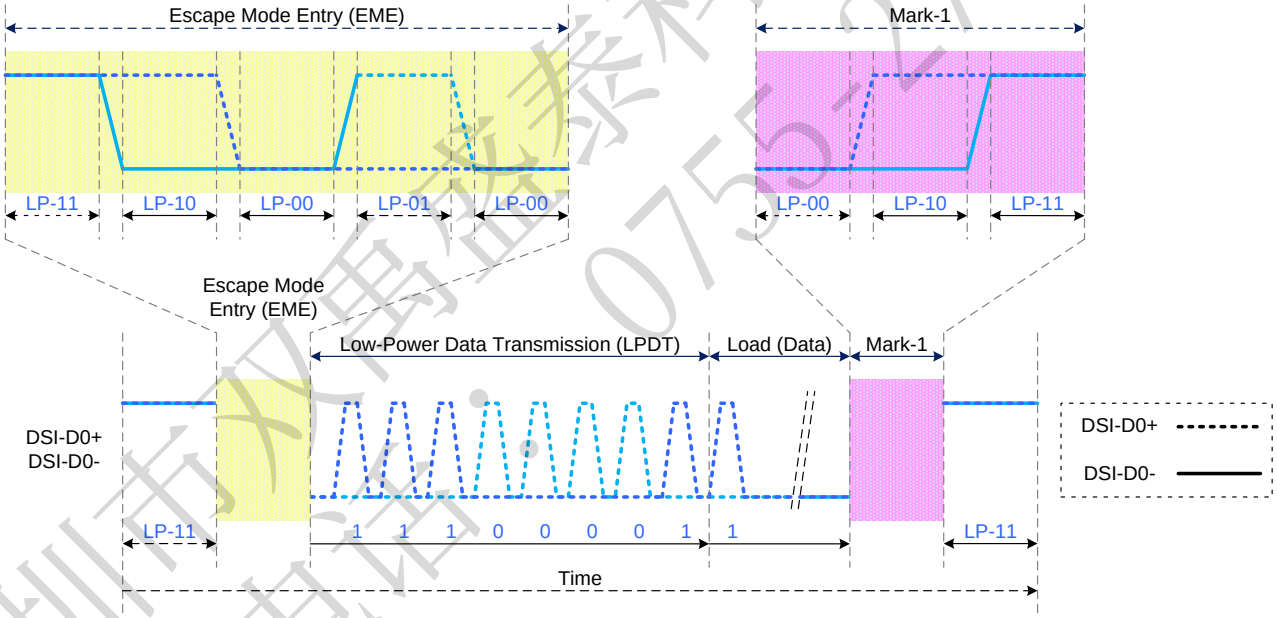



**Figure 24 Low-Power Data Transmission (LPDT)**

*Note:*

*Load(Data) is presenting that the first bit is logical '1' in this example*



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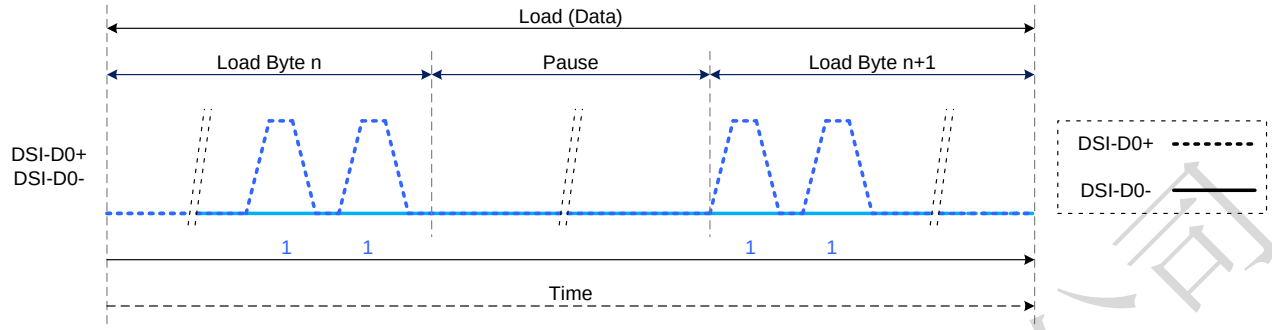



**Figure 25 Pause (Example)**

## Ultra-Low Power State (ULPS)

The MCU can force data lanes in Ultra-Low Power State (ULPS) mode when data lanes are entering in Escape Mode.

The Ultra-Low Power State (ULPS) is using a following sequence:

- Start: LP-11
- Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Ultra-Low Power State (ULPS) command in Escape Mode: 0001 1110 (First to Last bit)
- Ultra-Low Power State (ULPS) when the MCU is keeping data lanes low
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:

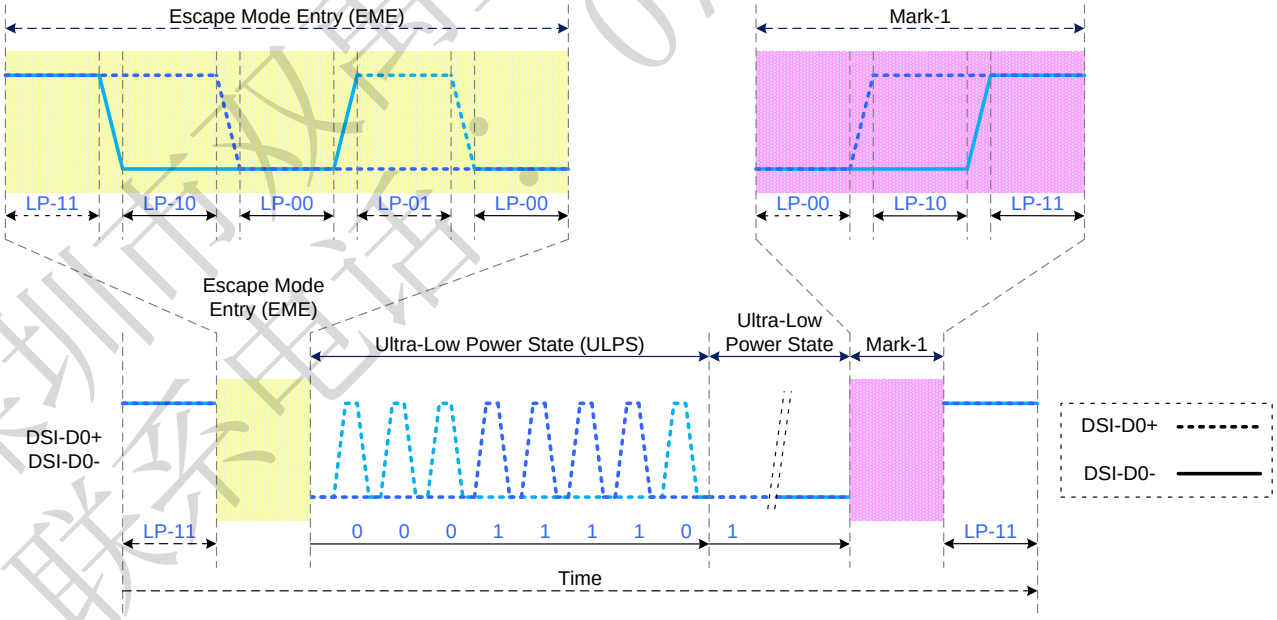



**Figure 26 Ultra-Low Power State (ULPS)**



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**Remote Application Reset (RAR)**

The MCU can inform to the display module that it should be reset in Remote Application Reset (RAR) trigger when data lanes are entering in Escape Mode.

The Remote Application Reset (RAR) is using a following sequence:

• Start: LP-11

• Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00

• Remote Application Reset (RAR) command in Escape Mode: 0110 0010 (First to Last bit)

• Mark-1: LP-00 =>LP-10 =>LP-11

• End: LP-11

This sequence is illustrated for reference purposes below:

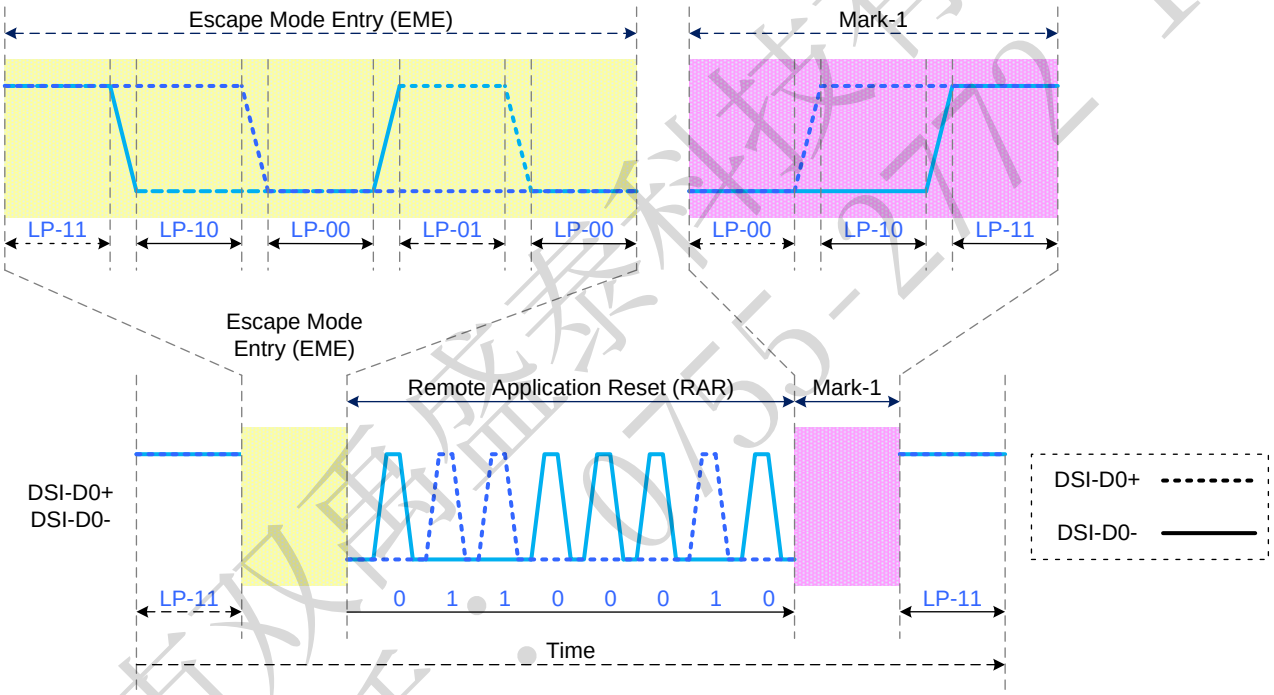



**Figure 27 Remote Application Reset (RAR)**

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**Tearing Effect (TEE)**

The display module can inform to the MCU when a tearing effect event (New V-synch) has been happen on the display module by Tearing Effect (TEE).

The Tearing Effect (TEE) is using a following sequence:

• Start: LP-11

• Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00

• Tearing Effect (TEE) trigger in Escape Mode: 0101 1101 (First to Last bit)

• Mark-1: LP-00 =>LP-10 =>LP-11

• End: LP-11

This sequence is illustrated for reference purposes below:

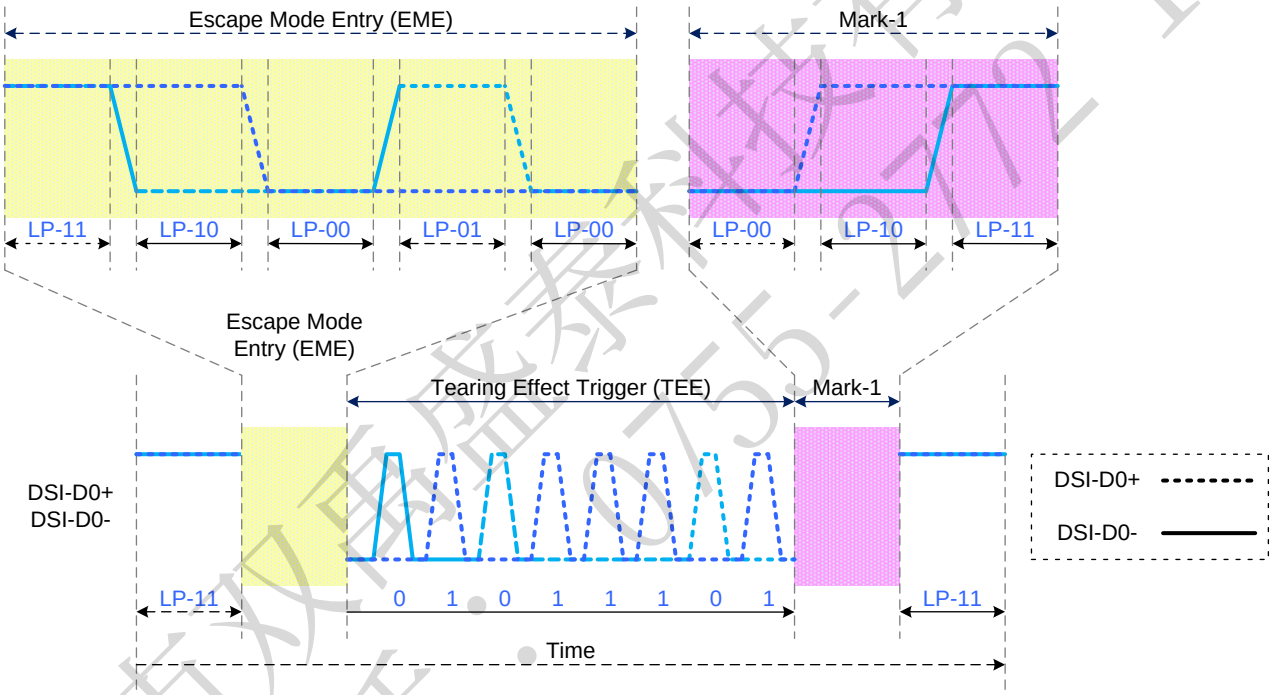



**Figure 28 Tearing Effect (TEE)**

*Note: Tearing Effect (TEE) can not be used in MIPI Video Mode*

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**Acknowledge (ACK)**

The display module can inform to the MCU when an error has not recognized on it by Acknowledge (ACK).

The Acknowledge (ACK) is using a following sequence:

• Start: LP-11

• Escape Mode Entry (EME): LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00

• Acknowledge (ACK) command in Escape Mode: 0010 0001 (First to Last bit)

• Mark-1: LP-00 =>LP-10 =>LP-11

• End: LP-11

This sequence is illustrated for reference purposes below:

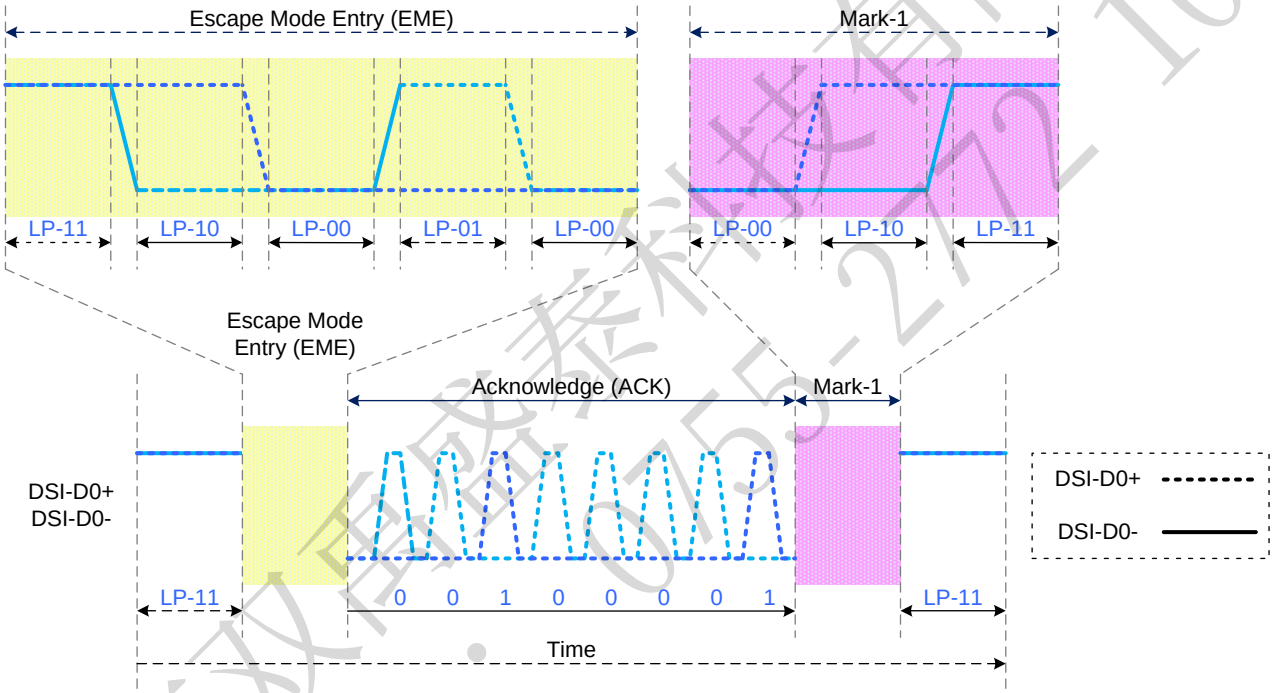



**Figure 29 Acknowledge (ACK)**

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#### 5.4.2.2.3.3 High-Speed Data Transmission (HSDT)

**Entering High-Speed Data Transmission ($T_{SOT}$ of HSDT)**

The display module is entering High-Speed Data Transmission (HSDT) when Clock lanes DSI-CLK+/- have already been entered in the High-Speed Clock Mode (HSCM) by the MCU. See more information on chapter “High-Speed Clock Mode (HSCM)”.

Data lanes of the display module are entering ($T_{SOT}$) in the High-Speed Data Transmission (HSDT) as follows

- Start: LP-11
- HS-Request: LP-01
- HS-Settle: LP-00 => HS-0 (Rx: Lane Termination Enable)
- Rx Synchronization: 011101 (Tx (= MCU) Synchronization: 0001 1101)
- End: High-Speed Data Transmission (HSDT) – Ready to receive High-Speed Data Load

This same entering High-Speed Data Transmission ($T_{SOT}$ of HSDT) sequence is illustrated below

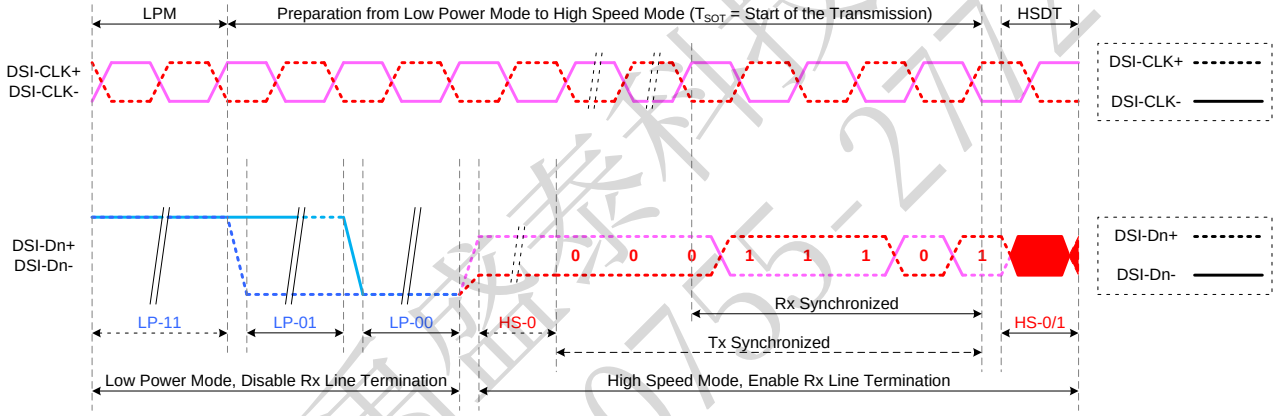



**Figure 30 Entering High-Speed Data transmission ($T_{SOT}$ of HSDT)**

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### Leaving High-Speed Data Transmission ($T_{EOT}$ of HSDT)

The display module is leaving the High-Speed Data Transmission ($T_{EOT}$ of HSDT) when Clock lanes DSI-CLK+/- are in the High-Speed Clock Mode (HSCM) by the MCU and this HSCM is kept until data lanes are in LP-11 mode. See more information on chapter “High-Speed Clock Mode (HSCM)”.

Data lanes of the display module are leaving from the High-Speed Data Transmission ($T_{EOT}$ of HSDT) as follows

• Start: High-Speed Data Transmission (HSDT)

• Stops High-Speed Data Transmission

MCU changes to HS-1, if the last load bit is HS-0

MCU changes to HS-0, if the last load bit is HS-1

• End: LP-11 (Rx: Lane Termination Disable)

This same leaving High-Speed Data Transmission ($T_{EOT}$ of HSDT) sequence is illustrated below

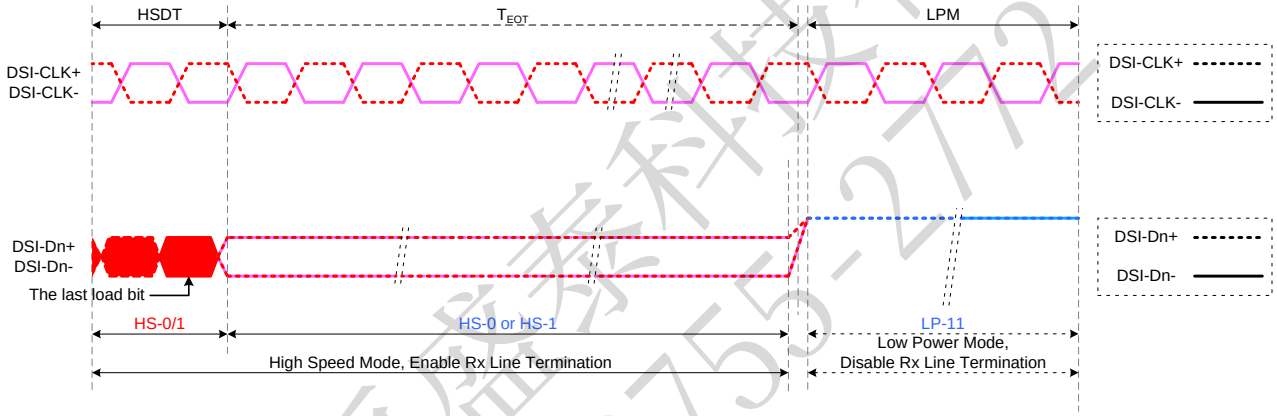



**Figure 31 Leaving High-Speed data Transmission ($T_{EOT}$ of HSDT)**

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**Burst of the High-Speed Data Transmission (HSDT)**

The burst of the High-Speed Data Transmission (HSDT) can consist of one data packet or several data packets. These data packets can be Long (LPa) or Short (SPa) packets.

These different burst of the High-Speed Data Transmission (HSDT) cases are illustrated for reference purposes below.

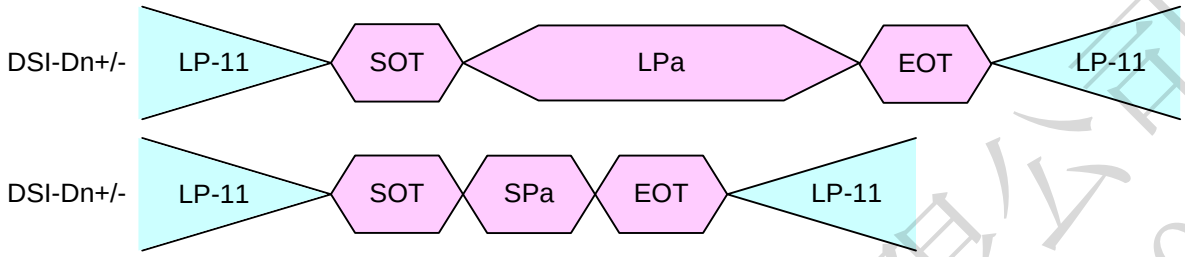



**Figure 32 Single Packet in High-Speed Data Transmission with EoT packet disabled**

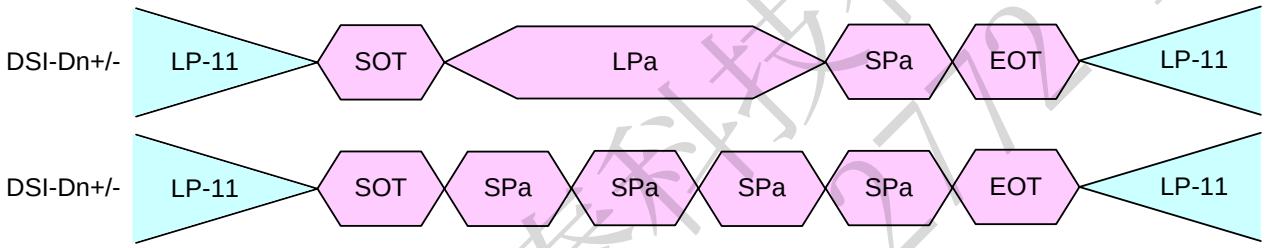



**Figure 33 Multiple Packets in High-Speed Data Transmission with EoT packet disabled**

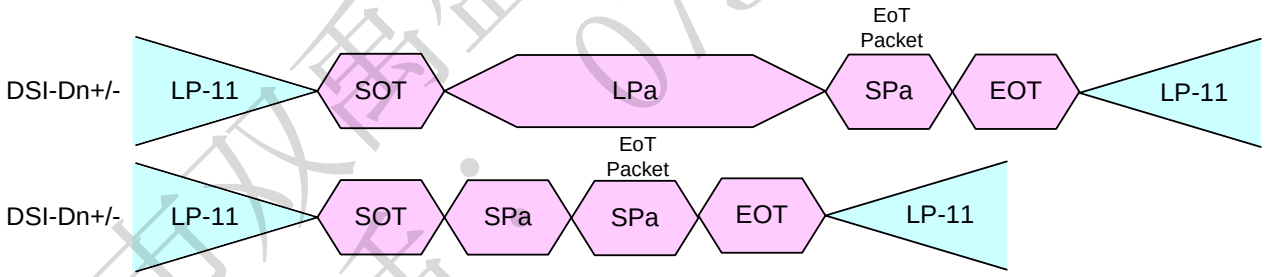



**Figure 34 Single Packet in High-Speed Data Transmission with EoT packet enable**

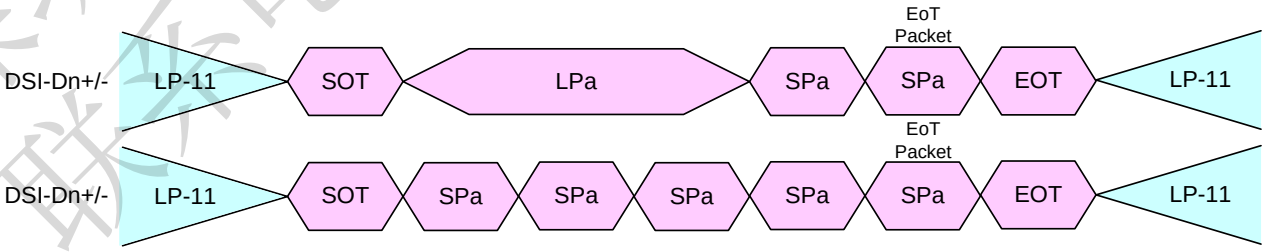



**Figure 35 Multiple Packets in High-Speed Data Transmission with EoT packet enable**



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| Abbreviation | Explanation |
|---|---|
| EoT | End of the Transmission |
| LPa | Long Packet |
| LP-11 | Low-Power Mode, Data lanes are'1's (Stop Mode) |
| SPa | Short Packet |
| SoT | Start of the Transmission |

#### 5.4.2.2.3.4 Bus Turnaround (BTA)

The MCU or display module, which is controlling DSI-D0+/- Data Lanes, can start a bus turnaround procedure when it wants information from a receiver, which can be the MCU or display module.

The MCU or display module is using the same sequence when this bus turnaround procedure is used.

This sequence is described for reference purposes, when the MCU wants to do the bus turnaround procedure to the display module, as follow.

• Start (MCU): LP-11

• Turnaround Request (MCU): LP-11 => LP-10 => LP-00 => LP-10 => LP-00

• The MCU wait until the display module is starting to control DSI-D0+/- data lanes and the MCU stop to control DSI-D0+/- data lanes (=High-Z)

• The display module changes to the stop mode: LP-00 => LP-10 => LP-11

The same bus turnaround .procedure (From the MCU to the display module) is illustrated below.

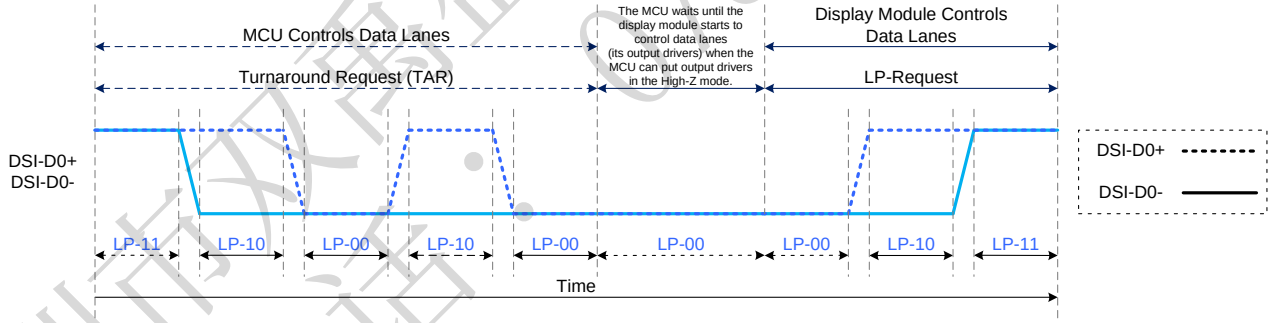



**Figure 36 Bus Turnaround Procedure**

MCU and the display module terms are switched on above figure, if the Bus Turnaround (BTA) is from the display module to the MCU..



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#### 5.4.2.3 Packer Level Communication

##### 5.4.2.3.1 Short Packet (SPa) and Long Packet (LPa) Structure

Short Packet (SPa) and Long Packet (LPa) are always used when data transmission is done in Low-Power Data Transmission (LPDT) or High-Speed Data Transmission (HSDT) modes.

The lengths of the packets are

• Short Packet (SPa): 4 bytes

• Long Packet (LPa): From 6 to 65,541 bytes

The type (SPa or LPa) of the packet can be recognized from their package headers (PH).

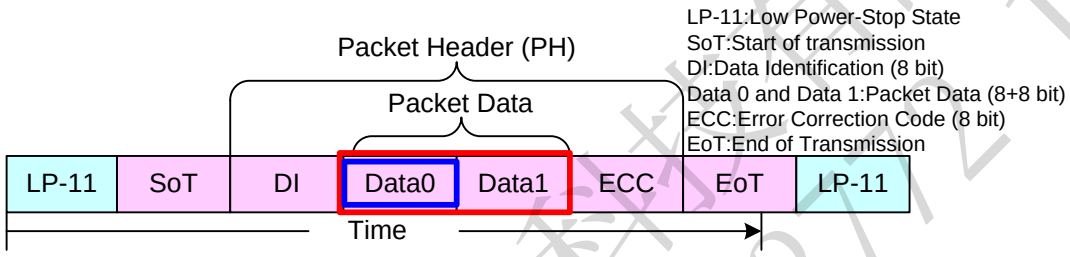



**Figure 37 Short Packet (SPa) Structure**

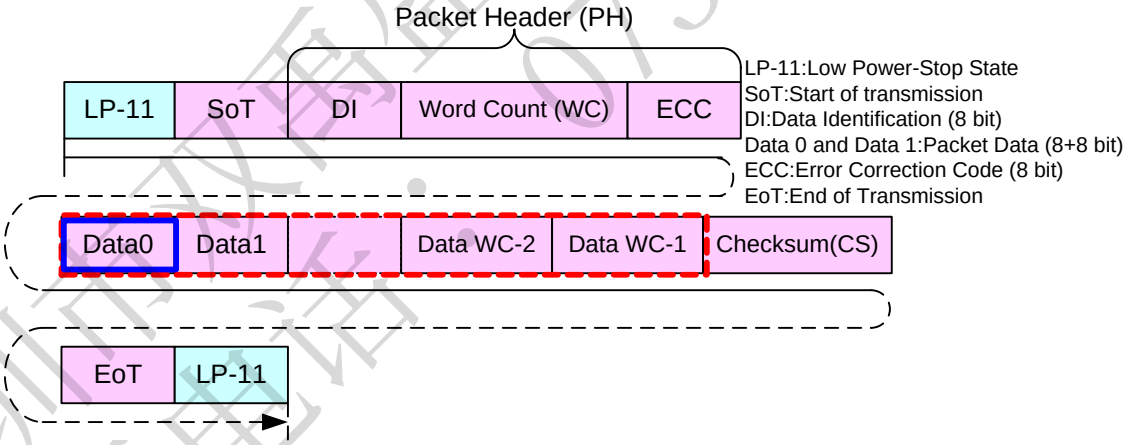



**Figure 38 Long Packet (LPa) Structure**

*Note:*

*Short Packet (SPa) Structure and Long Packet (LPa) Structure are presenting a single packet sending (= Includes LP-11, SoT and EoT for each packet sending).*

*The other possibility is that there is not needed SoT, EoT and LP-11 between packets if packets have sent in multiple*



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*packet format e.g.*

** LP-11 =>SoT =>SPa =>LPa =>SPa =>SPa =>EoT =>LP-11*

** LP-11 =>SoT =>SPa =>SPa =>SPa =>EoT =>LP-11*

** LP-11 =>SoT =>LPa =>LPa =>LPa =>EoT =>LP-11*

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#### 5.4.2.3.1.1 Bit Order of the Byte on Packets

The bit order of the byte, what is used on packets, is that the Least Significant Bit (LSB) of the byte is sent in the first and the Most Significant Bit (MSB) of the byte is sent in the last.

This same order is illustrated for reference purposes below.

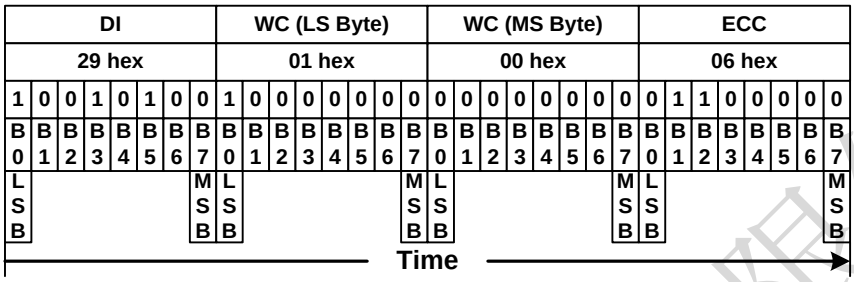



**Figure 39 Bit Order of Byte on Packets**

#### 5.4.2.3.1.2 Byte Order of the Multiple Byte Information on Packets

Byte order of the multiple bytes information, what is used on packets, is that the Least Significant (LS) Byte of the information is sent in the first and the Most Significant (MS) Byte of the information is sent in the last e.g. Word Count (WC) consists of 2 bytes (16 bits) when the LS byte is sent in the first and the MS byte is sent in the last.

This same order is illustrated for reference purposes below.

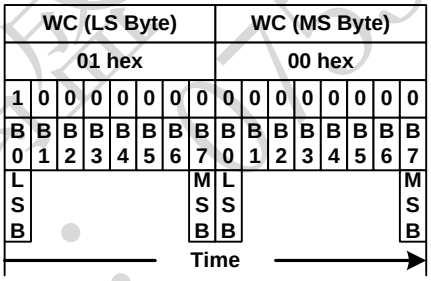



**Figure 40 Byte Order of the Multiple Byte on Packets**

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#### 5.4.2.3.1.3 Packet Header (PH)

The packet header is always consisting of 4 bytes. The content of these 4 bytes are different if it is used to Short Packet (SPa) or Long Packet (LPa).

Short Packet (SPa):

• 1st byte: Data Identification (DI) => Identification that this is Short Packet (SPa)

• 2nd and 3rd bytes: Packet Data (PD), Data 0 and Data 1

• 4th byte: Error Correction Code (ECC)

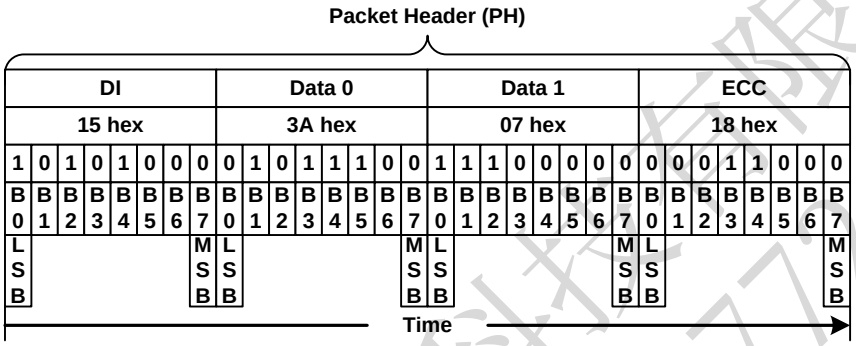



**Figure 41 Packet Header (PH) on the Short Packet(SPa)**

Long Packet (LPa):

• 1st byte: Data Identification (DI) => Identification that this is Long Packet (LPa)

• 2nd and 3rd bytes: Word Count (WC)

• 4th byte: Error Correction Code (ECC)

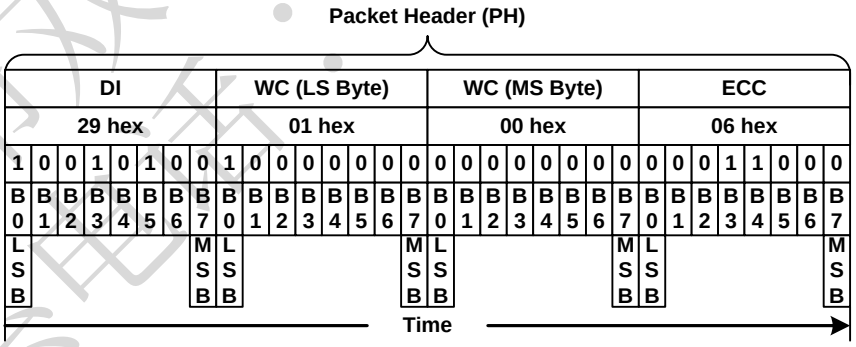



**Figure 42 Packet Header (PH) on the Long Packet (LPa)**



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- **Data Identification (DI)**

Data Identification (DI) is a part of Packet Header (PH) and it consists of 2 parts:

• Virtual Channel (VC), 2 bits, DI[7...6]

• Data Type (DT), 6 bits, DI[5…0]

The Data Identification (DI) structure is illustrated on a table below.

| Data Identification (DI) | | | | | | | |
|---|---|---|---|---|---|---|---|
| Virtual Channel (VC) | | Data Type (DT) | | | | | |
| Bit 7 | Bit 6 | Bit 5 | Bit 4 | Bit 3 | Bit 2 | Bit 1 | Bit 0 |

**Figure 43 Data Identification (DI) Structure**

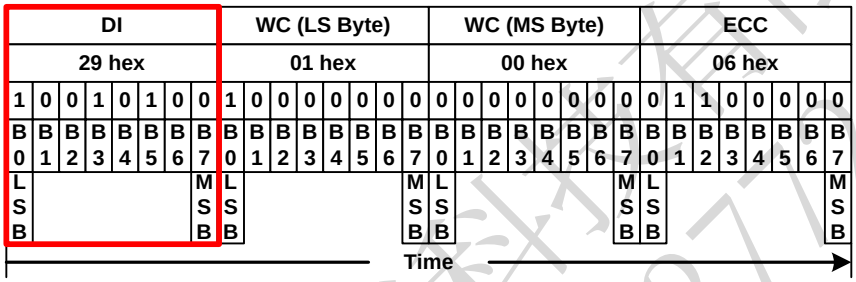



**Figure 44 Data Identification (DI) on the Packet Header(PH)**

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- **Virtual Channel (VC)**

Virtual Channel (VC) is a part of Data Identification (DI[7…6]) structure and it is used to address where a packet is wanted to send from the MCU.

Bits of the Virtual Channel (VC) are illustrated for reference purposes below.

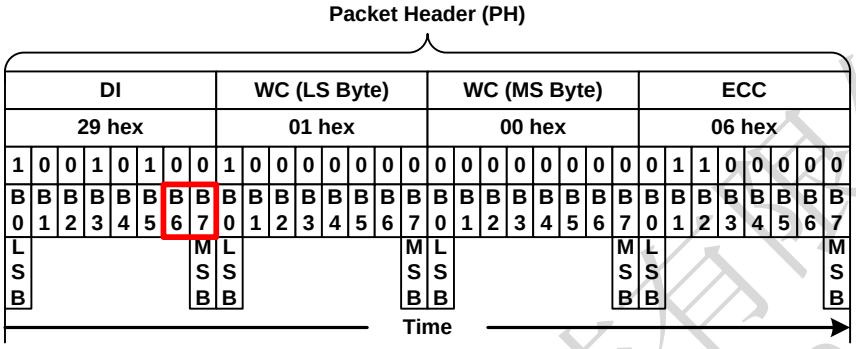



**Figure 45 Virtual Channel (VC) on the Packet Header (PH)**

Virtual Channel (VC) can address 4 different channels for e.g. 4 different display modules. Devices are using the same virtual channel what the MCU is using to send packets to them e.g.

• The MCU is using the virtual channel 0 when it sends packets to this display module

• This display module is also using the virtual channel 0 when it sends packets to the MCU

This functionality is illustrated below.

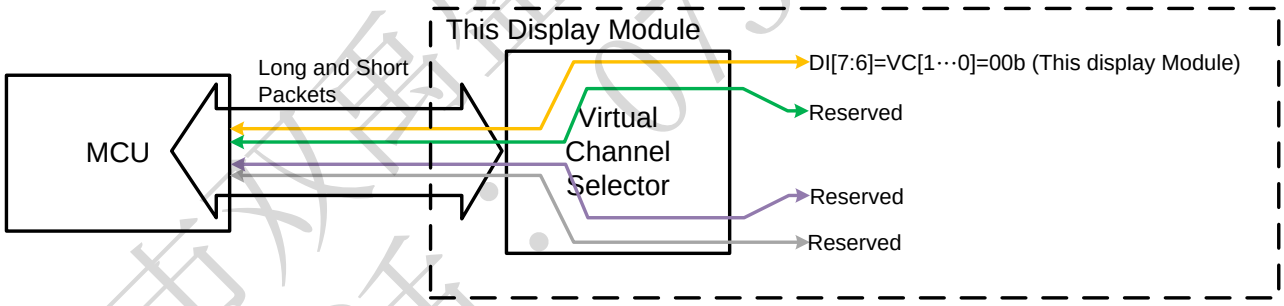



**Figure 46 Virtual Channel (VC) Configuration**

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- **Data Type (DT)**

Data Type (DT) is a part of Data Identification (DI[5…0]) structure and it is used to define a type of the used data on a packet.

Bits of the Data Type (DT) are illustrated for reference purposes below.

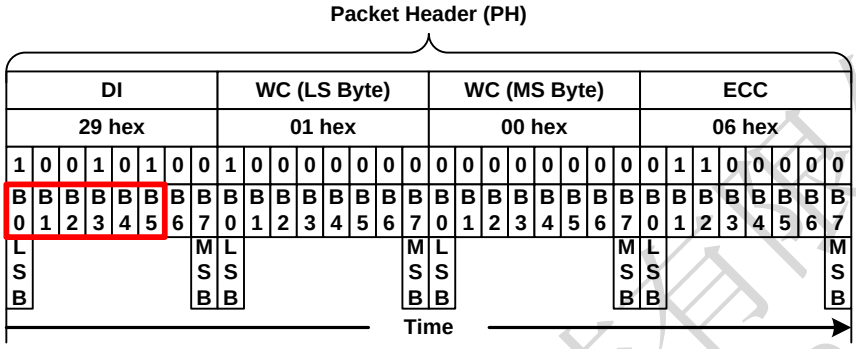



**Figure 47 Data Type (DT) on the Packet Header (PH)**

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This Data Type (DT) also defines what the used packet is: Short Packet (SPa) or Long Packet (LPa). Data Types (DT) are different from the MCU to the display module (or other devices) and vice versa.

These Data Type (DT) are defined on tables below.

| Data Type Hex | Data Type Binary | Description | Packet Size |
|---|---|---|---|
| 01h | 00 0001 | Sync Event, V Sync Start. | Short |
| 21h | 10 0001 | Sync Event, H Sync Start. | Short |
| 08h | 00 1000 | End of Transmission (EoT) packet. | Short |
| 02h | 00 0010 | Color Mode (CM) Off Command. | Short |
| 12h | 01 0010 | Color Mode (CM) On Command. | Short |
| hh22h | 10 0010 | Shut Down Peripheral Command. | Short |
| 32h | 11 0010 | Turn On Peripheral Command. | Short |
| 13h | 01 0011 | Generic Short WRITE, 1 parameter. | Short |
| 23h | 10 0011 | Generic Short WRITE, 2 parameters. | Short |
| 14h | 01 0100 | Generic READ, 1 parameter. | Short |
| 24h | 10 0100 | Generic READ, 2 parameters. | Short |
| 05h | 00 0101 | DCS WRITE, no parameter. | Short |
| 15h | 01 0101 | DCS WRITE, 1 parameter. | Short |
| 06h | 00 0110 | DCS READ, no parameter. | Short |
| 37h | 11 0111 | Set Maximum Return Packet Size. | Short |
| 09h | 00 1001 | Null Packet, no data. | Long |
| 19h | 01 1001 | Blanking Packet, no data. | Long |
| 29h | 10 1001 | Generic Long Write. | Long |
| 39h | 11 1001 | DCS Long Write/write_LUT Command Packet. | Long |
| 3Eh | 11 1110 | Packed Pixel Stream,24-bit RGB,8-8-8 Format. | Long |

**Data Type (DT) from MCU to the Display Module (or Other Devices)**

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| From the Display Module (or Other Devices) to the MCU | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Data Type Hex | B 5 | B 4 | B 3 | B 2 | B 1 | B 0 | Description | Packet | Abbreviation |
| 02h | 0 | 0 | 0 | 0 | 1 | 0 | Acknowledge and Error Report | Short | AwER |
| 11h | 0 | 1 | 0 | 0 | 0 | 1 | Generic Short READ Response,1 byte returned | Short | GENRR1-S |
| 12h | 0 | 1 | 0 | 0 | 1 | 0 | Generic Short READ Response,2 bytes returned | Short | GENRR2-S |
| 1Ah | 0 | 1 | 1 | 0 | 1 | 0 | Generic Long READ Response | Short | GENRR-L |
| 1Ch | 0 | 1 | 1 | 1 | 0 | 0 | DCS Long READ Response | Short | DCSRR_L |
| 21h | 1 | 0 | 0 | 0 | 0 | 1 | DCS Short READ Response, 1 byte returned | Short | DCSRR1_S |
| 22h | 1 | 0 | 0 | 0 | 1 | 0 | DCS Short READ Response, 2 bytes returned | Short | DCSRR2_S |

**Data Type (DT) from the Display Module (or Other Devices) to the MCU**

The receiver will ignore other Data Type (DT) if they are not defined on tables: “Data Type (DT) from the MCU to the Display Module (or Other Devices)” or “ Data Type (DT) from the Display Module (or Other Devices) to the MCU”.

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- **Packet Data (PD) on the Short Packet (SPa)**

Packet Data (PD) of the Short Packet (SPa) is defined after Data Type (DT) of the Data Identification (DI) has indicated that Short Packet (SPa) is wanted to send.

The Word Count (WC) indicates the number of Bytes of Packet of Packet Data (PD) send after the Packet Header.

Packet Data (PD) of the Short Packet (SPa) consists of 2 data bytes: Data 0 and Data 1.

Packet Data (PD) sending order is that Data 0 is sent in the first and the Data 1 is sent in the last.

Bits of Data 1 are set to '0' if the information length is 1 byte.

Packet Data (PD) of the Short Packet (SPa), when the length of the information is 1 or 2 bytes are illustrated for reference purposes below, when Virtual Channel (VC) is 0.

Packet Data (PD) information:

• Data 0: 35hex (Display Command Set (DCS) with 1 Parameter => DI(Data Type (DT)) = 15hex)

• Data 1: 01hex (DCS's parameter)

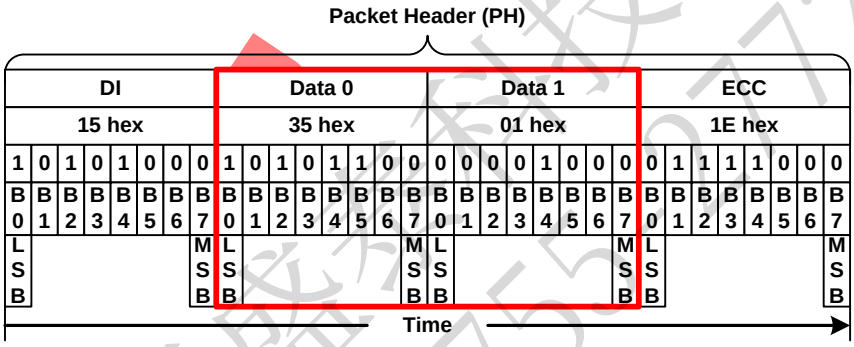



**Figure 48 Packet Data (PD) for Short Packet (SPa), 2 Bytes Information**

Packet Data (PD) information:

• Data 0: 10hex (DCS without parameter => DI(Data Type (DT)) = 05hex)

• Data 1: 00hex (Null)

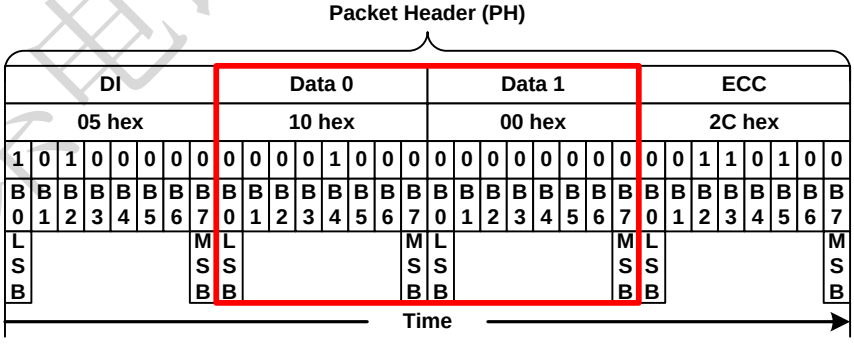



**Figure 49 Packet Data (PD) for Short Packet (SPa), 1 Bytes Information**



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- **Word Count (WC) on the Long Packet (LPa)**

Word Count (WC) of the Long Packet (LPa) is defined after Data Type (DT) of the Data Identification (DI) has indicated that Long Packet (LPa) is wanted to send.

Word Count (WC) indicates a number of the data bytes of the Packet Data (PD) what is wanted to send after Packet Header (PH) versus Packet Data (PD) of the Short Packet (SPa) is placed in the Packet Header (PH).

Word Count (WC) of the Long Packet (LPa) consists of 2 bytes.

These 2 bytes of the Word Count (WC) sending order is that the Least Significant (LS) Byte is sent in the first and the Most Significant (MS) Byte is sent in the last.

Word Count (WC) of the Long Packet (LPa) is illustrated for reference purposes below.

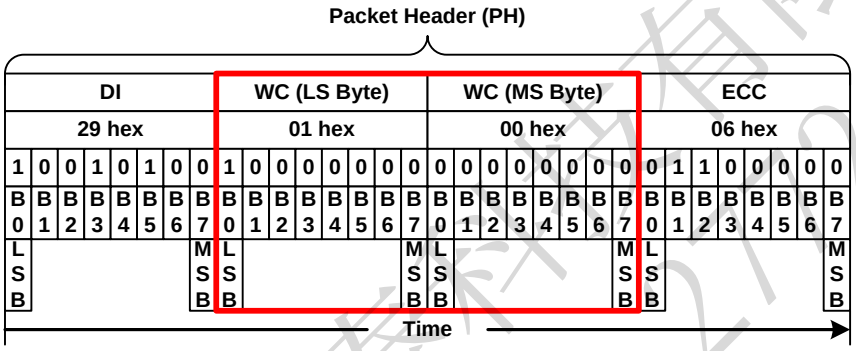



**Figure 50 Word Count (WC) on the Long Packet (LPa)**

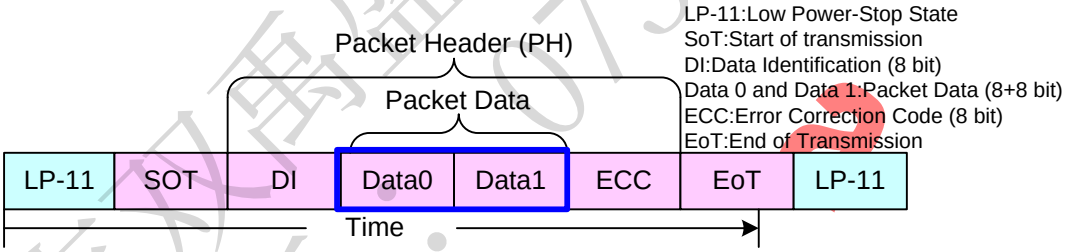



**Figure 51 Packet Data (PD) on the Short Packet (SPa)**

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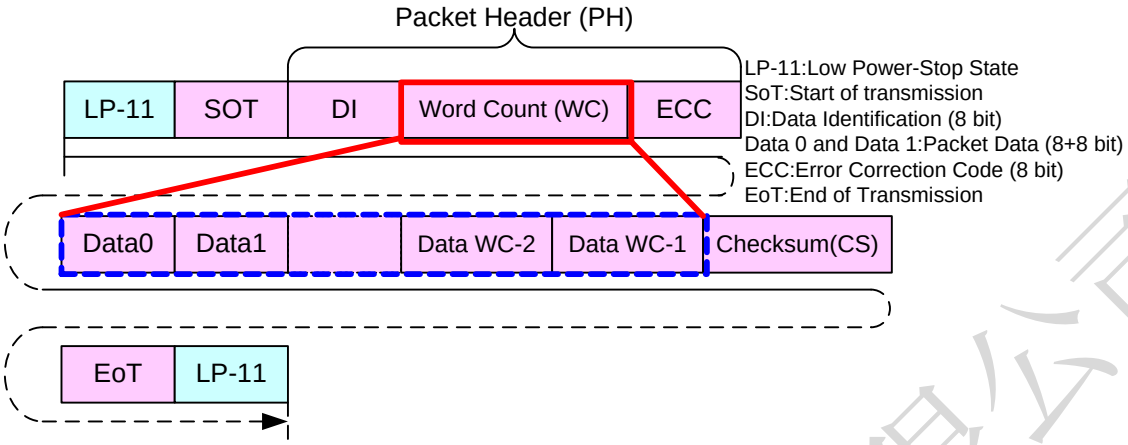



**Figure 52 Packet Data (PD) on the Long Packet (LPa)**

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- **Error Correction Code (ECC)**

Error Correction Code (ECC) is a part of Packet Header (PH) and its purpose is to identify an error or errors on the Packet Header (PH):

The ECC protects the following field”

• Short Packet (SPa): Data Identification (DI) byte (8 bits, D[0...7]), Packet Data (PD) bytes (16 bits, D[8...23]) and ECC(8 bits: P[0…7])

• Long Packet (LPa): Data Identification (DI) byte (8 bits, D[0…7]), Word Count (WC) bytes (16 bits: D[8…23]) and ECC (8 bits, P[0…7])

D[23…0] and P[7…0] are illustrated for reference purposes below.

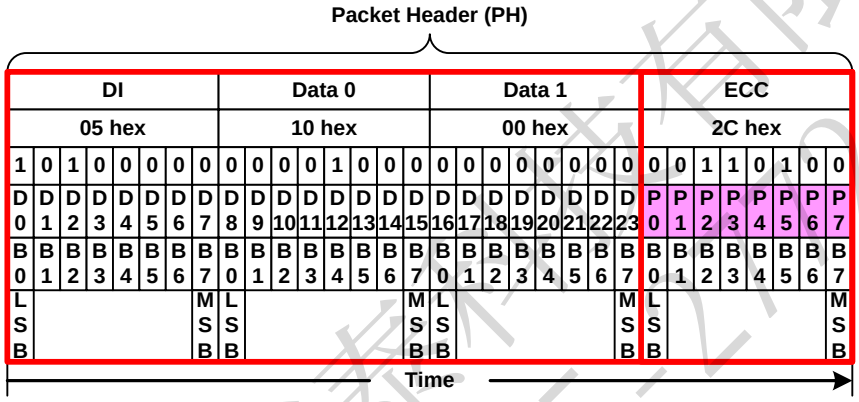



**Figure 53 D[23..0] and P[7…0] on the Short Packet (SPa)**

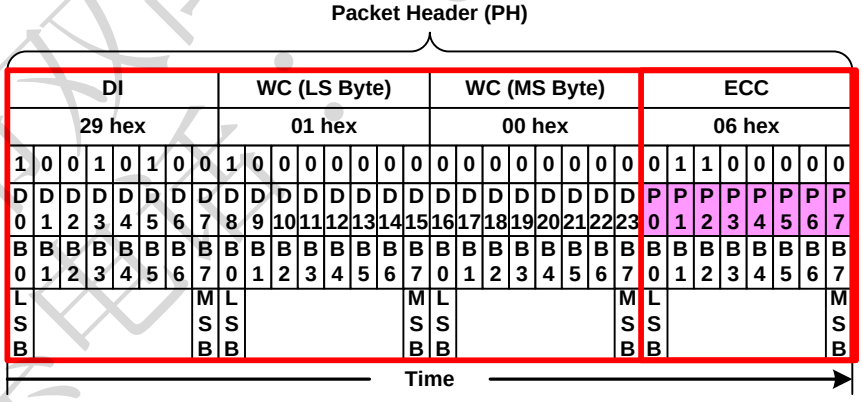



**Figure 54 D[23…0] and P[7…0] on the Long Packet (LPa)**

Error Correction Code (ECC) can recognize one error or several errors and makes correction in one bit error case

Bits (P[7…0]) of the Error Correction Code (ECC) are defined, where the symbol ‘^’ is presenting XOR function



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(Pn is '1' if there is odd number of '1's and Pn is '0' if there is even number of '1's), as follows.

• P7 = 0

• P6 = 0

• P5 = D10^D11^D12^D13^D14^D15^D16^D17^D18^D19^D21^D22^D23

• P4 = D4^D5^D6^D7^D8^D9^D16^D17^D18^D19^D20^D22^D23

• P3 = D1^D2^D3^D7^D8^D9^D13^D14^D15^D19^D20^D21^D23

• P2 = D0^D2^D3^D5^D6^D9^D11^D12^D15^D18^D20^D21^D22

• P1 = D0^D1^D3^D4^D6^D8^D10^D12^D14^D17^D20^D21^D22^D23

• P0 = D0^D1^D2^D4^D5^D7^D10^D11^D13^D16^D20^D21^D22^D23

P7 and P6 are set to '0' because Error Correction Code (ECC) is based on 64 bit value ([D63…0]), but this implementation is based on 24 bit value (D[23…0]). Therefore, there is only needed 6 bits (P[5…0]) for Error Correction Code (ECC).

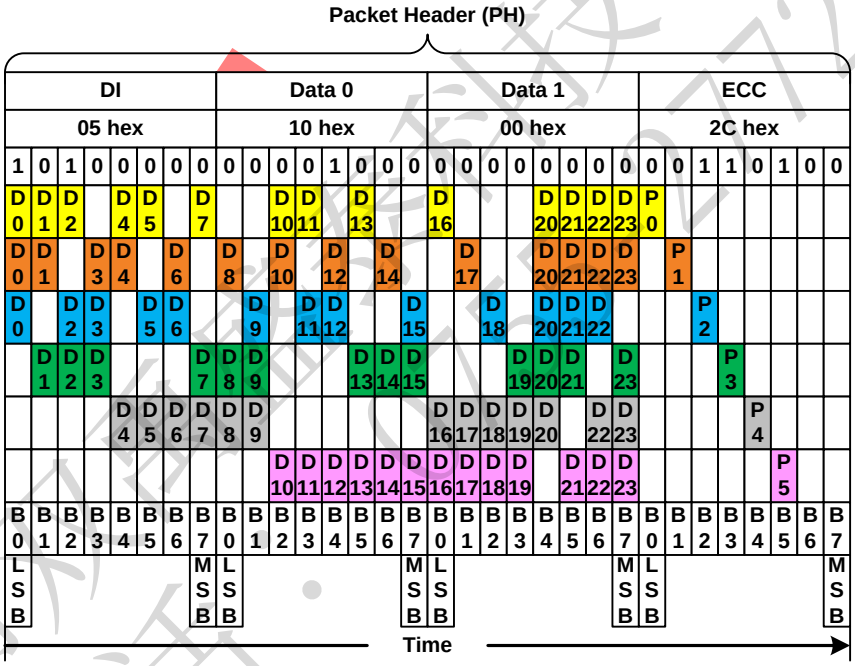



**Figure 55 XOR Functionality on the Short Packet (SPa)**



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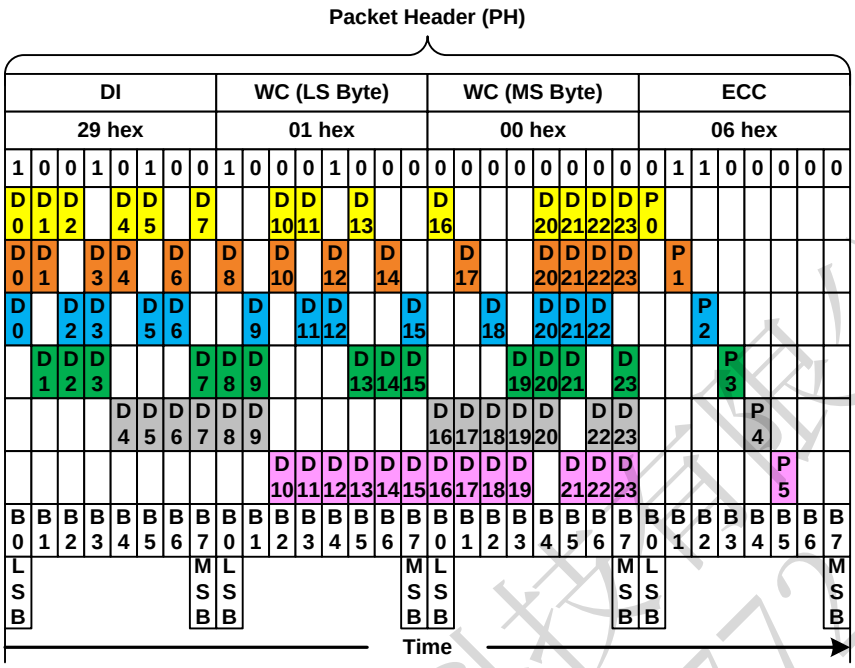



Figure 56 XOR Functionality on the Long Packet (LPa)

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The transmitter (The MCU or the Display Module) is sending data bits D[23…0] and Error Correction Code (ECC) P[7…0]. The receiver (The Display module or the MCU) is calculate an Internal Error Correction Code (IECC) and compares the received Error Correction Code (ECC) and the Internal Error Correction Code (IECC). This comparison is done when each power bit of ECC and IECC have been done XOR function. The result of this function is PO[7…0].

This functionality, where the transmitter is the MCU and the receiver is the display module, is illustrated for reference purposes below.

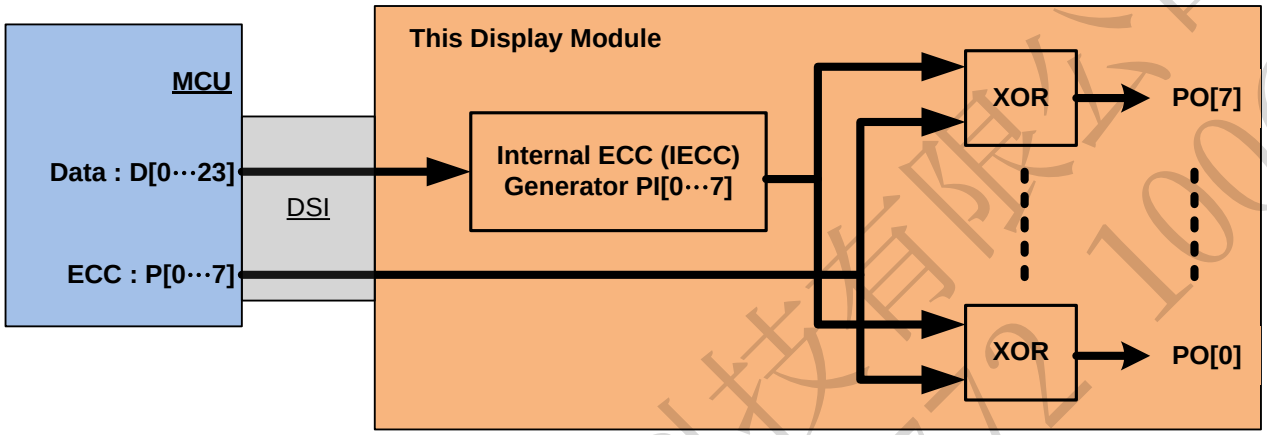



**Figure 57 Internal Error Correction Code (IECC) on the Display Module (The Receiver)**

The sent data bits (D[23…0]) and ECC (P[7…0]) are received correctly, if a value of the PO[7…0]) is 0 0h. The sent data bits (D[23…0]) and ECC (P[7…0]) are not received correctly, if a value of the PO[7…0]) is not 00h.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| ECC P[7…0] | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 03h |
| IECC PI[7…0] | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 03h |
| XOR(ECC,IECC) =>PO[7…0] | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | =00h=>No Error |
| | L | | | | | | | M | |
| | S | | | | | | | S | |
| | B | | | | | | | B | |

**Internal XOR Calculation between ECC and IECC Values-No Error**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| ECC P[7…0] | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 03h |
| IECC PI[7…0] | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0Fh |
| XOR(ECC,IECC) =>PO[7…0] | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | =0Ch=> Error |
| | L | | | | | | | M | |
| | S | | | | | | | S | |
| | B | | | | | | | B | |

**Internal XOR Calculation between ECC and IECC Values- Error**

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The received Error Correction Code (ECC) can be 00h when the Error Correction Code (ECC) functionality is not used for data values D[23…0] on the transmitter side.

The number of the errors (one or more) can be defined when the value of the PO[7…0] is compared to values on the following table.

| Data Bit | PO7 | PO6 | PO5 | PO4 | PO3 | PO2 | PO1 | PO0 | Hex |
|---|---|---|---|---|---|---|---|---|---|
| D[0] | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 07h |
| D[1] | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0Bh |
| D[2] | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0Dh |
| D[3] | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0Eh |
| D[4] | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 13h |
| D[5] | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 15h |
| D[6] | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 16h |
| D[7] | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 19h |
| D[8] | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 1Ah |
| D[9] | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1Ch |
| D[10] | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 23h |
| D[11] | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 25h |
| D[12] | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 26h |
| D[13] | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 29h |
| D[14] | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 2Ah |
| D[15] | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 2Ch |
| D[16] | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 31h |
| D[17] | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 32h |
| D[18] | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 34h |
| D[19] | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 38h |
| D[20] | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1Fh |
| D[21] | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 2Fh |
| D[22] | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 37h |
| D[23] | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 3Bh |

One error is detected if the value of the PO[7…0] is on : One Bit Error Value of the Error Correction Code (ECC) and the receiver can correct this one bit error because this found value also defines what is a location of the corrupt bit e.g.

• PO[7…0] = 0Eh

• The bit of the data (D[23…0]), what is not correct, is D[3]

More than one error is detected if the value of the PO[7…0] is not on: One Bit Error Value of the Error Correction Code (ECC) e.g. PO[7…0] = 0Ch.

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#### 5.4.2.3.1.4 Packet Date (PD) on the Long Packet (LPa)

Packet Data (PD) of the Long Packet (LPa) is defined after Packet Header (PH) of the Long Packet (LPa). The number of the data bytes is defined on chapter “Word Count (WC) on the Long Packet (LPa)”.

#### 5.4.2.3.1.5 Packet Footer (PF) on the Long Packet (LPa)

Packet Footer (PF) of the Long Packet (LPa) is defined after the Packet Data (PD) of the Long Packet (LPa). The Packet Footer (PF) is a checksum value what is calculated from the Packet Data of the Long Packet (LPa).

The checksum is using a 16-bit Cyclic Redundancy Check (CRC) value which is generated with a polynomial X16+X12+X5+X0 as it is illustrated below.

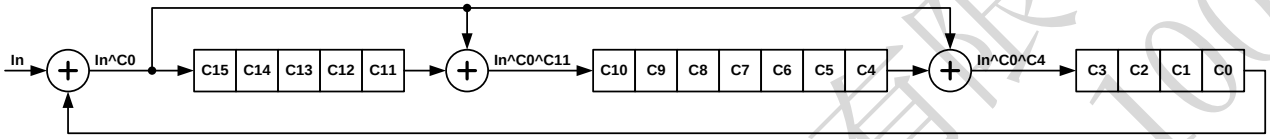



**Figure 58 16-bit Cyclic Redundancy Check (CRC) Calculation**

The 16-bit Cyclic Redundancy Check (CRC) generator is initialized to FFFFh before calculations. The Least Significant Bit (LSB) of the data byte of the Packet Data (PD) is the first bit what is inputted into the 16-bit Cyclic Redundancy Check (CRC).

An example of the 16-bit Cyclic Redundancy Check (CRC), where the Packet Data (PD) of the Long Packet (LPa) is 01h, is illustrated (step-by-step) below.

| Step | In | In^C0 | C15 | C14 | C13 | C12 | C11 | In^C0^C11 | C10 | C9 | C8 | C7 | C6 | C5 | C4 | In^C0^C4 | C3 | C2 | C1 | C0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1(LSB) | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |
| 2 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 3 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |
| 4 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 5 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 6 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| 7 | 0(MSB) | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 8 | X | X | 0 | 0 | 0 | 1 | 1 | X | 1 | 1 | 0 | 0 | 0 | 0 | 0 | X | 1 | 1 | 1 | 0 |
| CRC Result: | | | 0 | 0 | 0 | 1 | 1 | | 1 | 1 | 0 | 0 | 0 | 0 | 0 | | 1 | 1 | 1 | 0 |
| | | | MSB | | | | | | | | | | | | | | | | | LSB |

**Figure 59 CRC Calculation – Packet Data (PD) is 01h**



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A value of the Packet Footer (PF) is 1E0Eh in this example. This example (Command 01h has been sent) is Illustrated below.

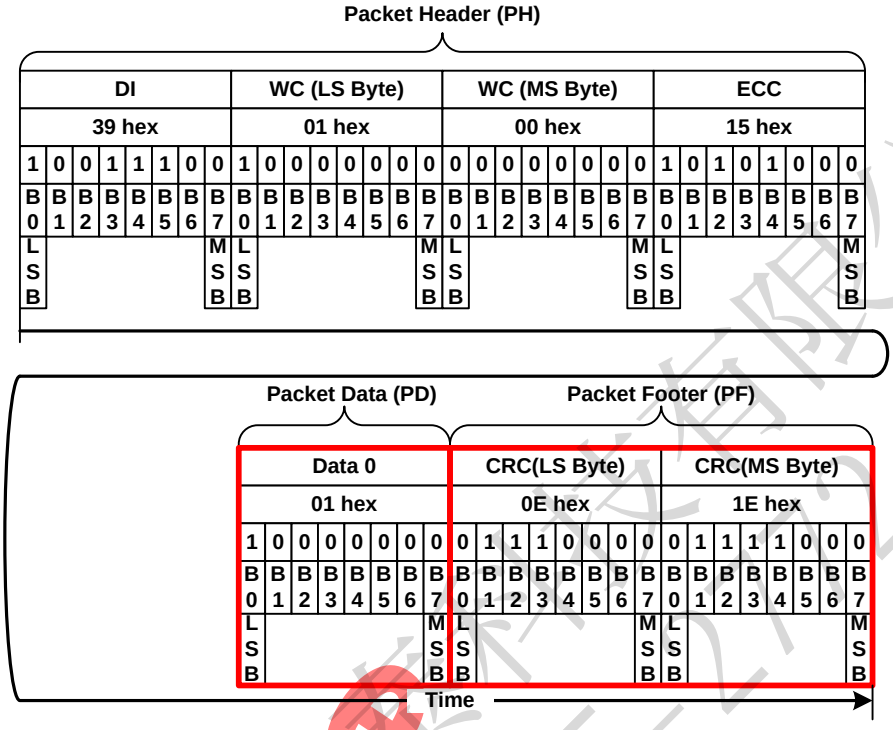



**Figure 60 Packet Footer (PF) Example**

The receiver is calculated own checksum value from received Packet Data (PD). The receiver compares own checksum and the Packet Footer (PF) what the transmitter has sent. The received Packet Data (PD) and Packet Footer (PF) are correct if the own checksum of the receiver and Packet Footer (PF) are equal and vice versa the received Packet Data (PD) and Packet Footer (PF) are not correct if the own checksum of the receiver and Packet Footer (PF) are not equal.

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#### 5.4.2.3.2 Packet Transmissions

##### 5.4.2.3.2.1 Packet from the MCU to the Display Module

- **Display Command Set (DCS)**

Display Command Set (DCS), which is defined on chapter “Command Description”, is used from the MCU to the display module. This Display Command Set (DCS) is always defined on the Data 0 of the Packet Data (PD),which is included in Short Packet (SPa) and Long Packet (LPa) as these are illustrated below.

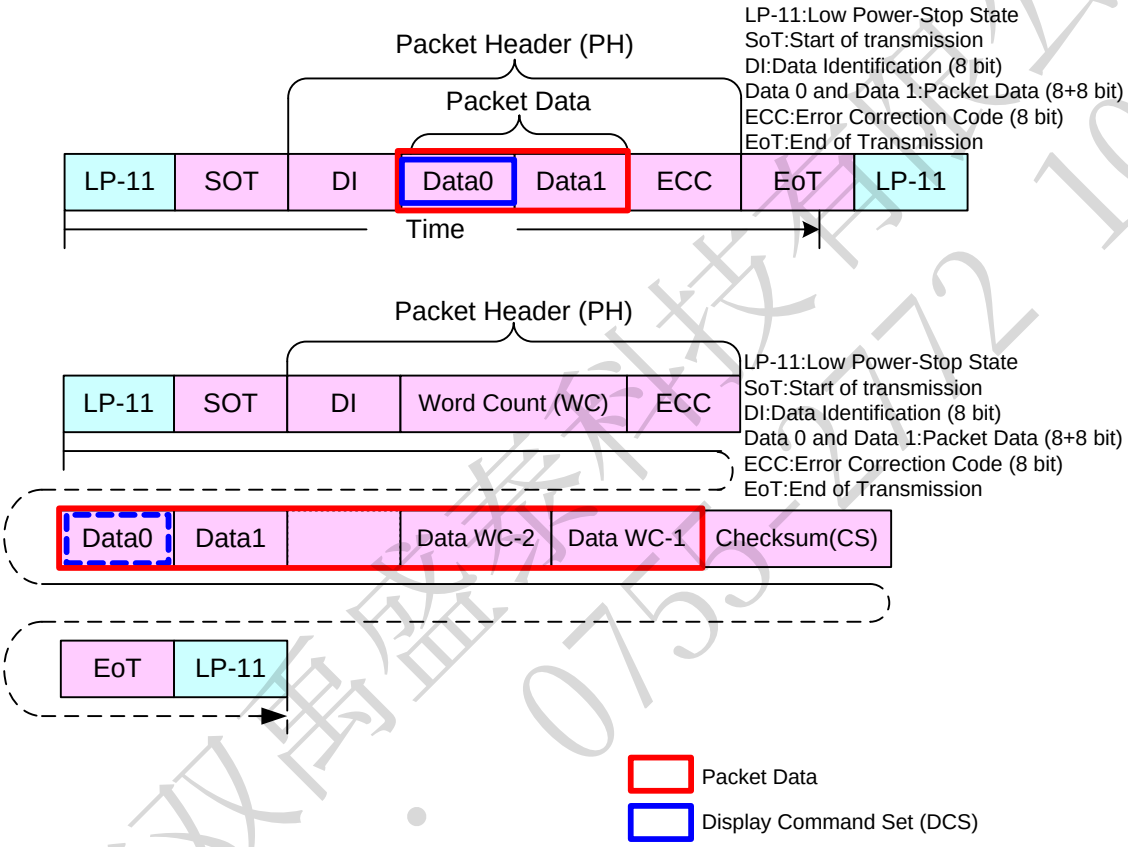



**Figure 61 Display Command Set (DCS) on Short Packet (SPa) and Long Packet (LPa)**

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#### 5.4.2.3.2.2 Packet from the Display Module to the MCU

- **Used Packet Types**

The display module is always using Short Packet (SPa) or Long Packet (LPa), when it is returning information to the MCU after the MCU has requested information from the Display Module. This information can be a response of the Display Command Set (DCS) or an Acknowledge with Error Report.

The used packet type is defined on Data Type (DT).

A number of the return bytes are more than the maximum size of the Packet Data (PD) on Long Packet (LPa) or Short Packet (SPa) when the display module is sending return bytes in several packets until all return bytes have been sent from the display module to the MCU.

It is not possible that the display module is sending return bytes in several packets even if the maximum size of the Packet Data (PD) could be sent on a packet.

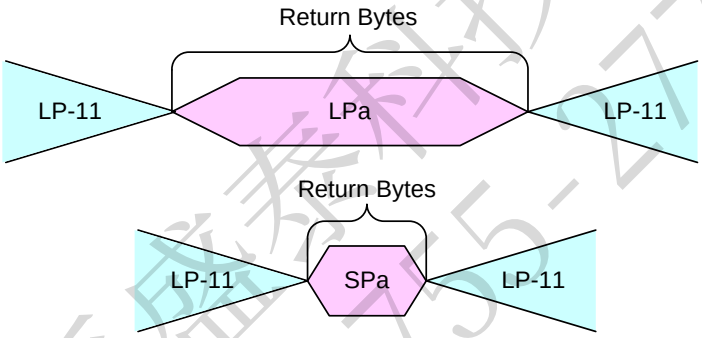



**Figure 62 Return Bytes on Signal Packet**

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| From the Display Module (or Other Devices) to the MCU | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Data Type Hex | B 5 | B 4 | B 3 | B 2 | B 1 | B 0 | Description | Packet | Abbreviation |
| 02h | 0 | 0 | 0 | 0 | 1 | 0 | Acknowledge and Error Report | Short | AwER |
| 11h | 0 | 1 | 0 | 0 | 0 | 1 | Generic Short READ Response,1 byte returned | Short | GENRR1-S |
| 12h | 0 | 1 | 0 | 0 | 1 | 0 | Generic Short READ Response,2 bytes returned | Short | GENRR2-S |
| 1Ah | 0 | 1 | 1 | 0 | 1 | 0 | Generic Long READ Response | Short | GENRR-L |
| 1Ch | 0 | 1 | 1 | 1 | 0 | 0 | DCS Long READ Response | Short | DCSRR_L |
| 21h | 1 | 0 | 0 | 0 | 0 | 1 | DCS Short READ Response, 1 byte returned | Short | DCSRR1_S |
| 22h | 1 | 0 | 0 | 0 | 1 | 0 | DCS Short READ Response, 2 bytes returned | Short | DCSRR2_S |

**Data Type for Display Module-sourced Packets**

The display module is return 2 packets (1st packet: Data, 2nd packet: Acknowledge with Error Report ) to the MCU when the display module has received a read command. These return packets are illustrated for reference purpose below.

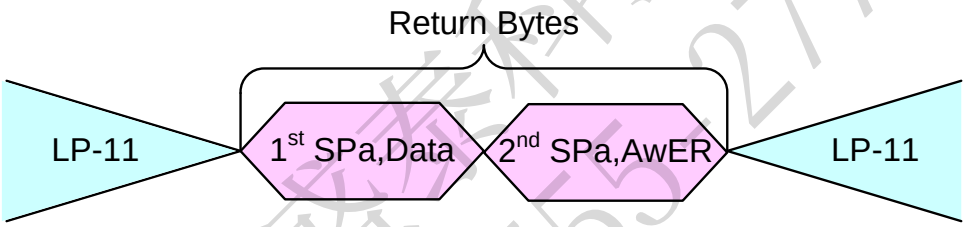



**Figure 63 Exception When Return Bytes on Several Packet**

*Note:*

*1. AwER=Acknowledge with Error Report*

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- **Acknowledge with Error Report (AwER), Data Type = 00 0010(02h)**

"Acknowledge with Error Report" (AwER) is always using a Short Packet (SPa), what is defined on Data Type (DT,00 0010b), from the display module to the MCU.

The Packet Data (PD) can include bits, which are defining the current error, when a corresponding bit is set to '1', as they are defined on the following table.

| Bit | Description | Sitronix LCD Driver Implementation |
|---|---|---|
| 0 | SoT Error | NO |
| 1 | SoT Sync Error | NO |
| 2 | EoT Sync Error | NO |
| 3 | Escape Mode Entry Command Error | YES |
| 4 | Low-Power Transmit Sync Error | YES |
| 5 | Any Protocol Timer Time-Out | NO |
| 6 | False Control Error | YES |
| 7 | Contention is Detected on the Display Module | NO |
| 8 | ECC Error, single-bit (detected and corrected) | YES |
| 9 | ECC Error, multi-bit (detected, not corrected) | YES |
| 10 | Set to "0" internally (Only for Long Packet (LP)) | YES |
| 11 | DSI Data Type (DT) Not Recognized | YES |
| 12 | DSI Virtual Channel (VC) ID Invalid | YES |
| 13 | Invalid Transmission Length | NO |
| 14 | Reserved, Set to '0' internally | NO |
| 15 | DSI Protocol Violation | NO |

**Acknowledge with Error Report (AwER) for Short Packet (SPa) Response**

*Note*

*AwER will return1-bit zero if the item is no implementation.*

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These errors are only included on the last packet, which has been received from the MCU to the display module before Bus Turnaround (BTA).

The display module ignores the received packet which includes error or errors

Acknowledge with Error Report (AwER) of the Short Packet (SPa) is defined e.g.

• Data Identification (DI)

Virtual Channel (VC, DI[7…6]): 00b

Data Type (DT, DI[5…0]): 00 0010b

• Packet Data (PD):

Bit 8: ECC Error, single-bit (detected and corrected)

AwER: 0100h

• Error Correction Code (ECC)

This is defined on the Short Packet (SPa) as follows.

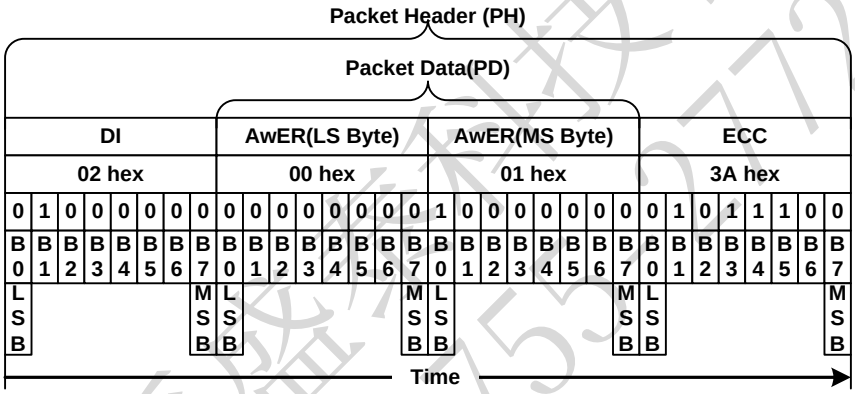



**Figure 64 Acknowledge with Error Report (AwER)-Example**

It is possible that the display module receivers several packets, which include error, from the MPU before the MPU performs the Bus Turnaround (BTA).Some examples are illustrated below for reference purpose.

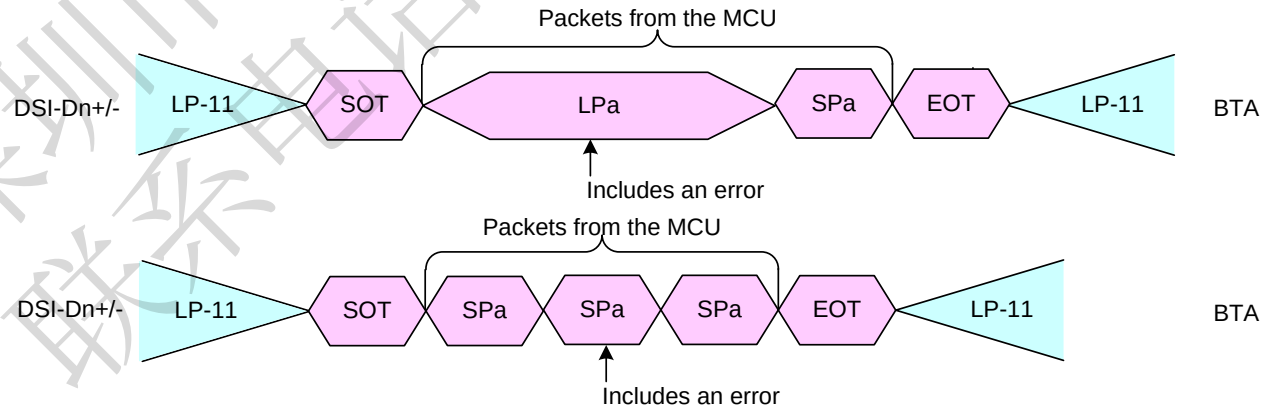



**Figure 65 Error Packet**



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Therefore, there is needed a method to check if there has been errors on the previous packets. These errors of the previous packets can check “Read Number of the Errors on DSI (05h)" command.

The number of the packets, which are including an ECC or CRC error, are calculated on the RDNUMED register, which can read “Read Number of the Errors on DSI (05h)” command. This command also sets the RDNUMED register to 00h after the MCU has read the RDNUMED register from the display module.

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### 5.5 Serial Interface_ Control Bus ( RGB_video / MIPI video)

The serial interface is either SPI9, SPI16 or SPI4/8-bits interface for communication between the micro controller and the LCD driver. The SPI9,SPI16 serial interface use: CSX (chip enable), SCL (serial clock) , SDA (serial data input) and SDO(serial data output), and the SPI8 serial interface use: CSX (chip enable), D/CX (data/ command flag), SCL (serial clock) , SDA (serial data input) and SDO(serial data output). Serial clock (SCL) is used for interface with MCU only, so it can be stopped when no communication is necessary.

Pin description

SPI9_3-line serial interface (9 bits) & SPI16_3-line serial interface (16 bits)

| Pin Name | Description |
|---|---|
| CSX | Chip selection signal |
| SCL | Serial input CLK |
| SDA | Serial input data |
| SDO | Serial output data |

SPI8_4-line serial interface (8 bits)

| Pin Name | Description |
|---|---|
| CSX | Chip selection signal |
| DCX | Data is regarded as a command when SCL is low<br>Data is regarded as a parameter or data when SCL is high |
| SCL | Clock signal |
| SDA | Serial input data |
| SDO | Serial output data |

#### 5.5.1 SPI Write mode

##### 5.5.1.1 SPI8 & SPI9

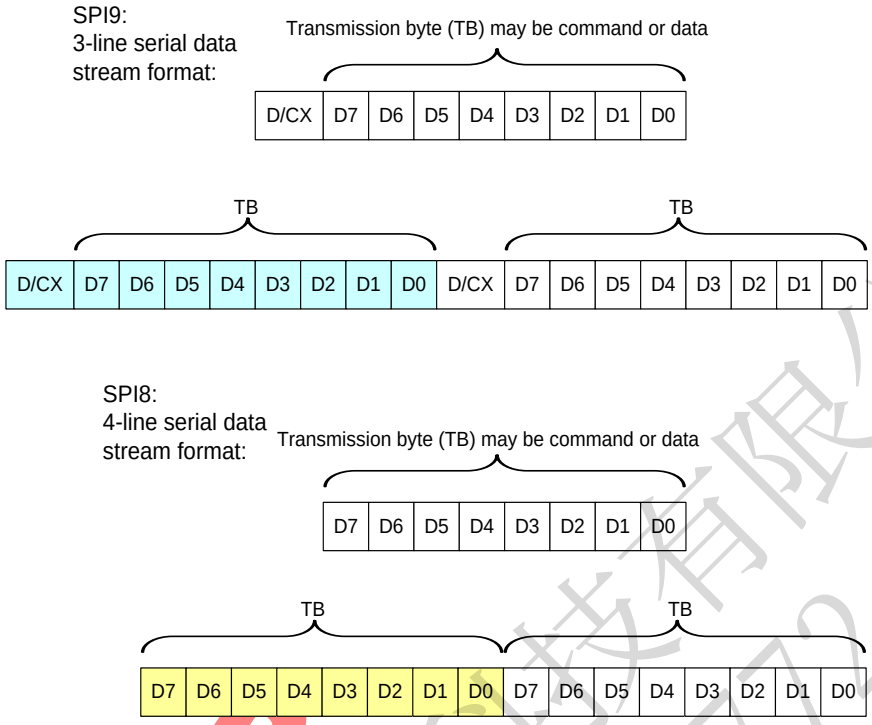
The write mode of the interface means the micro controller writes commands and data to the LCD driver. 3-lines serial data packet contains a control bit D/CX and a transmission byte. In 4-lines serial interface, data packet contains just transmission byte and control bit D/CX is transferred by the D/CX pin. If D/CX is “low”, the transmission byte is interpreted as a command byte. If D/CX is “high”, the transmission byte is command register as parameter.

Any instruction can be sent in any order to the driver. The MSB is transmitted first. The serial interface is initialized when CSX is high. In this state, SCL clock pulse or SDA data have no effect. A falling edge on CSX enables the serial interface and indicates the start of data transmission.

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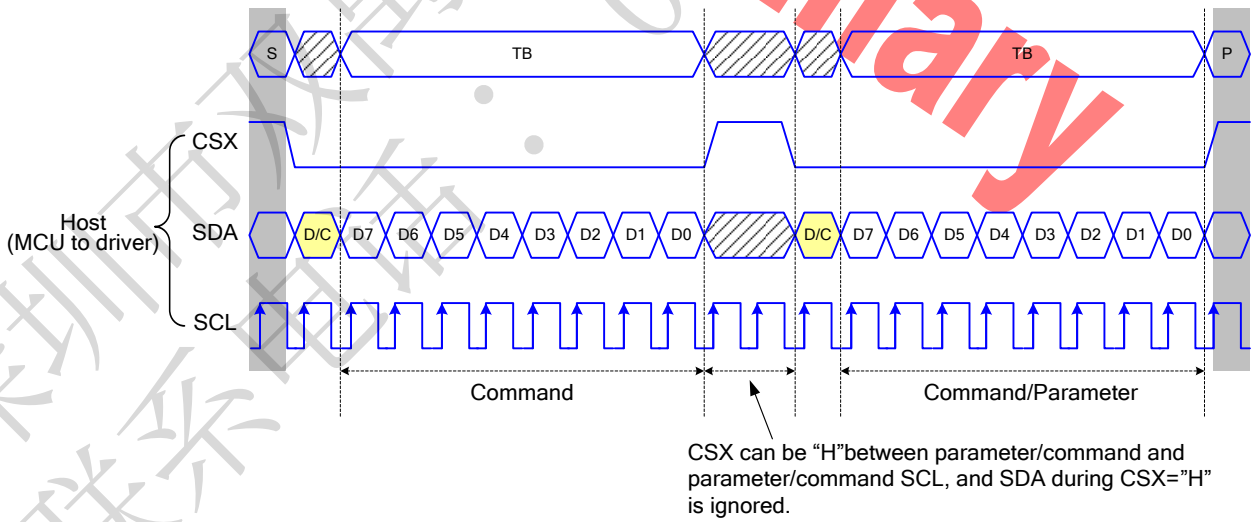


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**Serial interface data stream format**

When CSX is “high”, SCL clock is ignored. During the high period of CSX the serial interface is initialized. At the falling edge of CSX, SCL can be high or low. SDA is sampled at the rising edge of SCL. D/CX indicates whether the byte is command (D/CX=’0’) or parameter data (D/CX=’1’). D/CX is sampled when first rising edge of SCL (SPI9:3-line serial interface) or 8th rising edge of SCL (SPI8:4-line serial interface). If CSX stays low after the last bit of command/data byte, the serial interface expects the D/CX bit (SPI9:3-line serial interface) or D7 (SPI8:4-line serial interface) of the next byte at the next rising edge of SCL..

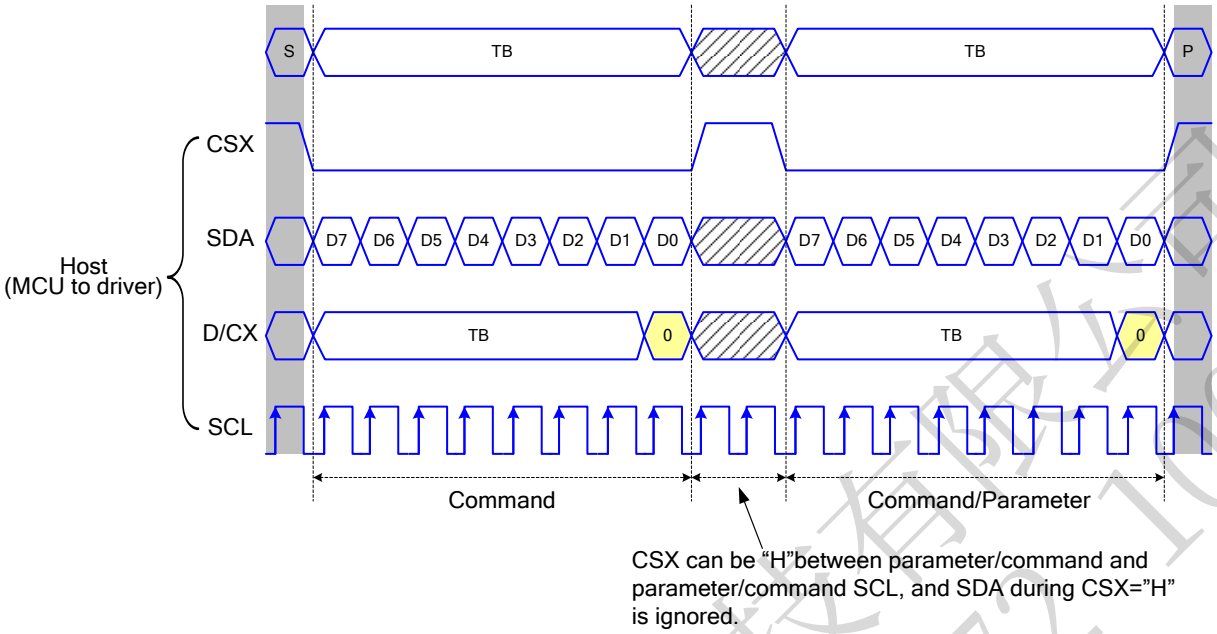



**SPI9:3-line serial interface write protocol (write to register with control bit in transmission)**

Preliminary



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**SPI8:4-line serial interface write protocol (write to register with control bit in transmission)**

#### 5.5.1.2 SPI16

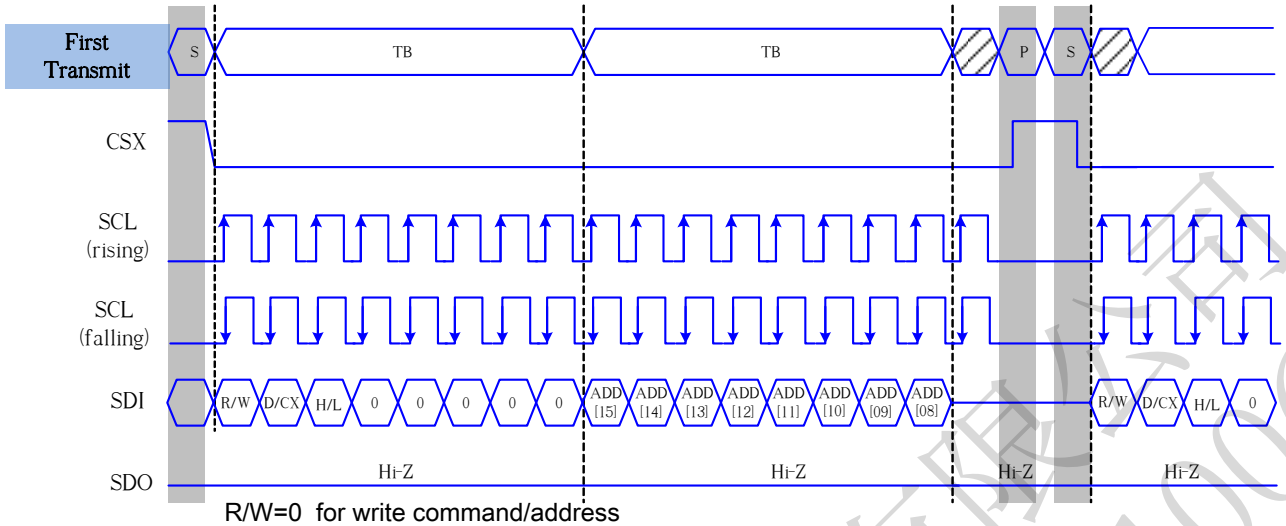
The write mode of the interface means the micro controller writes commands and data to the ST7701S. The serial interface is initialized when CSX is high. In this state, SCL clock pulse or SDI data have no effect. A falling edge on CSX enables the serial interface and indicates the start of data transmission.

When CSX is high, SCL clock is ignored. During the high time of CSX the serial interface is initialized. At the falling CSX edge, SCL can be high or low. SDI/SDO are sampled at the rising edge of SCL. R/W indicates, whether the byte is read command (R/W = '1') or write command (R/W = '0'). It is sampled when first rising SCL edge. If CSX stays low after the last bit of command/data byte, the serial interface expects the R/W bit of the next byte at the next rising edge of SCL

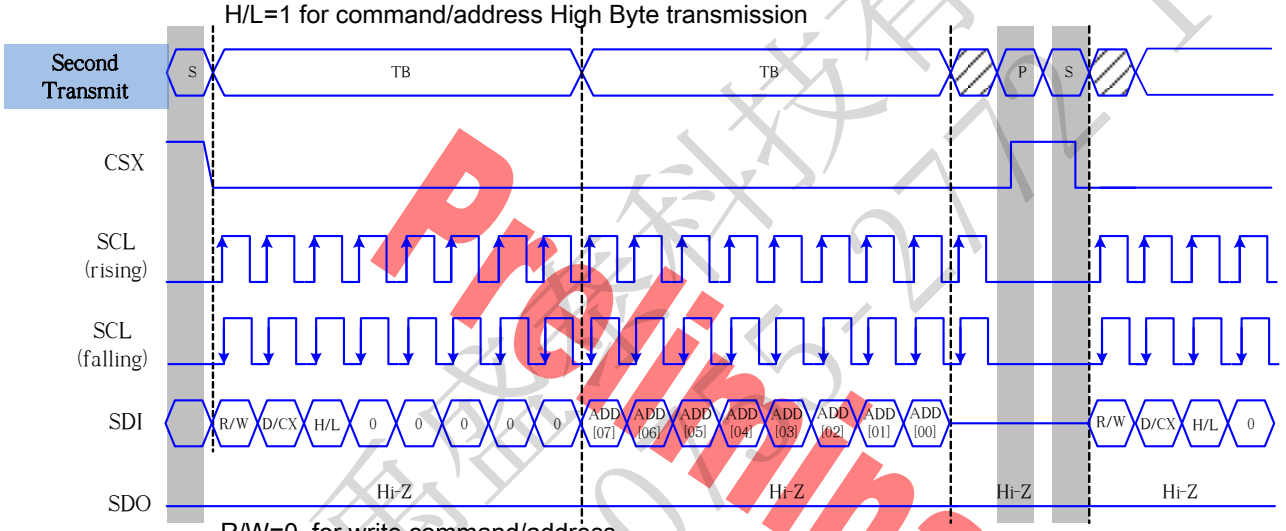
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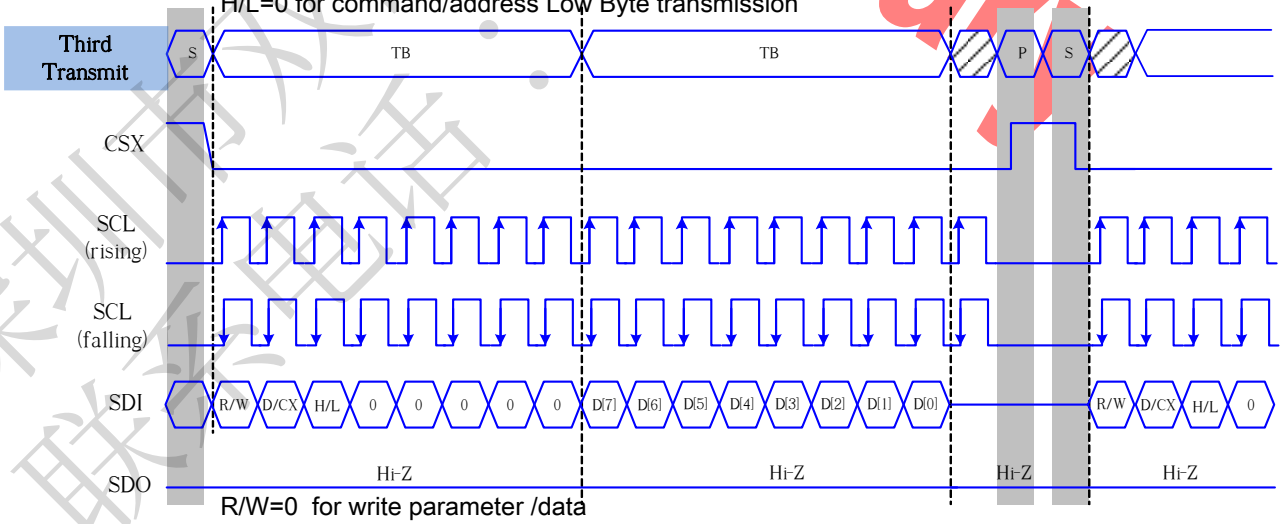
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R/W=0 for write command/address
D/CX=0 for command /address transmission
H/L=1 for command/address High Byte transmission




R/W=0 for write command/address
D/CX=0 for command /address transmission
H/L=0 for command/address Low Byte transmission




R/W=0 for write parameter /data
D/CX=1 for parameter /data transmission
H/L=0 for parameter /data Low Byte transmission

**serial 16 bit interface write mode**

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### 5.5.2 SPI Read mode

#### 5.5.2.1 SPI8 & SPI9

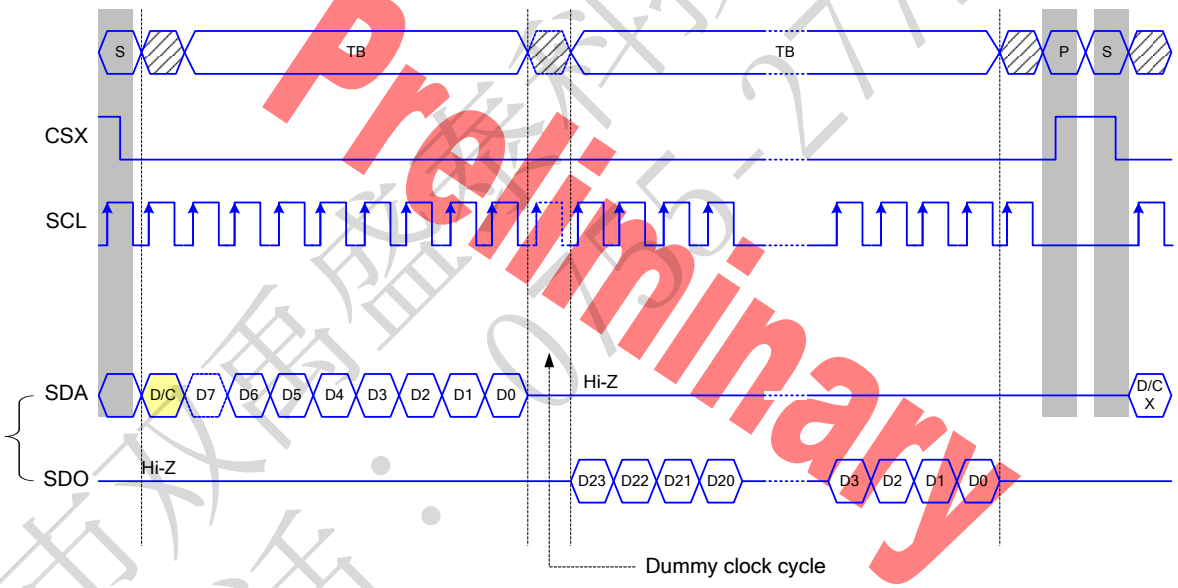
The read mode of the interface means that the micro controller reads register value from the driver. To achieve read function, the micro controller first has to send a command (read ID or register command) and then the following byte is transmitted in the opposite direction. After that CSX is required to go to high before a new command is send (see the below figure). The driver samples the SDA (input data) at rising edge of SCL, but shifts SDO (output data) at the falling edge of SCL. Thus the micro controller is supported to read at the rising edge of SCL.

After the read status command has been sent, the SDA line must be set to tri-state no later than at the falling edge of SCL of the last bit.

**SPI9:3-line serial interface protocol**

SPI9:3-line serial protocol (for RDID1/RDID2/RDID3/0Ah/0Bh/0Ch/0Dh/0Eh/0Fh command: 8-bit read):

SPI9:3-line serial protocol (for RDDID command: 24-bit read)

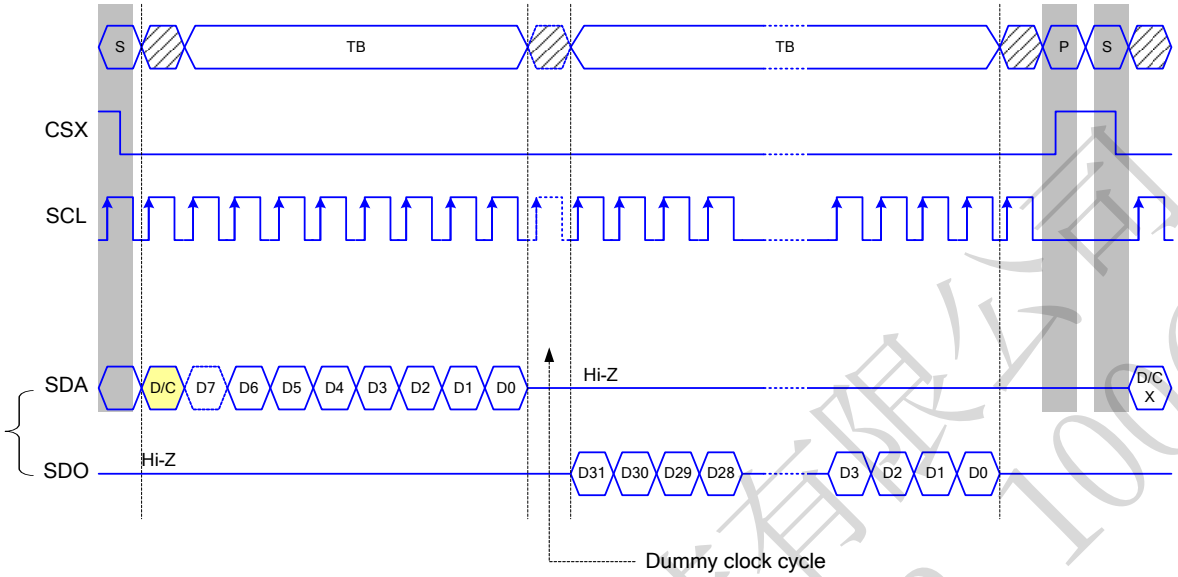



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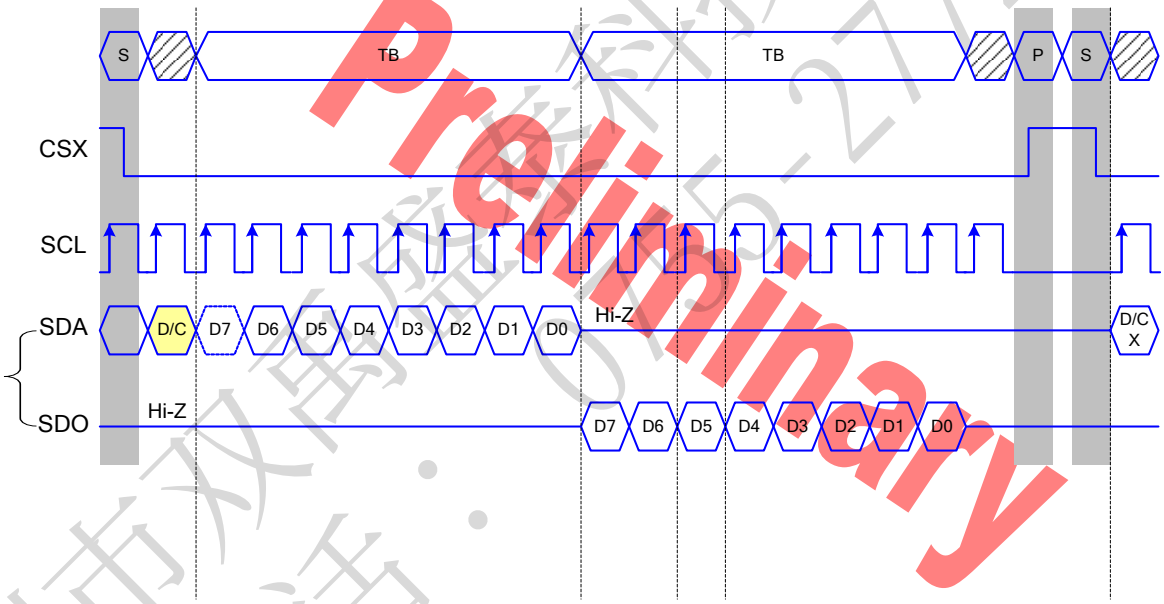


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SPI9:3-line Serial Protocol (for RDDST command: 32-bit read)




SPI9:3-line serial interface read protocol



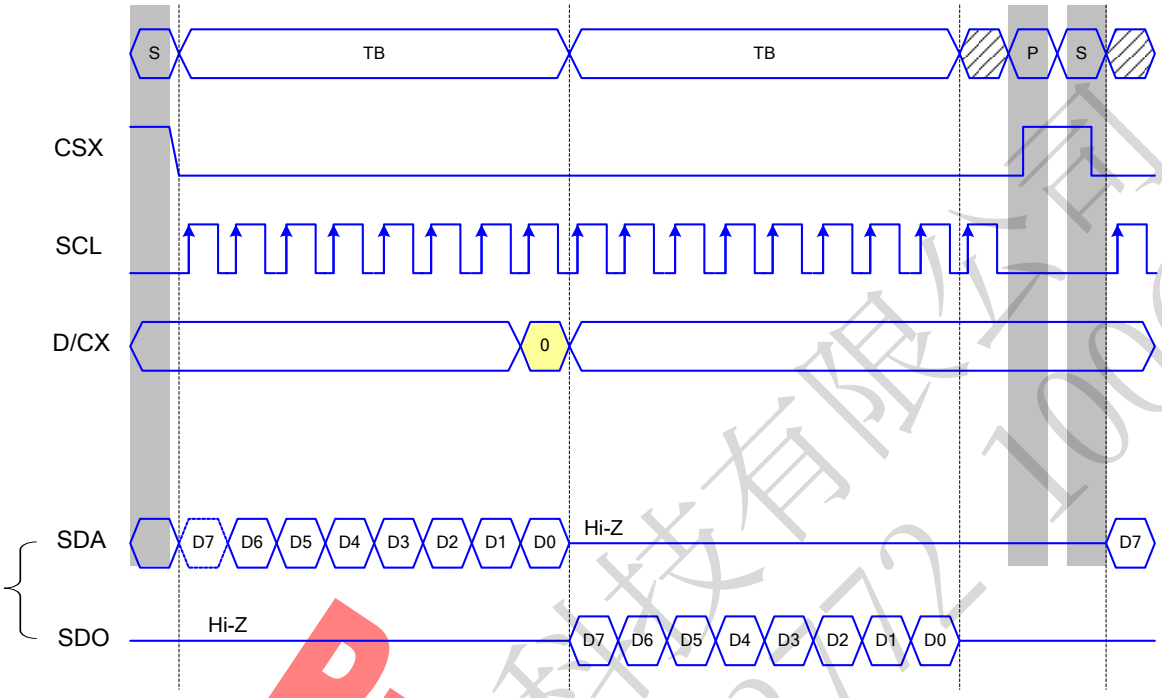

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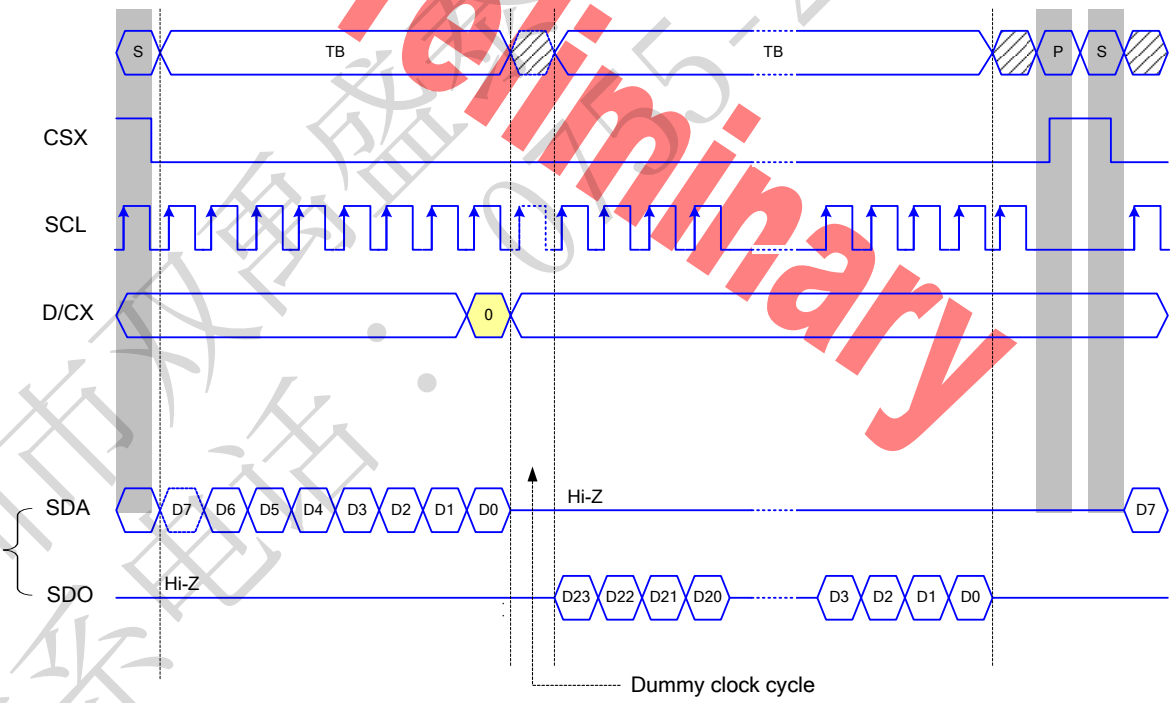
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## SPI8:4-line serial protocol

SPI8:4-line serial protocol (for RDID1/RDID2/RDID3/0Ah/0Bh/0Ch/0Dh/0Eh/0Fh command: 8-bit read):




SPI8:4-line serial protocol (for RDDID command: 24-bit read)

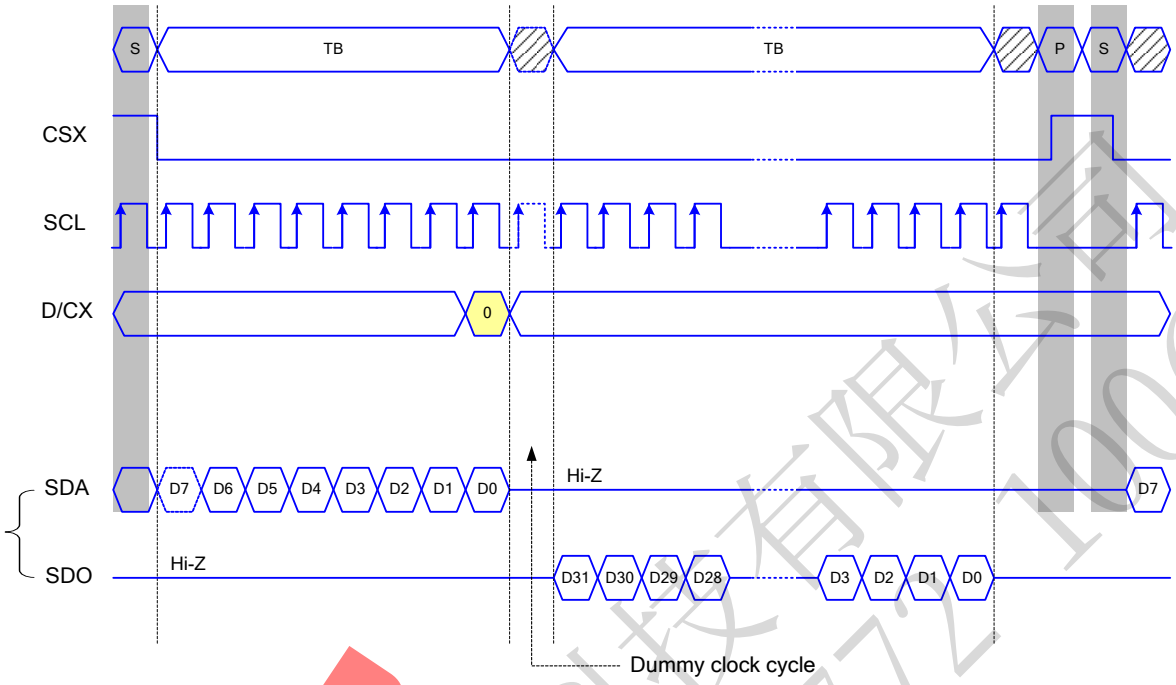



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SPI8:4-line Serial Protocol (for RDDST command: 32-bit read)




**SPI8:4-line serial interface read protocol**

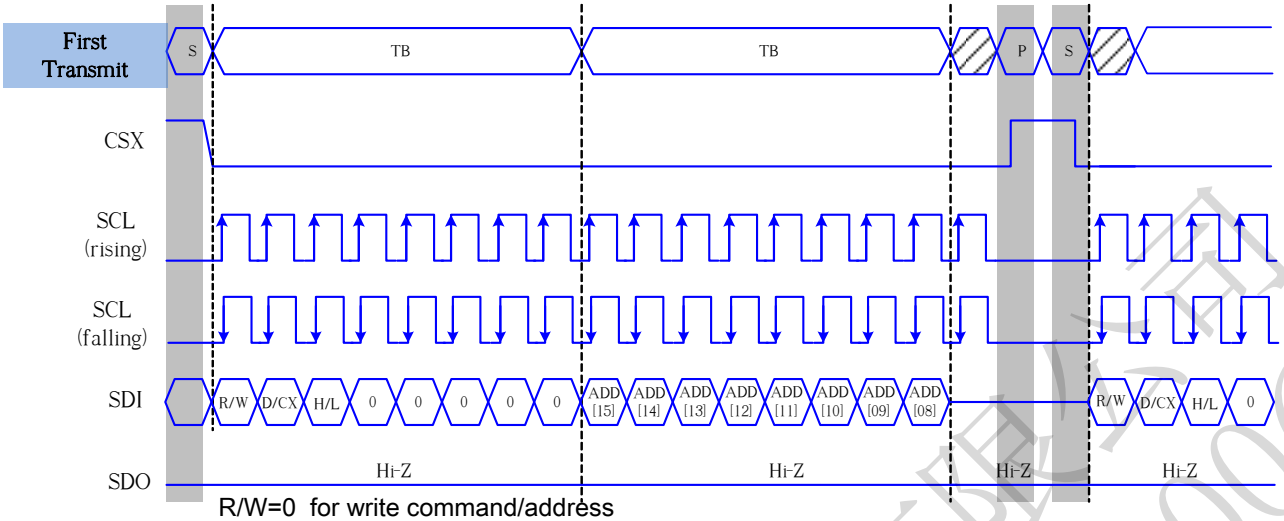
#### 5.5.2.1 SPI16

The read mode of the interface means that the micro controller reads register value from the IC. To do so the micro controller first has to send a command and then the following byte is transmitted in the opposite direction. After that CSX is required to go high before a new command is send. The IC samples the SDI (input data) at the rising edges, but shifts SDO (output data) at the falling SCL edges. Thus the micro controller is supported to read data at the rising SCL edges. After the read status command has been sent, the SDI line must be set to tri-state no later than at the falling SCL edge of the last bit. It doesn't need any dummy clock when execute the command data read.

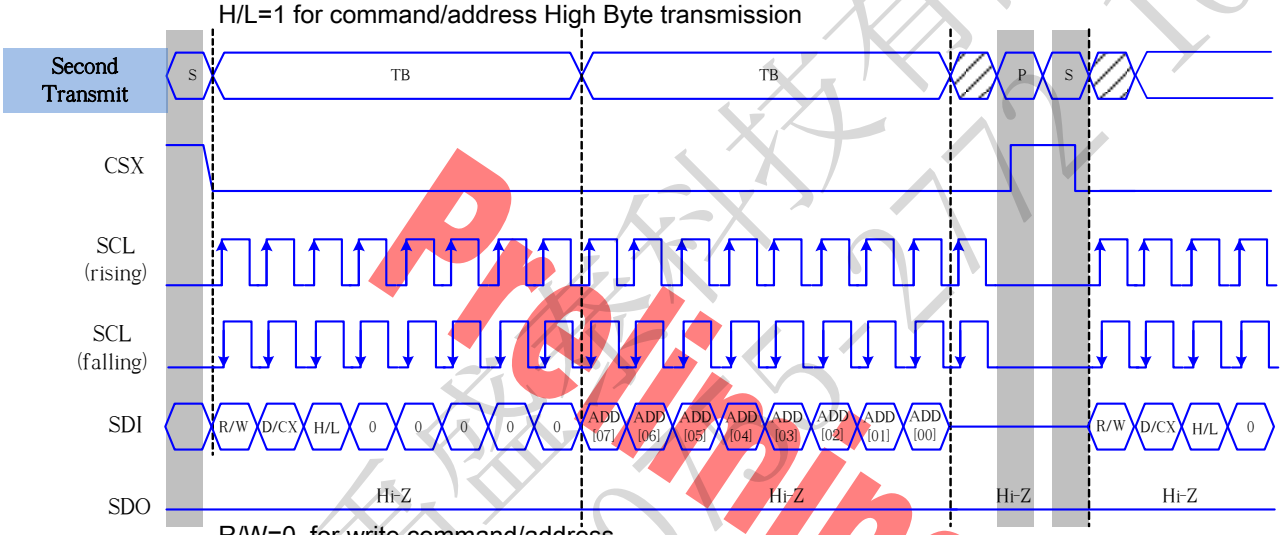
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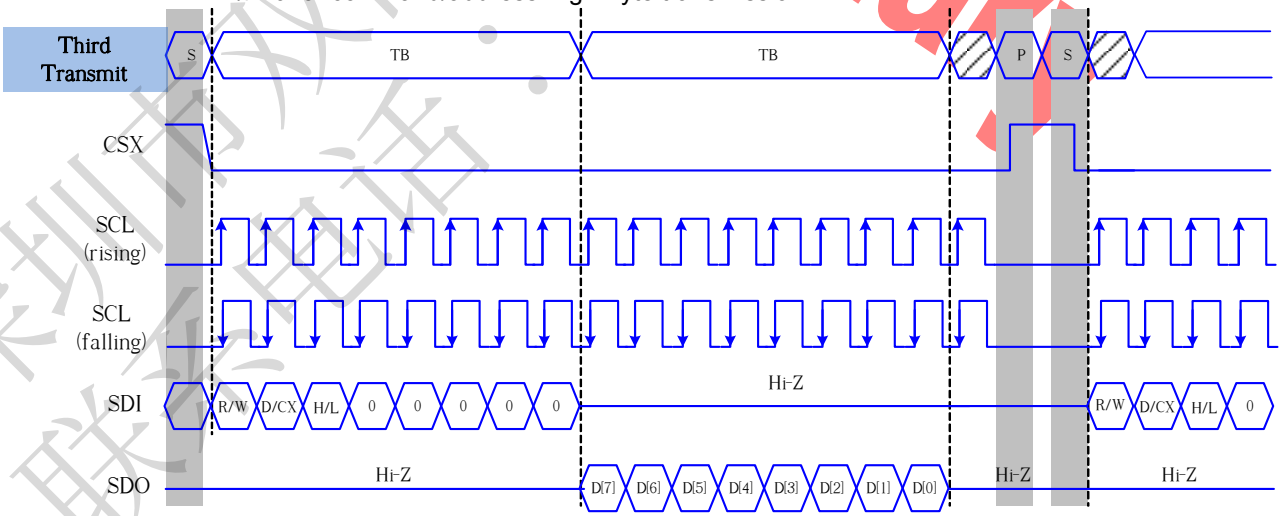
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R/W=0 for write command/address
D/CX=0 for command /address transmission
H/L=1 for command/address High Byte transmission




R/W=0 for write command/address
D/CX=0 for command /address transmission
H/L=0 for command/address High Byte transmission




R/W=1 for read command/address
D/CX=0 for command /address transmission
H/L=0 for command/address High Byte transmission

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### 5.6 RGB Interface

The ST7102 support RGB interface Mode 1 and Mode 2. The interface signals as shown in following table.

The Mode 1 and Mode 2 function is select by setting in the Command 2, please reference application note.

In RGB Mode 1, writing data to line buffer is done by PCLK and Video Data Bus (D[23:0]), when Enable is high state. The external clocks (PCLK, VS and HS) are used for internal displaying clock. So, controller must always transfer PCLK, VS and HS signal to ST7102.

In RGB Mode 2, back porch of Vsync is defined by VBP_HVRGB [7:0] of RGBCTR command. And back porch of Hsync is defined by HBP_HVRGB [7:0] of RGBCTR command. Front porch of Vsync are not setting by this mode

| RGB I/F Mode | PCLK | DE | VS | HS | DB[23:0] | Register for Blanking Porch setting |
|---|---|---|---|---|---|---|
| RGB Mode 1 | Used | Used | Used | Used | Used | Not Used |
| RGB Mode 2 | Used | Not Used | Used | Used | Used | Used |

| Symbol | Name | Description |
|---|---|---|
| PCLK | Pixel clock | Pixel clock for capturing pixels at display interface |
| HS | Horizontal sync | Horizontal synchronization timing signal |
| VS | Vertical sync | Vertical synchronization timing signal |
| DE | Data enable | Data enable signal (assertion indicates valid pixels) |
| DB[23:0] | Pixel data | Pixel data in 16-bit,18-bit and 24-bit format |

The interface signals of RGB interface

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### 5.6.1 RGB Color Format

ST7701S supports two kinds of RGB interface, DE mode (mode 1) and HV mode (mode 2), and 16bit/18bit and 24 bit data format. When DE mode is selected and the VSYNC, HSYNC, DOTCLK, ENABLE, D[23:0] pins can be used; when HV mode is selected and the VSYNC, HSYNC, PCLK, D[23:0] pins can be used. When using RGB interface, only serial interface can be selected.

| Pad name | 24 bits configuration VIPF[3:0]=0111 | 18 bits configuration VIPF[3:0]=0110 | | 16 bits configuration VIPF[3:0]=0101 |
|---|---|---|---|---|
| | | MDT=0 | MDT=1 | |
| DB[23] | R7 | Not used | Not used | Not used |
| DB[22] | R6 | Not used | Not used | Not used |
| DB[21] | R5 | R5 | Not used | Not used |
| DB[20] | R4 | R4 | Not used | R4 |
| DB[19] | R3 | R3 | Not used | R3 |
| DB[18] | R2 | R2 | Not used | R2 |
| DB[17] | R1 | R1 | R5 | R1 |
| DB[16] | R0 | R0 | R4 | R0 |
| DB[15] | G7 | Not used | R3 | Not used |
| DB[14] | G6 | Not used | R2 | Not used |
| DB[13] | G5 | G5 | R1 | G5 |
| DB[12] | G4 | G4 | R0 | G4 |
| DB[11] | G3 | G3 | G5 | G3 |
| DB[10] | G2 | G2 | G4 | G2 |
| DB[09] | G1 | G1 | G3 | G1 |
| DB[08] | G0 | G0 | G2 | G0 |
| DB[07] | B7 | Not used | G1 | Not used |
| DB[06] | B6 | Not used | G0 | Not used |
| DB[05] | B5 | B5 | B5 | Not used |
| DB[04] | B4 | B4 | B4 | B4 |
| DB[03] | B3 | B3 | B3 | B3 |
| DB[02] | B2 | B2 | B2 | B2 |
| DB[01] | B1 | B1 | B1 | B1 |
| DB[00] | B0 | B0 | B0 | B0 |

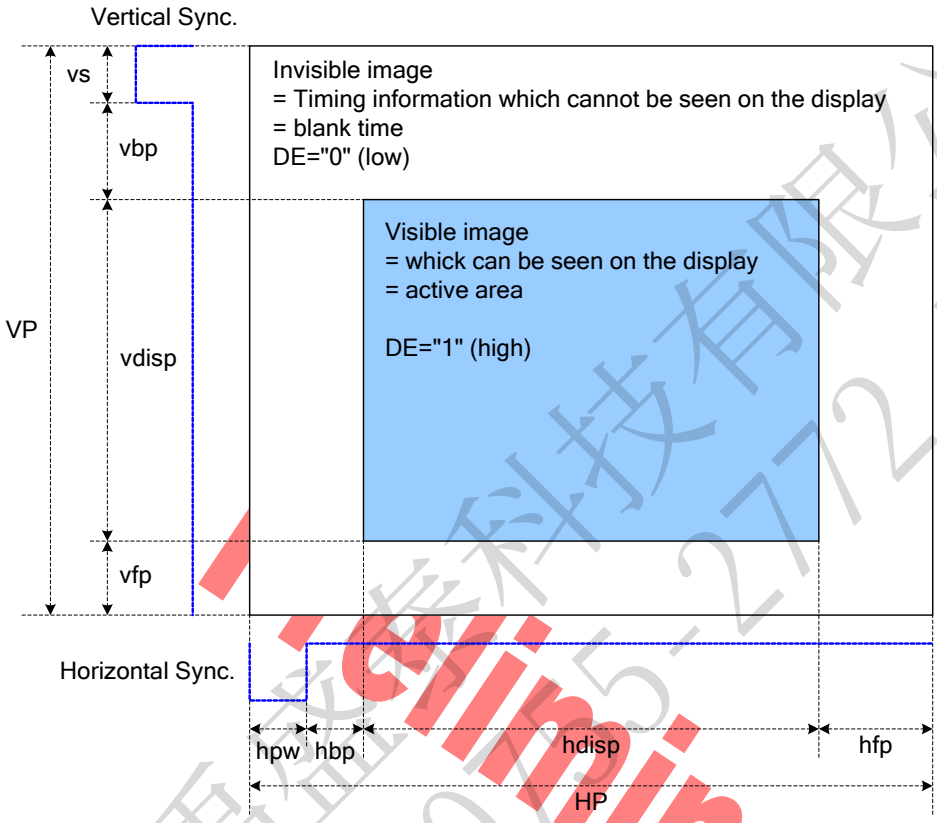
The interface color mapping of RGB interface



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### 5.6.2 RGB Interface Definition

The display operation via the RGB interface is synchronized with the VSYNC, HSYNC, and PCLK signals. The data can be written only within the specified area with low power consumption by using window address function. The back porch and front porch are used to set the RGB interface timing.




**Access Area by RGB Interface**

Please refer to the following table for the setting limitation of RGB interface signals.

| Parameter | Symbol | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| Horizontal Sync. Width | hpw | 2 | - | TBD | Clock |
| Horizontal Sync. Back Porch | hbp | 2 | -- | TBD | Clock |
| Horizontal Sync. Front Porch | hfp | 2 | -- | - | Clock |
| Vertical Sync. Width | vs | 2 | -- | TBD | Line |
| Vertical Sync. Back Porch | vbp | 2 | -- | TBD | Line |
| Vertical Sync. Front Porch | vfp | 2 | -- | -- | Line |

*Note:*

*1、Typical value are related to the setting frame rate is 60Hz..*

*2、VS+VBP<=* TBD, *HPW+HBP<=* TBD

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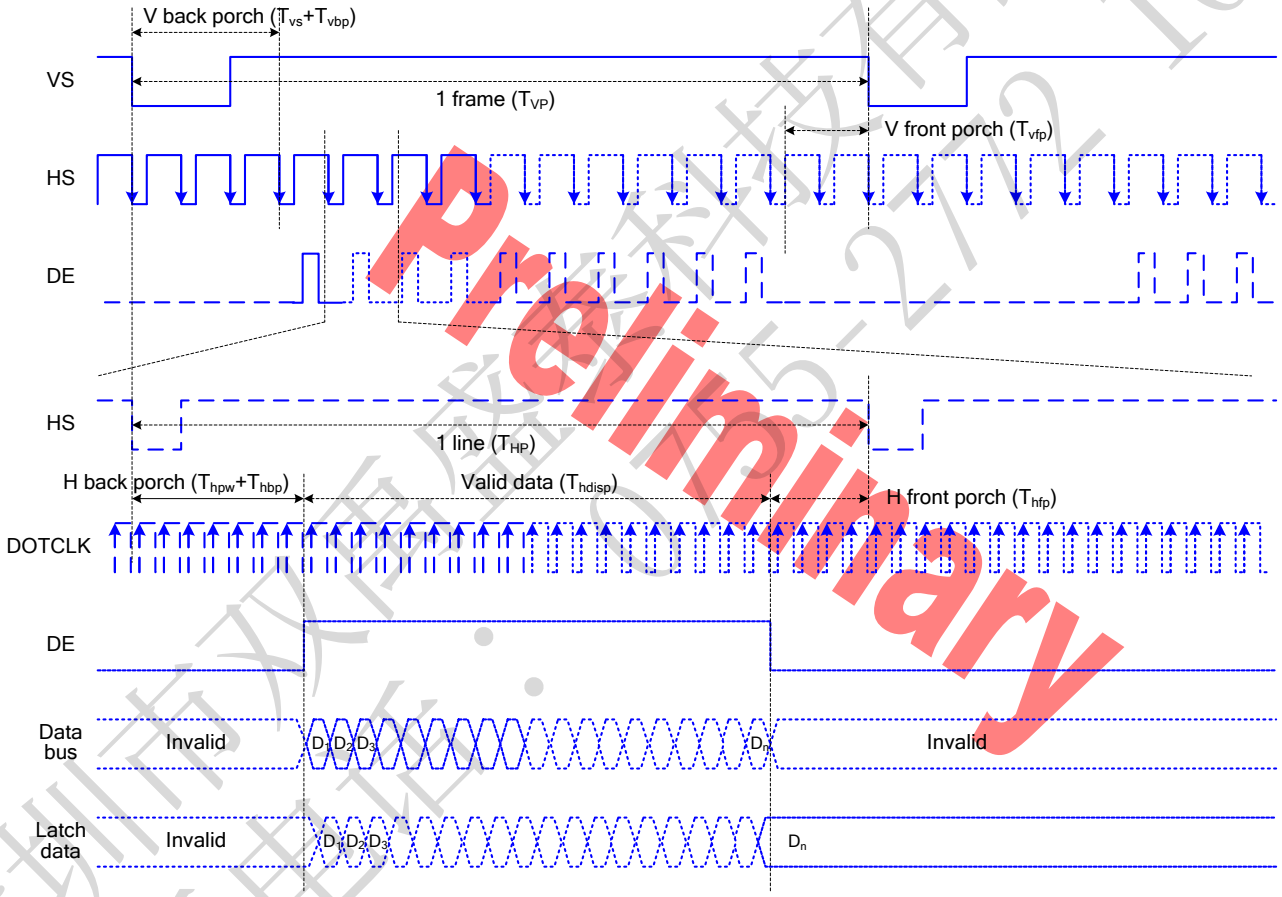
### 5.6.1 RGB Interface Mode Selection

ST7102 supports two kinds of RGB interface, DE mode and HV mode. The table shown below uses command C3h to select RGB interface mode.

| DE/HV(Sync) | RGB Mode |
|---|---|
| 0 | DE mode |
| 1 | HV mode |

## RGB Interface Timing

The timing chart of RGB interface DE mode is shown as follows.




*Note: The setting of front porch and back porch in host must match that in IC as this mode.*

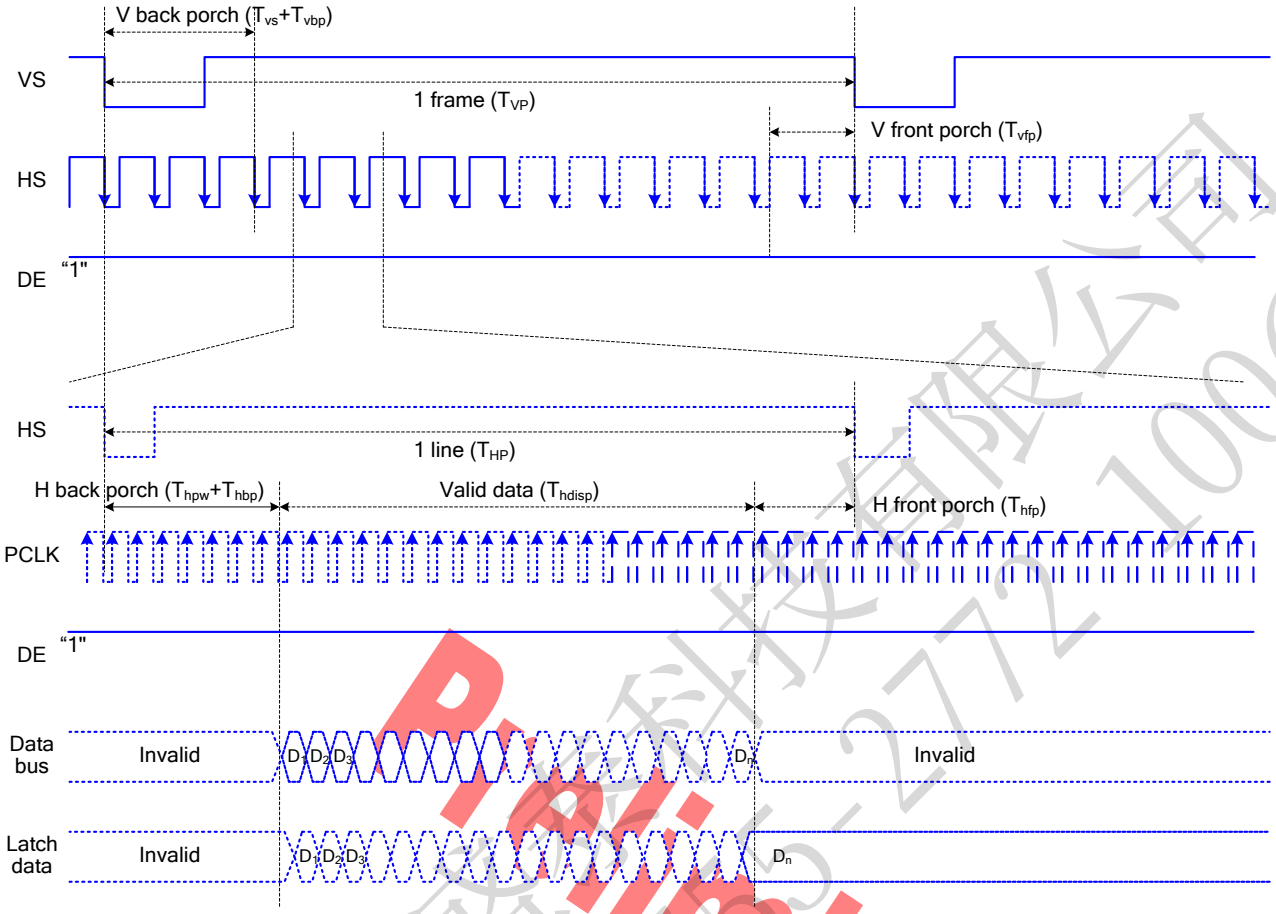
**Timing Chart of Signals in RGB Interface DE Mode**

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The timing chart of RGB interface HV mode is shown as follows.




**Timing chart of RGB interface HV mode**

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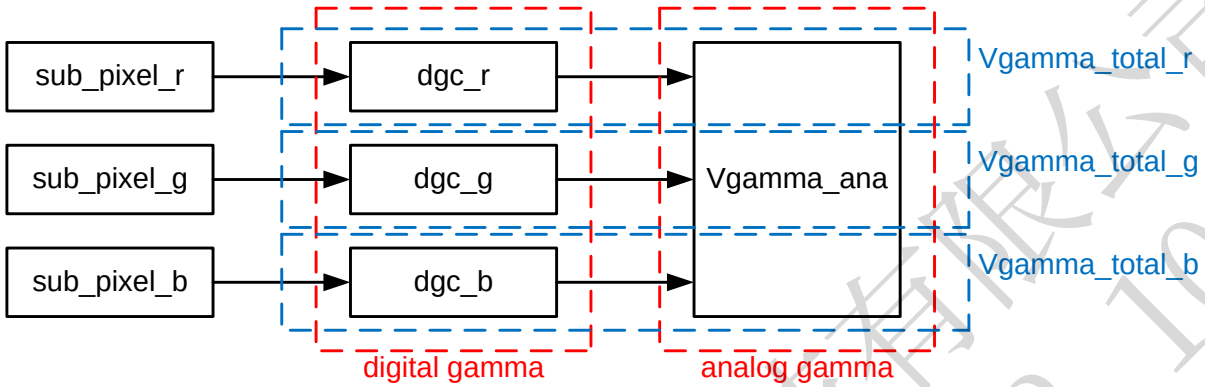


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### 5.7 Digital Gamma

Digital gamma correct makes three sub-pixel (Red, Green, Blue) gamma curve by using one analog gamma curve.

By setting the digital gamma correction registers we can make three equivalent gamma curve Vgamma_total_r, Vgamma_total_g, Vgamma_total_b as desired.




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## 5.8 Gamma Correction Function

ST7102 incorporates the gamma correction function to display 16,777,216 colors for the LCD panel. There is a analog gamma correction is performed with 2 registers to set both positive and negative polarity voltage for gamma curve.

### 5.8.1 Gamma Correction Registers

The all of grayscale reference levels are shown blow.

| Register Groups | Gamma Polarity | | Description |
|---|---|---|---|
| | Positive | Negative | |
| Grayscale adjustment | VGMP0[9:0] | VGMN0[9:0] | 1023-to-1 selector (voltage level of R grayscale 0) |
| | VGMP1[9:0] | VGMN1[9:0] | 1023-to-1 selector (voltage level of R grayscale 1) |
| | VGMP2[9:0] | VGMN2[9:0] | 1023-to-1 selector (voltage level of R grayscale 2) |
| | VGMP3[9:0] | VGMN3[9:0] | 1023-to-1 selector (voltage level of R grayscale 4) |
| | VGMP4[9:0] | VGMN4[9:0] | 1023-to-1 selector (voltage level of R grayscale 8) |
| | VGMP5[5:0] | VGMN5[5:0] | 1023-to-1 selector (voltage level of R grayscale 12) |
| | VGMP6[9:0] | VGMN6[9:0] | 1023-to-1 selector (voltage level of R grayscale 16) |
| | VGMP7[5:0] | VGMN7[5:0] | 1023-to-1 selector (voltage level of R grayscale 24) |
| | VGMP8[9:0] | VGMN8[9:0] | 1023-to-1 selector (voltage level of R grayscale 32) |
| | VGMP9[5:0] | VGMN9[5:0] | 1023-to-1 selector (voltage level of R grayscale 48) |
| | VGMP10[9:0] | VGMN10[9:0] | 1023-to-1 selector (voltage level of R grayscale 64) |
| | VGMP11[3:0] | VGMN11[3:0] | 1023-to-1 selector (voltage level of R grayscale 80) |
| | VGMP12[9:0] | VGMN12[9:0] | 1023-to-1 selector (voltage level of R grayscale 96) |
| | VGMP13[3:0] | VGMN13[3:0] | 1023-to-1 selector (voltage level of R grayscale 112) |
| | VGMP14[9:0] | VGMN14[9:0] | 1023-to-1 selector (voltage level of R grayscale 128) |
| | VGMP15[3:0] | VGMN15[3:0] | 1023-to-1 selector (voltage level of R grayscale 144) |
| | VGMP16[9:0] | VGMN16[9:0] | 1023-to-1 selector (voltage level of R grayscale 160) |
| | VGMP17[3:0] | VGMN17[3:0] | 1023-to-1 selector (voltage level of R grayscale 176) |
| | VGMP18[9:0] | VGMN18[9:0] | 1023-to-1 selector (voltage level of R grayscale 192) |
| | VGMP19[3:0] | VGMN19[3:0] | 1023-to-1 selector (voltage level of R grayscale 208) |
| | VGMP20[9:0] | VGMN20[9:0] | 1023-to-1 selector (voltage level of R grayscale 224) |
| | VGMP21[3:0] | VGMN21[3:0] | 1023-to-1 selector (voltage level of R grayscale 232) |
| | VGMP22[9:0] | VGMN22[9:0] | 1023-to-1 selector (voltage level of R grayscale 240) |
| | VGMP23[3:0] | VGMN23[3:0] | 1023-to-1 selector (voltage level of R grayscale 244) |
| | VGMP24[9:0] | VGMN24[9:0] | 1023-to-1 selector (voltage level of R grayscale 248) |
| | VGMP25[3:0] | VGMN25[3:0] | 1023-to-1 selector (voltage level of R grayscale 250) |

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| | | | |
|---|---|---|---|
| | VGMP26[9:0] | VGMN26[9:0] | 1023-to-1 selector (voltage level of R grayscale 252) |
| | VGMP27[9:0] | VGMN27[9:0] | 1023-to-1 selector (voltage level of R grayscale 254) |
| | VGMP28[9:0] | VGMN28[9:0] | 1023-to-1 selector (voltage level of R grayscale 255) |

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### 5.8.2 Gamma function architecture

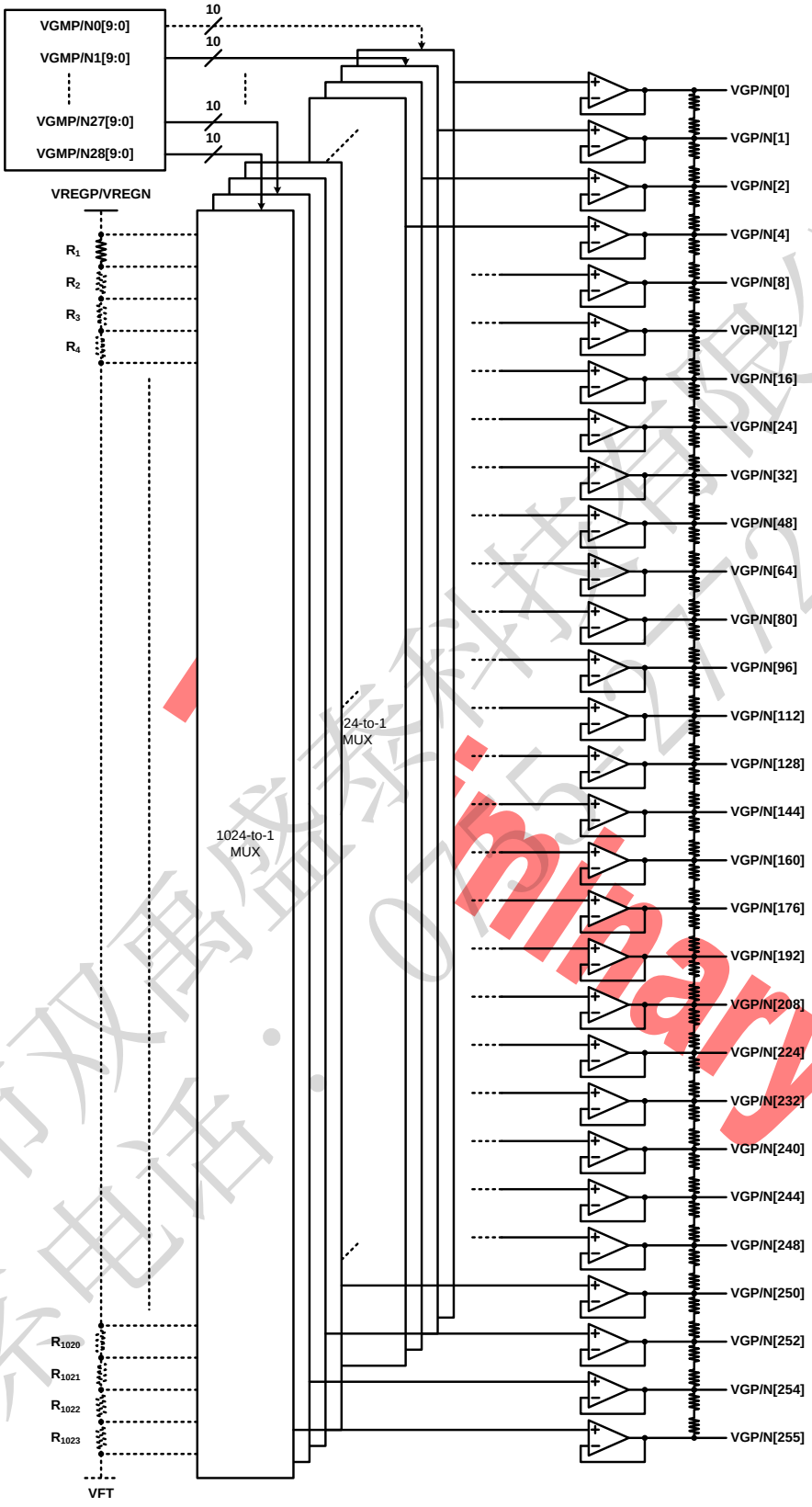



**Figure 66 positive and Negative grayscale voltage generation**



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*Note1. VFT: Feed through voltage*

*Note2.VREGP: VFT+VGPAMP=feed through voltage + positive gamma amplitude.*

*Note3.VREGN: VFT+VGNAMP=feed through voltage + negative gamma amplitude.*

### 5.8.3 Grayscale voltage formula

| Grayscale Voltage | Formula | Grayscale Voltage | Formula |
|---|---|---|---|
| V0 | VGP/N0 | V64 | VGP/N10 |
| V1 | VGP/N1 | V65 | 15/16 * VGP/N10 + 1/16 * VGP/N11 |
| V2 | VGP/N2 | V66 | 14/16 * VGP/N10 + 2/16 * VGP/N11 |
| V3 | 1/2 * VGP/N2 + 1/2* VGP/N3 | V67 | 13/16 * VGP/N10 + 3/16 * VGP/N11 |
| V4 | VGP/N3 | V68 | 12/16 * VGP/N10 + 4/16 * VGP/N11 |
| V5 | 3/4 * VGP/N3 + 1/4* VGP/N4 | V69 | 11/16 * VGP/N10 + 5/16 * VGP/N11 |
| V6 | 2/4 * VGP/N3 + 2/4* VGP/N4 | V70 | 10/16 * VGP/N10 + 6/16 * VGP/N11 |
| V7 | 1/4 * VGP/N3 + 3/4* VGP/N4 | V71 | 9/16 * VGP/N10 + 7/16 * VGP/N11 |
| V8 | VGP/N4 | V72 | 8/16 * VGP/N10 + 8/16 * VGP/N11 |
| V9 | 3/4 * VGP/N4 + 1/4* VGP/N5 | V73 | 7/16 * VGP/N10 + 9/16 * VGP/N11 |
| V10 | 2/4 * VGP/N4 + 2/4* VGP/N5 | V74 | 6/16 * VGP/N10 + 10/16 * VGP/N11 |
| V11 | 1/4 * VGP/N4 + 3/4* VGP/N5 | V75 | 5/16 * VGP/N10 + 11/16 * VGP/N11 |
| V12 | VGP/N5 | V76 | 4/16 * VGP/N10 + 12/16 * VGP/N11 |
| V13 | 3/4 * VGP/N5 + 1/4* VGP/N6 | V77 | 3/16 * VGP/N10 + 13/16 * VGP/N11 |
| V14 | 2/4 * VGP/N5 + 2/4* VGP/N6 | V78 | 2/16 * VGP/N10 + 14/16 * VGP/N11 |
| V15 | 1/4 * VGP/N5 + 3/4* VGP/N6 | V79 | 1/16 * VGP/N10 + 15/16 * VGP/N11 |
| V16 | VGP/N6 | V80 | VGP/N11 |
| V17 | 7/8 * VGP/N6 + 1/8 * VGP/N7 | V81 | 15/16 * VGP/N11 + 1/16 * VGP/N12 |
| V18 | 6/8 * VGP/N6 + 2/8 * VGP/N7 | V82 | 14/16 * VGP/N11 + 2/16 * VGP/N12 |
| V19 | 5/8 * VGP/N6 + 3/8 * VGP/N7 | V83 | 13/16 * VGP/N11 + 3/16 * VGP/N12 |
| V20 | 4/8 * VGP/N6 + 4/8 * VGP/N7 | V84 | 12/16 * VGP/N11 + 4/16 * VGP/N12 |
| V21 | 3/8 * VGP/N6 + 5/8 * VGP/N7 | V85 | 11/16 * VGP/N11 + 5/16 * VGP/N12 |
| V22 | 2/8 * VGP/N6 + 6/8 * VGP/N7 | V86 | 10/16 * VGP/N11 + 6/16 * VGP/N12 |
| V23 | 1/8 * VGP/N6 + 7/8 * VGP/N7 | V87 | 9/16 * VGP/N11 + 7/16 * VGP/N12 |
| V24 | VGP/N7 | V88 | 8/16 * VGP/N11 + 8/16 * VGP/N12 |
| V25 | 7/8 * VGP/N7 + 1/8 * VGP/N8 | V89 | 7/16 * VGP/N11 + 9/16 * VGP/N12 |
| V26 | 6/8 * VGP/N7 + 2/8 * VGP/N8 | V90 | 6/16 * VGP/N11 + 10/16 * VGP/N12 |
| V27 | 5/8 * VGP/N7 + 3/8 * VGP/N8 | V91 | 5/16 * VGP/N11 + 11/16 * VGP/N12 |

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| | | | |
|---|---|---|---|
| V28 | 4/8 * VGP/N7 + 4/8 * VGP/N8 | V92 | 4/16 * VGP/N11 + 12/16 * VGP/N12 |
| V29 | 3/8 * VGP/N7 + 5/8 * VGP/N8 | V93 | 3/16 * VGP/N11 + 13/16 * VGP/N12 |
| V30 | 2/8 * VGP/N7 + 6/8 * VGP/N8 | V94 | 2/16 * VGP/N11 + 14/16 * VGP/N12 |
| V31 | 1/8 * VGP/N7 + 7/8 * VGP/N8 | V95 | 1/16 * VGP/N11 + 15/16 * VGP/N12 |
| V32 | VGP/N8 | V96 | VGP/N12 |
| V33 | 15/16 * VGP/N8 + 1/16 * VGP/N9 | V97 | 15/16 * VGP/N12 + 1/16 * VGP/N13 |
| V34 | 14/16 * VGP/N8 + 2/16 * VGP/N9 | V98 | 14/16 * VGP/N12 + 2/16 * VGP/N13 |
| V35 | 13/16 * VGP/N8 + 3/16 * VGP/N9 | V99 | 13/16 * VGP/N12 + 3/16 * VGP/N13 |
| V36 | 12/16 * VGP/N8 + 4/16 * VGP/N9 | V100 | 12/16 * VGP/N12 + 4/16 * VGP/N13 |
| V37 | 11/16 * VGP/N8 + 5/16 * VGP/N9 | V101 | 11/16 * VGP/N12 + 5/16 * VGP/N13 |
| V38 | 10/16 * VGP/N8 + 6/16 * VGP/N9 | V102 | 10/16 * VGP/N12 + 6/16 * VGP/N13 |
| V39 | 9/16 * VGP/N8 + 7/16 * VGP/N9 | V103 | 9/16 * VGP/N12 + 7/16 * VGP/N13 |
| V40 | 8/16 * VGP/N8 + 8/16 * VGP/N9 | V104 | 8/16 * VGP/N12 + 8/16 * VGP/N13 |
| V41 | 7/16 * VGP/N8 + 9/16 * VGP/N9 | V105 | 7/16 * VGP/N12 + 9/16 * VGP/N13 |
| V42 | 6/16 * VGP/N8 + 10/16 * VGP/N9 | V106 | 6/16 * VGP/N12 + 10/16 * VGP/N13 |
| V43 | 5/16 * VGP/N8 + 11/16 * VGP/N9 | V107 | 5/16 * VGP/N12 + 11/16 * VGP/N13 |
| V44 | 4/16 * VGP/N8 + 12/16 * VGP/N9 | V108 | 4/16 * VGP/N12 + 12/16 * VGP/N13 |
| V45 | 3/16 * VGP/N8 + 13/16 * VGP/N9 | V109 | 3/16 * VGP/N12 + 13/16 * VGP/N13 |
| V46 | 2/16 * VGP/N8 + 14/16 * VGP/N9 | V110 | 2/16 * VGP/N12 + 14/16 * VGP/N13 |
| V47 | 1/16 * VGP/N8 + 15/16 * VGP/N9 | V111 | 1/16 * VGP/N12 + 15/16 * VGP/N13 |
| V48 | VGP/N9 | V112 | VGP/N13 |
| V49 | 15/16 * VGP/N9 + 1/16 * VGP/N10 | V113 | 15/16 * VGP/N13 + 1/16 * VGP/N14 |
| V50 | 14/16 * VGP/N9 + 2/16 * VGP/N10 | V114 | 14/16 * VGP/N13 + 2/16 * VGP/N14 |
| V51 | 13/16 * VGP/N9 + 3/16 * VGP/N10 | V115 | 13/16 * VGP/N13 + 3/16 * VGP/N14 |
| V52 | 12/16 * VGP/N9 + 4/16 * VGP/N10 | V116 | 12/16 * VGP/N13 + 4/16 * VGP/N14 |
| V53 | 11/16 * VGP/N9 + 5/16 * VGP/N10 | V117 | 11/16 * VGP/N13 + 5/16 * VGP/N14 |
| V54 | 10/16 * VGP/N9 + 6/16 * VGP/N10 | V118 | 10/16 * VGP/N13 + 6/16 * VGP/N14 |
| V55 | 9/16 * VGP/N9 + 7/16 * VGP/N10 | V119 | 9/16 * VGP/N13 + 7/16 * VGP/N14 |
| V56 | 8/16 * VGP/N9 + 8/16 * VGP/N10 | V120 | 8/16 * VGP/N13 + 8/16 * VGP/N14 |
| V57 | 7/16 * VGP/N9 + 9/16 * VGP/N10 | V121 | 7/16 * VGP/N13 + 9/16 * VGP/N14 |
| V58 | 6/16 * VGP/N9 + 10/16 * VGP/N10 | V122 | 6/16 * VGP/N13 + 10/16 * VGP/N14 |
| V59 | 5/16 * VGP/N9 + 11/16 * VGP/N10 | V123 | 5/16 * VGP/N13 + 11/16 * VGP/N14 |
| V60 | 4/16 * VGP/N9 + 12/16 * VGP/N10 | V124 | 4/16 * VGP/N13 + 12/16 * VGP/N14 |
| V61 | 3/16 * VGP/N9 + 13/16 * VGP/N10 | V125 | 3/16 * VGP/N13 + 13/16 * VGP/N14 |
| V62 | 2/16 * VGP/N9 + 14/16 * VGP/N10 | V126 | 2/16 * VGP/N13 + 14/16 * VGP/N14 |
| V63 | 1/16 * VGP/N9 + 15/16 * VGP/N10 | V127 | 1/16 * VGP/N13 + 15/16 * VGP/N14 |

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| Grayscale Voltage | Formula | Grayscale Voltage | Formula |
|---|---|---|---|
| V128 | VGP/N14 | V192 | VGP/N18 |
| V129 | 15/16 * VGP/N14 + 1/16 * VGP/N15 | V193 | 15/16 * VGP/N18 + 1/16 * VGP/N19 |
| V130 | 14/16 * VGP/N14 + 2/16 * VGP/N15 | V194 | 14/16 * VGP/N18 + 2/16 * VGP/N19 |
| V131 | 13/16 * VGP/N14 + 3/16 * VGP/N15 | V195 | 13/16 * VGP/N18 + 3/16 * VGP/N19 |
| V132 | 12/16 * VGP/N14 + 4/16 * VGP/N15 | V196 | 12/16 * VGP/N18 + 4/16 * VGP/N19 |
| V133 | 11/16 * VGP/N14 + 5/16 * VGP/N15 | V197 | 11/16 * VGP/N18 + 5/16 * VGP/N19 |
| V134 | 10/16 * VGP/N14 + 6/16 * VGP/N15 | V198 | 10/16 * VGP/N18 + 6/16 * VGP/N19 |
| V135 | 9/16 * VGP/N14 + 7/16 * VGP/N15 | V199 | 9/16 * VGP/N18 + 7/16 * VGP/N19 |
| V136 | 8/16 * VGP/N14 + 8/16 * VGP/N15 | V200 | 8/16 * VGP/N18 + 8/16 * VGP/N19 |
| V137 | 7/16 * VGP/N14 + 9/16 * VGP/N15 | V201 | 7/16 * VGP/N18 + 9/16 * VGP/N19 |
| V138 | 6/16 * VGP/N14 + 10/16 * VGP/N15 | V202 | 6/16 * VGP/N18 + 10/16 * VGP/N19 |
| V139 | 5/16 * VGP/N14 + 11/16 * VGP/N15 | V203 | 5/16 * VGP/N18 + 11/16 * VGP/N19 |
| V140 | 4/16 * VGP/N14 + 12/16 * VGP/N15 | V204 | 4/16 * VGP/N18 + 12/16 * VGP/N19 |
| V141 | 3/16 * VGP/N14 + 13/16 * VGP/N15 | V205 | 3/16 * VGP/N18 + 13/16 * VGP/N19 |
| V142 | 2/16 * VGP/N14 + 14/16 * VGP/N15 | V206 | 2/16 * VGP/N18 + 14/16 * VGP/N19 |
| V143 | 1/16 * VGP/N14 + 15/16 * VGP/N15 | V207 | 1/16 * VGP/N18 + 15/16 * VGP/N19 |
| V144 | VGP/N15 | V208 | VGP/N19 |
| V145 | 15/16 * VGP/N15 + 1/16 * VGP/N16 | V209 | 15/16 * VGP/N19 + 1/16 * VGP/N20 |
| V146 | 14/16 * VGP/N15 + 2/16 * VGP/N16 | V210 | 14/16 * VGP/N19 + 2/16 * VGP/N20 |
| V147 | 13/16 * VGP/N15 + 3/16 * VGP/N16 | V211 | 13/16 * VGP/N19 + 3/16 * VGP/N20 |
| V148 | 12/16 * VGP/N15 + 4/16 * VGP/N16 | V212 | 12/16 * VGP/N19 + 4/16 * VGP/N20 |
| V149 | 11/16 * VGP/N15 + 5/16 * VGP/N16 | V213 | 11/16 * VGP/N19 + 5/16 * VGP/N20 |
| V150 | 10/16 * VGP/N15 + 6/16 * VGP/N16 | V214 | 10/16 * VGP/N19 + 6/16 * VGP/N20 |
| V151 | 9/16 * VGP/N15 + 7/16 * VGP/N16 | V215 | 9/16 * VGP/N19 + 7/16 * VGP/N20 |
| V152 | 8/16 * VGP/N15 + 8/16 * VGP/N16 | V216 | 8/16 * VGP/N19 + 8/16 * VGP/N20 |
| V153 | 7/16 * VGP/N15 + 9/16 * VGP/N16 | V217 | 7/16 * VGP/N19 + 9/16 * VGP/N20 |
| V154 | 6/16 * VGP/N15 + 10/16 * VGP/N16 | V218 | 6/16 * VGP/N19 + 10/16 * VGP/N20 |
| V155 | 5/16 * VGP/N15 + 11/16 * VGP/N16 | V219 | 5/16 * VGP/N19 + 11/16 * VGP/N20 |
| V156 | 4/16 * VGP/N15 + 12/16 * VGP/N16 | V220 | 4/16 * VGP/N19 + 12/16 * VGP/N20 |
| V157 | 3/16 * VGP/N15 + 13/16 * VGP/N16 | V221 | 3/16 * VGP/N19 + 13/16 * VGP/N20 |
| V158 | 2/16 * VGP/N15 + 14/16 * VGP/N16 | V222 | 2/16 * VGP/N19 + 14/16 * VGP/N20 |
| V159 | 1/16 * VGP/N15 + 15/16 * VGP/N16 | V223 | 1/16 * VGP/N19 + 15/16 * VGP/N20 |
| V160 | VGP/N16 | V224 | VGP/N20 |
| V161 | 15/16 * VGP/N16 + 1/16 * VGP/N17 | V225 | 7/8 * VGP/N20 + 1/8 * VGP/N21 |

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| | | | |
|---|---|---|---|
| V162 | 14/16 * VGP/N16 + 2/16 * VGP/N17 | V226 | 6/8 * VGP/N20 + 2/8 * VGP/N21 |
| V163 | 13/16 * VGP/N16 + 3/16 * VGP/N17 | V227 | 5/8 * VGP/N20 + 3/8 * VGP/N21 |
| V164 | 12/16 * VGP/N16 + 4/16 * VGP/N17 | V228 | 4/8 * VGP/N20 + 4/8 * VGP/N21 |
| V165 | 11/16 * VGP/N16 + 5/16 * VGP/N17 | V229 | 3/8 * VGP/N20 + 5/8 * VGP/N21 |
| V166 | 10/16 * VGP/N16 + 6/16 * VGP/N17 | V230 | 2/8 * VGP/N20 + 6/8 * VGP/N21 |
| V167 | 9/16 * VGP/N16 + 7/16 * VGP/N17 | V231 | 1/8 * VGP/N20 + 7/8 * VGP/N21 |
| V168 | 8/16 * VGP/N16 + 8/16 * VGP/N17 | V232 | VGP/N21 |
| V169 | 7/16 * VGP/N16 + 9/16 * VGP/N17 | V233 | 7/8 * VGP/N21 + 1/8 * VGP/N22 |
| V170 | 6/16 * VGP/N16 + 10/16 * VGP/N17 | V234 | 6/8 * VGP/N21 + 2/8 * VGP/N22 |
| V171 | 5/16 * VGP/N16 + 11/16 * VGP/N17 | V235 | 5/8 * VGP/N21 + 3/8 * VGP/N22 |
| V172 | 4/16 * VGP/N16 + 12/16 * VGP/N17 | V236 | 4/8 * VGP/N21 + 4/8 * VGP/N22 |
| V173 | 3/16 * VGP/N16 + 13/16 * VGP/N17 | V237 | 3/8 * VGP/N21 + 5/8 * VGP/N22 |
| V174 | 2/16 * VGP/N16 + 14/16 * VGP/N17 | V238 | 2/8 * VGP/N21 + 6/8 * VGP/N22 |
| V175 | 1/16 * VGP/N16 + 15/16 * VGP/N17 | V239 | 1/8 * VGP/N21 + 7/8 * VGP/N22 |
| V176 | VGP/N17 | V240 | VGP/N22 |
| V177 | 15/16 * VGP/N17 + 1/16 * VGP/N18 | V241 | 3/4 * VGP/N22 + 1/4* VGP/N23 |
| V178 | 14/16 * VGP/N17 + 2/16 * VGP/N18 | V242 | 2/4 * VGP/N22 + 2/4* VGP/N23 |
| V179 | 13/16 * VGP/N17 + 3/16 * VGP/N18 | V243 | 1/4 * VGP/N22 + 3/4* VGP/N23 |
| V180 | 12/16 * VGP/N17 + 4/16 * VGP/N18 | V244 | VGP/N23 |
| V181 | 11/16 * VGP/N17 + 5/16 * VGP/N18 | V245 | 3/4 * VGP/N23 + 1/4* VGP/N24 |
| V182 | 10/16 * VGP/N17 + 6/16 * VGP/N18 | V246 | 2/4 * VGP/N23 + 2/4* VGP/N24 |
| V183 | 9/16 * VGP/N17 + 7/16 * VGP/N18 | V247 | 1/4 * VGP/N23 + 3/4* VGP/N24 |
| V184 | 8/16 * VGP/N17 + 8/16 * VGP/N18 | V248 | VGP/N24 |
| V185 | 7/16 * VGP/N17 + 9/16 * VGP/N18 | V249 | 1/2 * VGP/N24 + 1/2* VGP/N25 |
| V186 | 6/16 * VGP/N17 + 10/16 * VGP/N18 | V250 | VGP/N25 |
| V187 | 5/16 * VGP/N17 + 11/16 * VGP/N18 | V251 | 1/2 * VGP/N25 + 1/2* VGP/N26 |
| V188 | 4/16 * VGP/N17 + 12/16 * VGP/N18 | V252 | VGP/N26 |
| V189 | 3/16 * VGP/N17 + 13/16 * VGP/N18 | V253 | 1/2 * VGP/N26 + 1/2* VGP/N27 |
| V190 | 2/16 * VGP/N17 + 14/16 * VGP/N18 | V254 | VGP/N27 |
| V191 | 1/16 * VGP/N17 + 15/16 * VGP/N18 | V255 | VGP/N28 |

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### 5.9 Reset function

The Reset function of ST7102 is triggered by RESX input. After reset function is triggered, the ST7102 is into a reset period, and the duration of this period must be at least 1ms. During this period, the ST7102 and its power circuit is initialized.

**Initial State Of Output Pins**

| Output Pins Name | Initial State |
|---|---|
| S[480:1] (source output) | GND |
| VGH | GND |
| VGL | GND |
| VCOM | GND |
| CGOUTL_R[16:1](GIP signal) | GND |

#### 5.9.1 Reset Timing Diagram

ST7102 provides Power On Reset and HWRST pin (RESX) for IC initialization and exit Deep Standby Mode (DSTB). For exiting DSTB Mode, HWRST pin should be tied to high at least 10ms. The timing diagram is located at below.

##### 5.9.1.1 Power On Reset & HWRST Reset

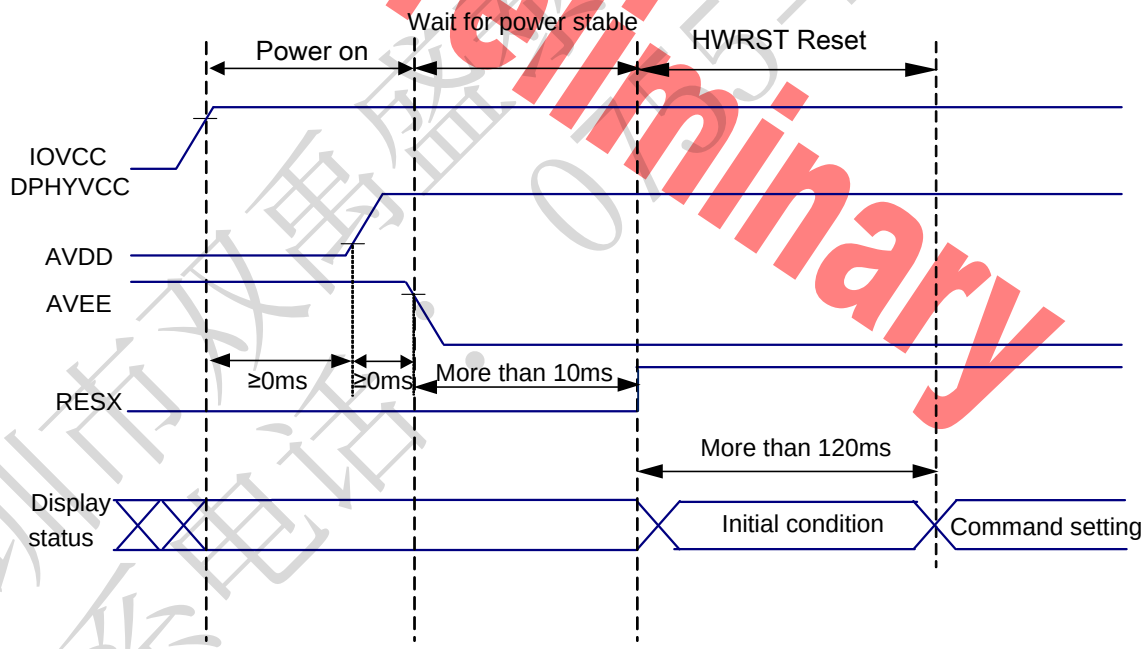



**Figure 67 Power on Reset and HWRST reset**

*Note: RESX should be tied from low to high when exiting DSTB Mode.*

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### 5.10 Abnormal Power off Function

ST7102 provides Power Drop detection. If external power drops lower than circuit detection voltage, then the system will enter into Sleep In Mode.

#### 5.10.1 Abnormal Power Off

Abnormal Power Off circuit can detect external voltage source IOVCC, AVDD, AVEE, if one of them is below detection voltage, then the system will be into Sleep In Mode. The following schematic is to show how the detection circuit produces abnormal signal.

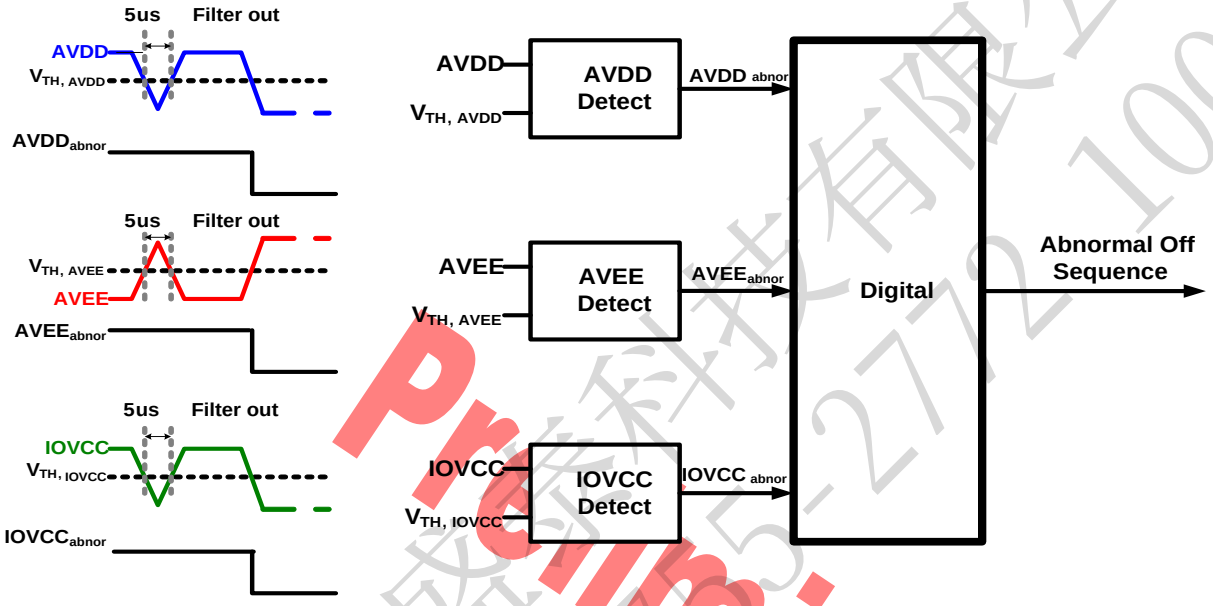



**Figure 68 Abnormal Off Function Block Diagram**

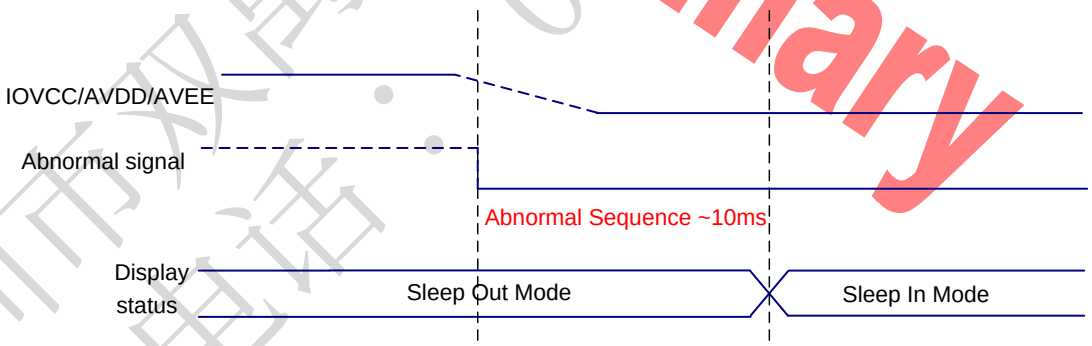



**Figure 69 Abnormal Off Sequence Diagram**

*Note1: Abnormal function is working only in Sleep Out Mode*

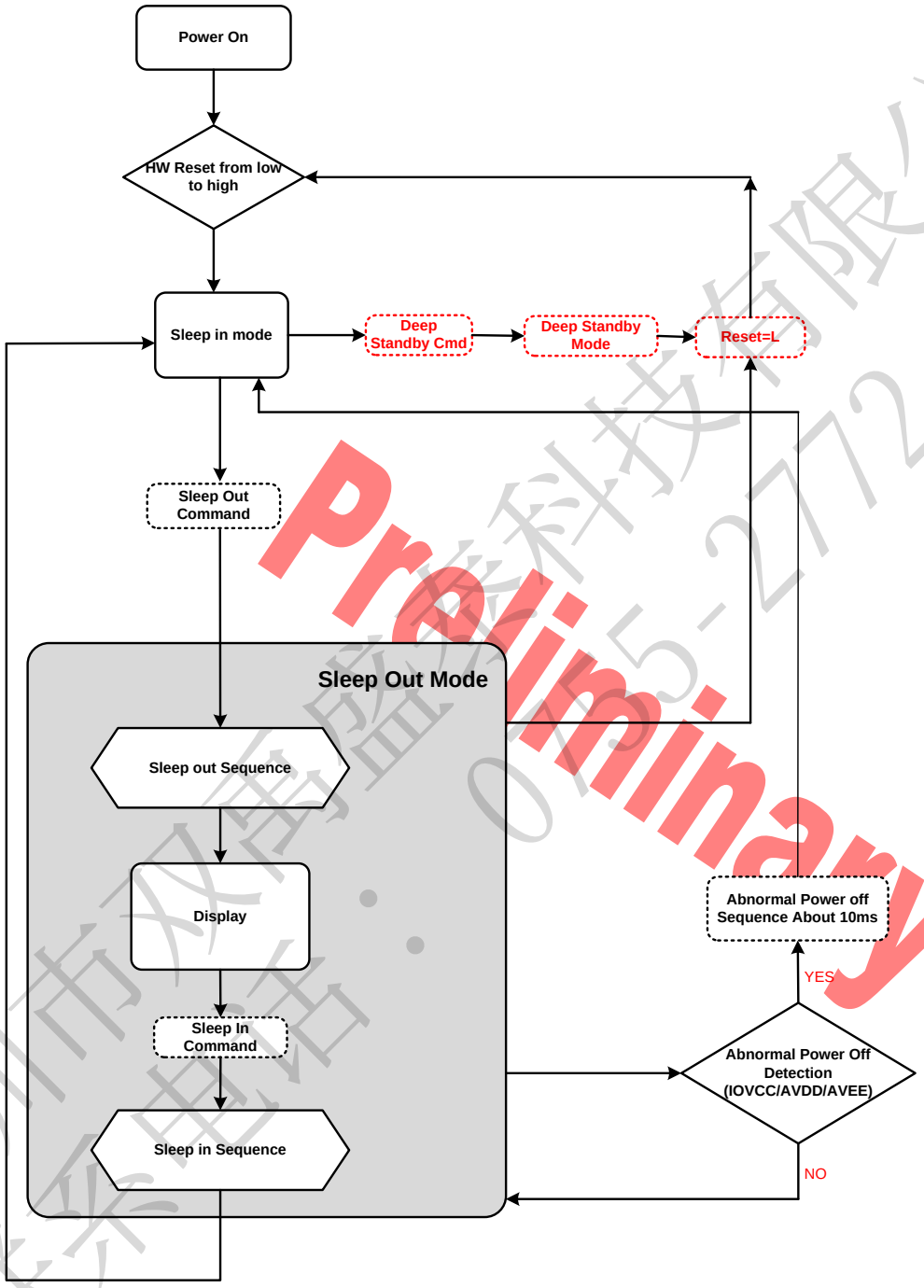
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### 5.11 Basic Operation Mode

The basic operation mode of ST7102 is illustrated below. When changing from one mode to another, make sure to follow the sequence indicated in below figure.




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**Figure 70 Basic Operation Mode**



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### 5.12 Power On/Off Sequence

The power On/Off sequence is illustrated below.

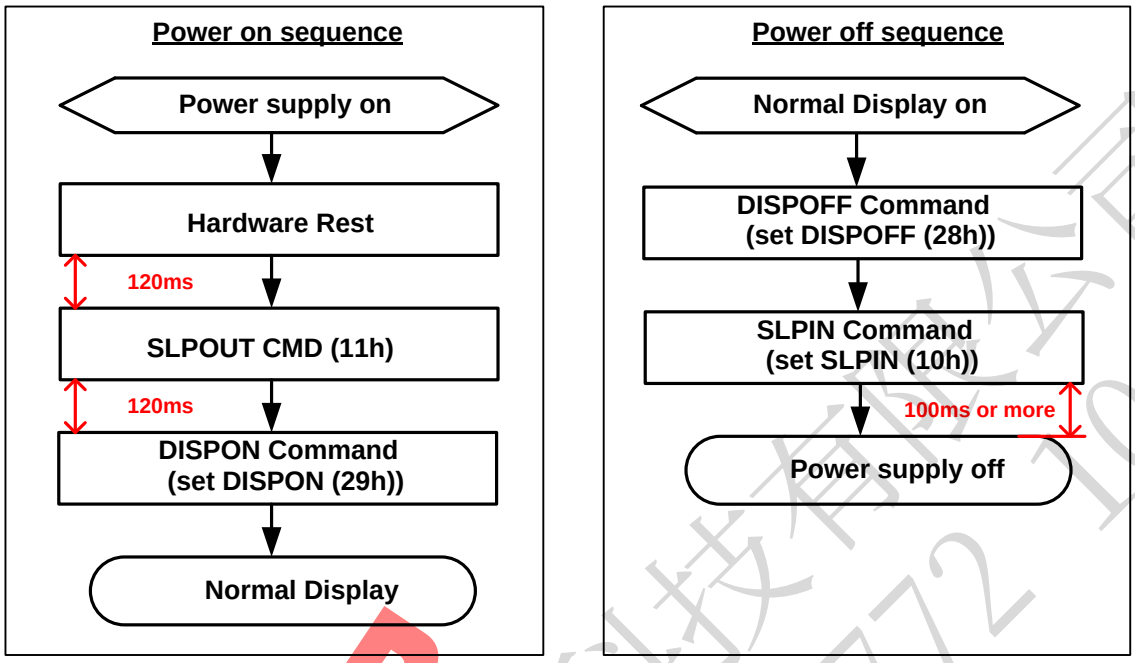



**Figure 71 The power On/Off sequence block diagram**

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### 5.12.1 Power On/Off Timing

The power On/Off timing diagram is illustrated below.

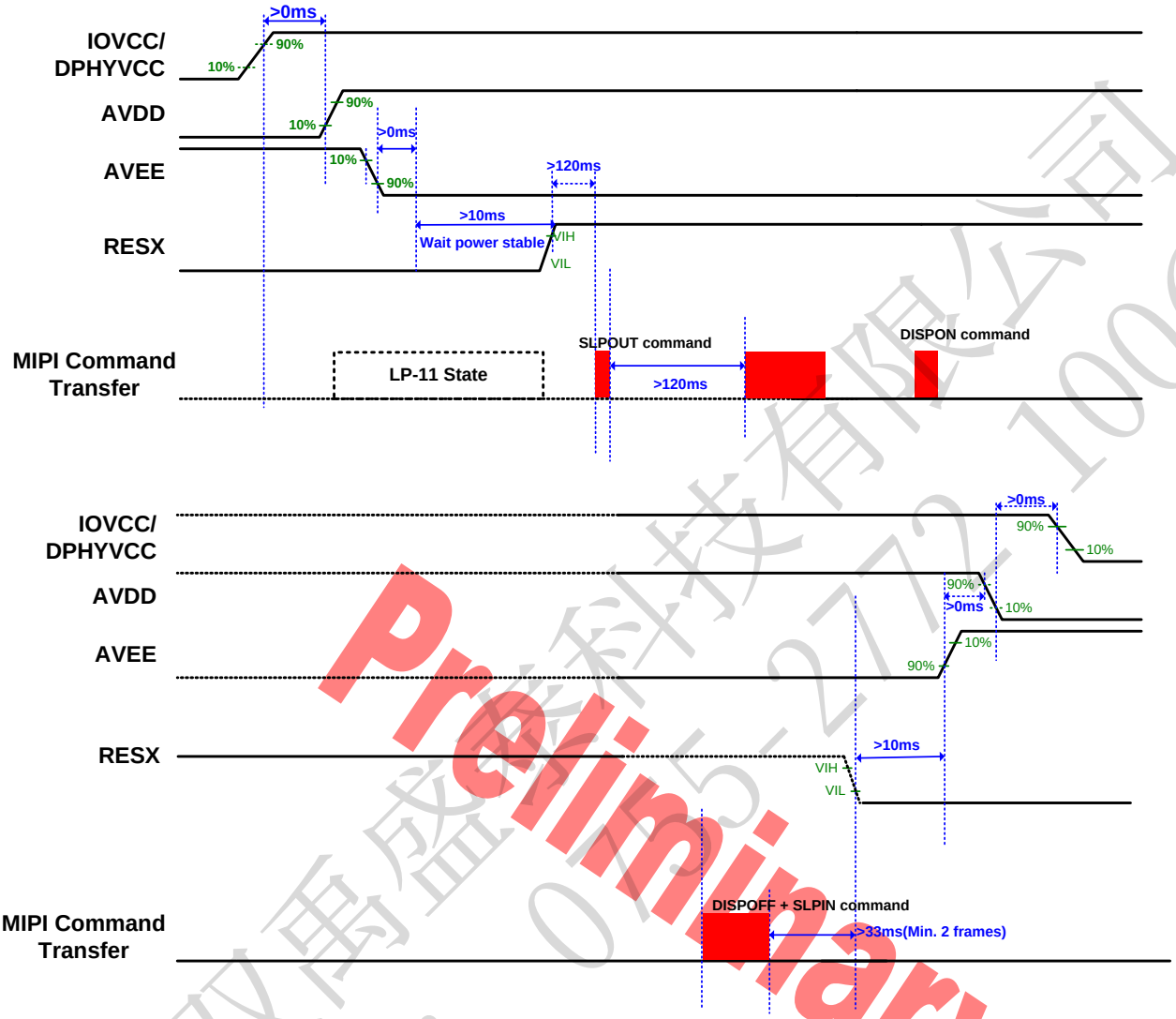



**Figure 72 The power on sequence timing**

*Note 1: MIPI lanes must go to LP11 after Power IOVCC/DPHYVCC is ready*

*Note 2: After SLPOUT command, driver IC will start internal power on action. Any other settings should be set after SLPOUT command with a minimum of 120mS.*

*Note 3: DISPOFF and SLPIN command should be set after SLPOUT command with a minimum delay time of 120mS.*

*Note 4: RESX tied low into deep standby mode with a minimum time of 10ms.*

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### 5.12.2 Power Ramp Up/Down Specifications

The power ramp up/down specifications are illustrate below

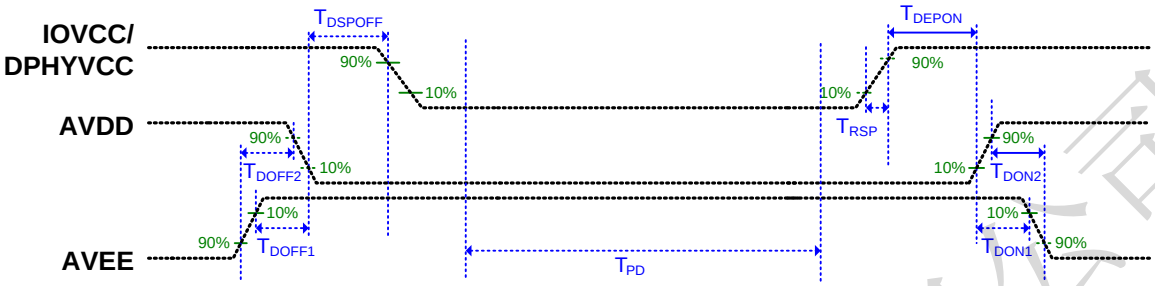



**Figure 73 The power ramp up/down timing**

| Item | Symbol | Unit | Min. | Max. |
|---|---|---|---|---|
| System power (IOVCC) rise time (10% to 90%) | $T_{RSP}$ | ms | - | 2 |
| System power (IOVCC) on to AVDD on time | $T_{DEPON}$ | ms | 0 | - |
| AVDD-AVEE on delay time (10% to 10%) | $T_{DON1}$ | ms | 0 | - |
| AVDD-AVEE on delay time (90% to 90%) | $T_{DON2}$ | ms | 0 | - |
| AVEE-AVDD off delay time (10% to 10%) | $T_{DOFF1}$ | ms | 0 | - |
| AVEE-AVDD off delay time (90% to 90%) | $T_{DOFF2}$ | ms | 0 | - |
| AVDD off to system power off time | $T_{DSPOFF}$ | ms | 0 | - |
| Power down time | $T_{PD}$ | ms | 10 | |

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### 5.13 Instruction Setting Sequence

#### 5.13.1 Sleep Enter/Exit Sequences

When setting instruction to the ST7102, the sequence shown in below figure must be followed to complete the instruction setting.

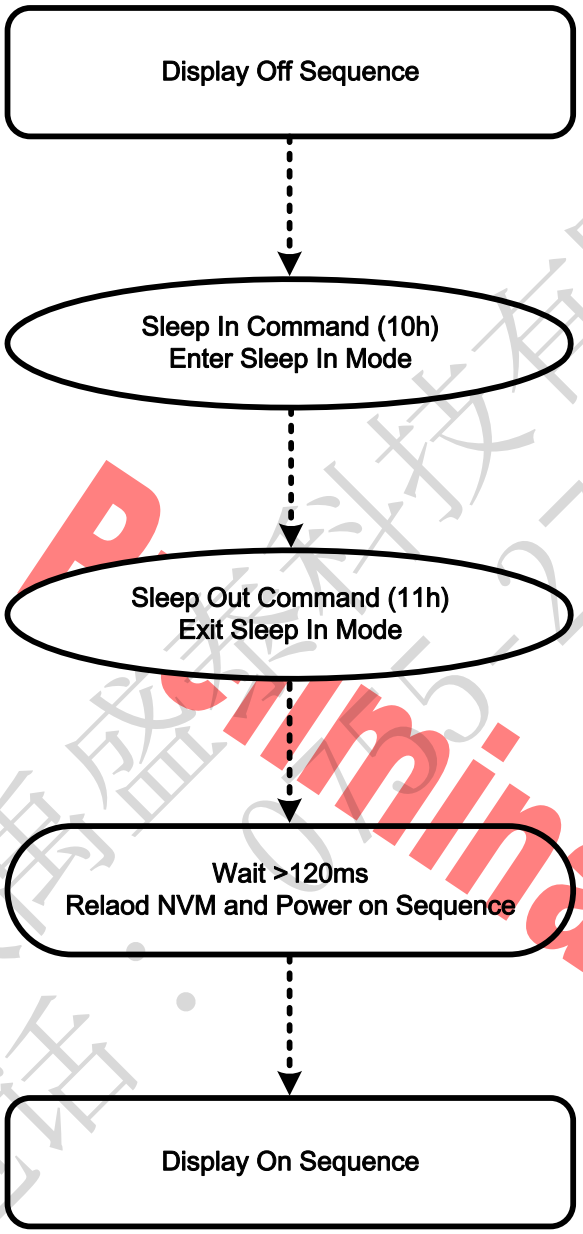



**Figure 74 Sleep Enter/Exit Setting Sequence**

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### 5.13.2 Deep Standby Mode Enter/Exit Sequences

When setting instruction to the ST7102, the sequence shown in below figure must be followed to complete the instruction setting.

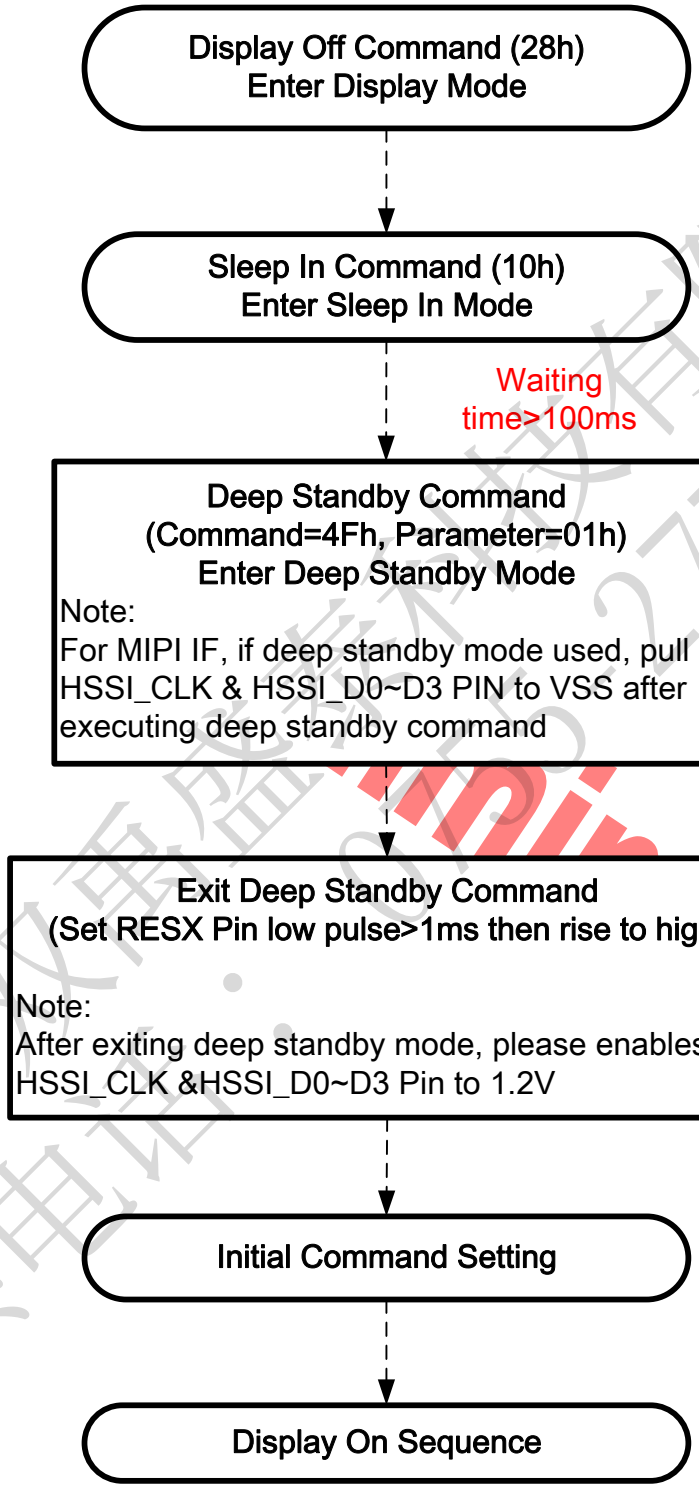



**Figure 75 Deep Standby Mode Enter/Exit Setting Sequence**



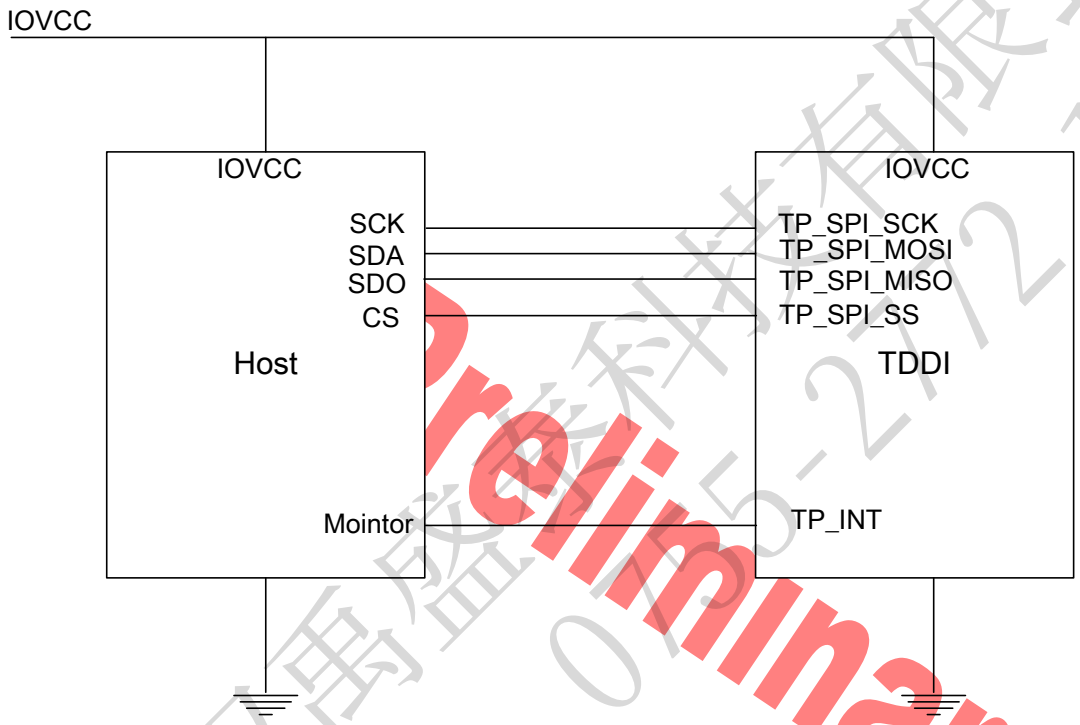
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### 5.14 Touch interface protocol

ST7102 supports SPI and I2C interface, which allows full-duplex, synchronous, serial communications with host controllers. ST7102 SPI equips four serial signals, SS, SCK, MISO and MOSI.

#### 5.14.1 SPI interface

ST7102 operates as a SPI slave device and data length is 16-bit. TP_SPI_SS, TP_SPI_SCK and TP_SPI_MOSI are Schmitt trigger inputs. The data on TP_SPI_MOSI is latched on rising edge of TP_SPI_SCK and the data on TP_SPI_MISO output on falling edge of TP_SPI_SCK. The maximum SPI clock rate is 16 MHz.




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The information transmitted by SPI can be divided into command packet and data packet. Command packet can access register and data packet can access internal RAM. The most significant bit, called ID bit, after TP_SPI_SS falling edge is used to identify command or data packet. And the following bit, called R/W bit, is used to identify write or read operation.

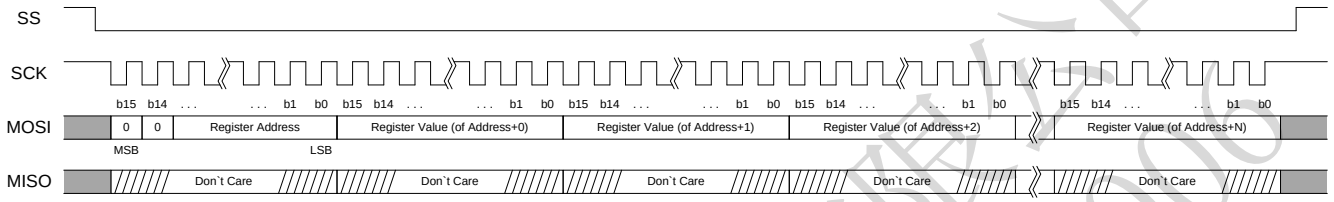
##### 5.14.1.1 Command Protocol

In command protocol, ID bit of 1st word after TP_SPI_SS falling edge is “0” to indicate register access. R/W bit is “0” to indicate write operation and “1” to indicate read operation. Bit13 to bit0 of 1st word, called ADD bits, are address of register to be accessed. If at write operation, the 2nd word on TP_SPI_MOSI will update the register which is addressed by ADD and afterward ADD will increase one automatically. If at read operation, the 2nd word on TP_SPI_MOSI is dummy

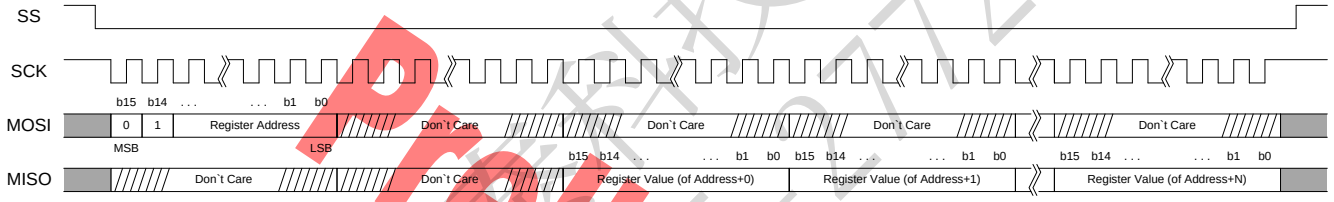


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word (value doesn't care). The 3rd word on TP_SPI_MISO will be outputted from register addressed by ADD and similarly ADD is increment afterwards. Keep writing and reading words can access the consecutive registers during one SPI transmission. If accessing non-consecutive register is happened, SS should return to "1" to end current transmission and start another SPI command protocol to assign new register address.



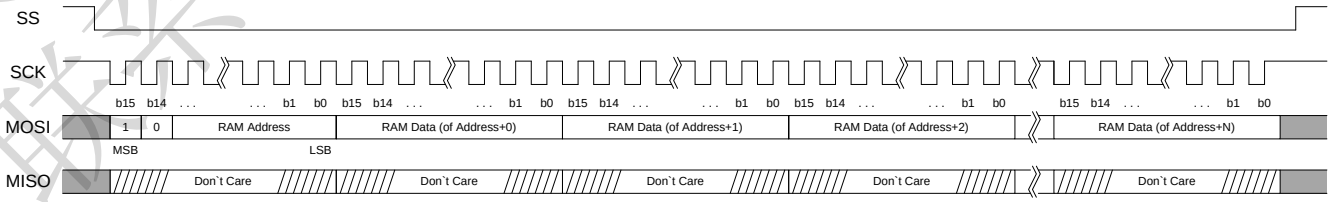

Host write register command




Host read register command

### 5.14.1.2 Data Protocol

In data protocol, ID bit of 1st word after SS falling edge is "1" to indicate RAM access. R/W bit is "0" to indicate write operation and "1" to indicate read operation. Bit13 to bit0 of 1st word, called ADD bits, are address of RAM to be accessed. If at write operation, the 2nd word on MOSI will update RAM which is addressed by ADD and afterward ADD will increase one automatically. If at read operation, the 2nd word on MISO is 16-bit header. The 3rd word on MISO will be outputted from RAM addressed by ADD and similarly ADD is increment afterwards. Keep writing and reading words can access the consecutive RAM addresses during one SPI transmission. If accessing non-consecutive RAM address is happened, SS should return to "1" to end current transmission and start another SPI data protocol to assign new RAM address.

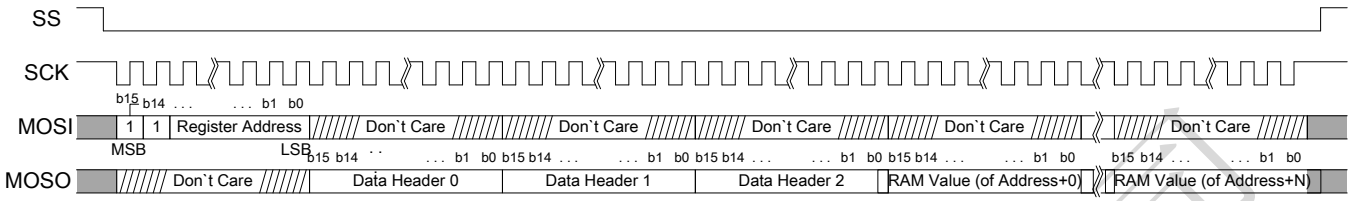



Host write RAM data

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Host read RAM data

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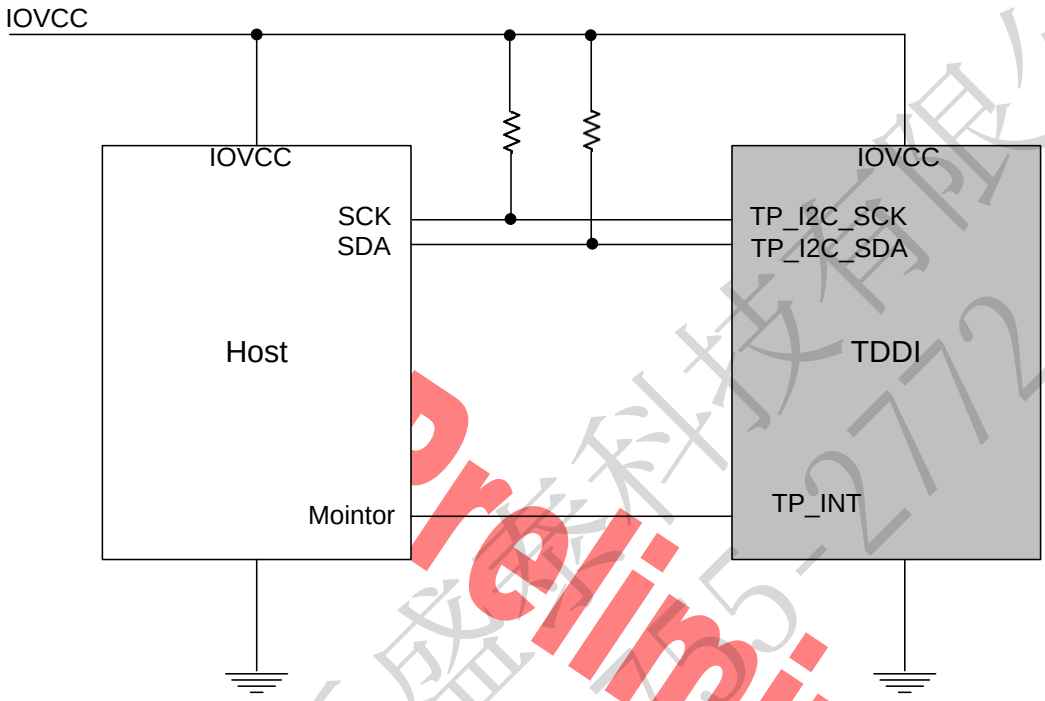


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### 5.14.2 I2C

ST7102 protocol supports operating speeds up to 400 kb/s with 7-bit addressing and 8-bit data bytes. The TP_I2C_SCK, TP_I2C_SDA, and TP_TSIX pins are typically used in an I2C interface.

The values of the pull-up resistors should be chosen to ensure that the rise times of the TP_I2C_SDA and TP_I2C_SCK signals are within the limits set by the I2C specification. These values depend on what other devices, if any, are on the I2C bus. Typical values fall within the range of 2 kΩ to 10 kΩ.




- **I2C Pin Definition**

| Name | I/O | Description | Connect Pin |
|---|---|---|---|
| IOVCC | I | Power supply for I/O system | IOVCC |
| SCK | I | I2C clock pin | TP_I2C_SCK |
| SDA | I/O | I2C data pin | TP_I2C_SDA |
| IRQ | O | Data ready interrupt pin | TP_INT |

- **I2C clock timing**

When a transaction contains the slave address and R/W bit (start condition), the sensor can hold SCL low and checks that the slave address matches. If the slave address fails to match, the sensor no longer clock stretches on subsequent byte transmission until it detects the next start condition. When the slave address matches, the sensor acknowledges and can continue to clock stretch at the end of subsequent bytes within the same transaction.

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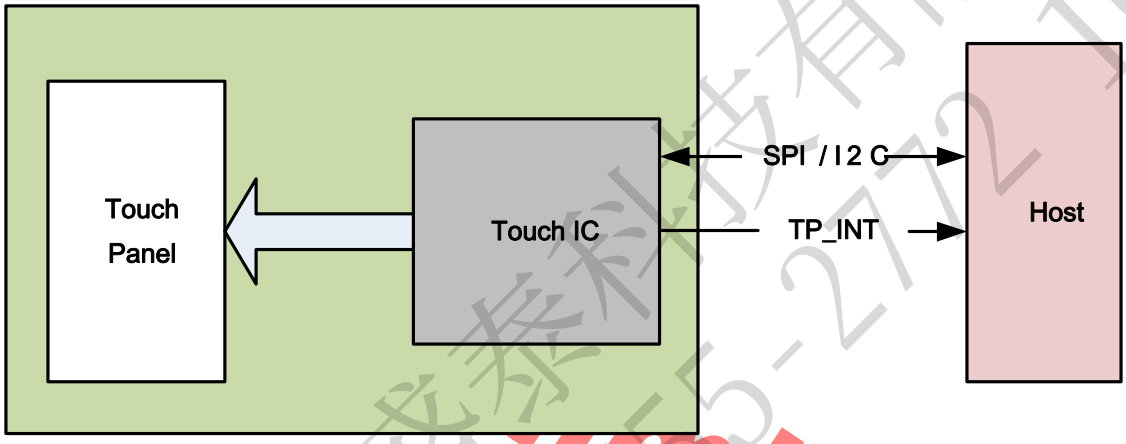


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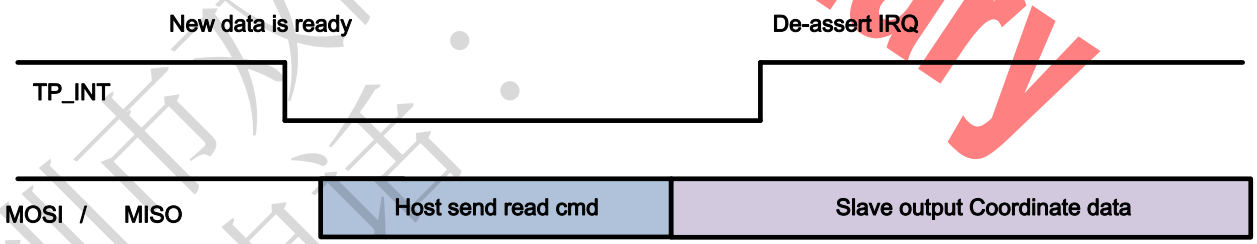



### 5.14.3 IRQ

ST7102 provides an interrupt pin(TP_TSIX) that is asserted to indicate that new data is available for reading by the host. The TP_TSIX signal is intended to be used as interrupt source to a host. IRQ functionality is added by MCU firmware. The IRQ to host connection as below.




The TP_INT pin behavior




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# 6 COMMAND DESCRIPTION

## 6.1 User Command Set (UCS) List

| Command | (Hex) | Write/Read /Command | NVM option | Function | Parameter Number | MIPI Transmission |
|---|---|---|---|---|---|---|
| NOP | 00 | C | No | No Operation | 0 | LPDT/HSDT |
| SWRESET | 01 | C | No | Software Reset | 0 | LPDT/HSDT |
| RDDID | 04 | R | No | Read Display Identification Information | 3 | LPDT/HSDT |
| RDNUMED | 05 | R | No | Read Number of Errors on DSI | 1 | LPDT/HSDT |
| RDDST | 09 | R | No | Read Display Status | 4 | LPDT/HSDT |
| RDDPM | 0A | R | No | Read Display Power Mode | 1 | LPDT/HSDT |
| RDDMADCTL | 0B | R | No | Read Display MADCTL | 1 | LPDT/HSDT |
| RDDIM | 0D | R | No | Read Display Image Mode | 1 | LPDT/HSDT |
| RDDSM | 0E | R | No | Read Display Signal Mode | 1 | LPDT/HSDT |
| RDDSDR | 0F | R | No | Read Display Self-Diagnostic Result | 1 | LPDT/HSDT |
| SLPIN | 10 | C | No | Enter Sleep In mode | 0 | LPDT/HSDT |
| SLPOUT | 11 | C | No | Enter Sleep Out mode | 0 | LPDT/HSDT |
| NORON | 13 | C | No | Enter Normal Display mode | 0 | LPDT/HSDT |
| INVOFF | 20 | C | No | Display Inversion Off | 0 | LPDT/HSDT |
| INVON | 21 | C | No | Display Inversion On | 0 | LPDT/HSDT |
| ALLPOFF | 22 | C | No | All pixel off | 0 | LPDT/HSDT |
| ALLPON | 23 | C | No | All pixel on | 0 | LPDT/HSDT |
| GAMSEL | 26 | C | No | Gamma curve select | 1 | LPDT/HSDT |
| DISPOFF | 28 | C | No | Display off | 0 | LPDT/HSDT |
| DISPON | 29 | C | No | Display on | 0 | LPDT/HSDT |
| TEOFF | 34 | C | No | Tearing Effect Line Off | 0 | LPDT/HSDT |
| TEEON | 35 | W | No | Tearing Effect Line On | 1 | LPDT/HSDT |
| MADCTL | 36 | W | Yes | Memory Access Direction Control | 1 | LPDT/HSDT |
| IDMOFF | 38 | W | No | Idle Mode OFF | 0 | LPDT/HSDT |
| IDMON | 39 | W | No | Idle Mode ON | 0 | LPDT/HSDT |
| TESLWR | 44 | W | No | Write TE Scan Line | 2 | LPDT/HSDT |
| TESLRD | 45 | R | No | Read Scan Line | 2 | LPDT/HSDT |
| DSTB | 4F | W | No | Deep standby mode on | 1 | LPDT/HSDT |
| WRDISBV | 51 | W | No | Write Display Brightness | 1 | LPDT/HSDT |
| RDDISBV | 52 | R | No | Read Display Brightness Value | 1 | LPDT/HSDT |
| WRCTRLD | 53 | W | No | Write CTRL Display | 1 | LPDT/HSDT |

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| | | | | | | |
|---|---|---|---|---|---|---|
| RDCTRLD | 54 | R | No | Read CTRL Display | 1 | LPDT/HSDT |
| WRCABC | 55 | W | No | Write Content Adaptive Brightness Control | 1 | LPDT/HSDT |
| RDCABC | 56 | R | No | Read Content Adaptive Brightness Control | 1 | LPDT/HSDT |
| WRCABCMB | 5E | W | No | Write CABC Minimum Brightness | 1 | LPDT/HSDT |
| RDCABCMB | 5F | R | No | Read CABC Minimum Brightness | 1 | LPDT/HSDT |
| RDID1 | DA | R | Yes | Read ID1 | 1 | LPDT/HSDT |
| RDID2 | DB | R | Yes | Read ID2 | 1 | LPDT/HSDT |
| RDID3 | DC | R | Yes | Read ID3 | 1 | LPDT/HSDT |
| RDDDBS | A1 | R | No | Read DDB Start | 8 | LPDT/HSDT |
| RDDDBC | A8 | R | Yes | Read DDB Continue | 8 | LPDT/HSDT |
| RDFCS | AA | R | No | Read First Checksum | 1 | LPDT/HSDT |
| RDCCS | AF | R | No | Read Continue Checksum | 1 | LPDT/HSDT |
| RDICCD | F4 | R | No | Read Sitronix IC ID | 2 | LPDT/HSDT |
| MIPIEXTFMAT | F9 | W/R | YES | MIPI Extension Format | 1 | LPDT/HSDT |

*Note: LPDT (Low Power Mode), HSDT (High Speed Mode)*

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## 6.2 User Command Set (UCS) DESCRIPTION

### 6.2.1 NOP (00H) : No Operation

| 00H | NOP (No Operation) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| NOP | Write | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | (00H) |
| Parameter | No Parameter | | | | | | | | | - |
| Description | - This command is empty command. It does not have effect on the display module. | | | | | | | | | |
| Restriction | - | | | | | | | | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.2 SWRESET (01H): Software Reset

| 01H | SWRESET (Software Reset) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| SWRESET | Write | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | (01H) |
| Parameter | No Parameter | | | | | | | | | - |
| Description | - When the Software Reset command is written, it causes a software reset. It resets the commands and parameters to their S/W Reset default values and all source & gate outputs are set to VSS (display off). | | | | | | | | | |
| Restriction | - | | | | | | | | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.3 RDDID (04H) Read Display Identification Information

| 04H | RDDID (Read Display Identification Information) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDID | Write | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | (04H) |
| 1st Parameter | Read | ID1[7:0] | | | | | | | | FFh |
| 2nd Parameter | Read | ID2[7:0] | | | | | | | | FFh |
| 3rd Parameter | Read | ID3[7:0] | | | | | | | | FFh |
| Description | - Read Display Identification Information<br>-ID1 : LCD module's manufacturer ID (FFh: not programmed)<br>-ID2 : LCD module/driver version ID (FFh: not programmed)<br>-ID3 : LCD module/driver ID (FFh: not programmed) | | | | | | | | | |
| Restriction | - | | | | | | | | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.4 RDNUMED (05H) Read Number of Errors on DSI

| 05H | RDNUMED | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDNUMED | Write | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | (05H) |
| 1st Parameter | Read | DSI_NUMBER[7:0] | | | | | | | | 00h |
| Description | - Read Number of Errors on DSI, DSI_NUMBER[7:0] is a number of the errors on DSI. | | | | | | | | | |
| Restriction | - | | | | | | | | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.5 RDDST (09H) Read Display Status

| 09H | RDNUMED(Read Display Status) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | (Default) |
| RDDST | Write | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | (09H) |
| 1st Parameter | Read | Booster_on | 0 | 0 | 0 | 0 | BGR | 0 | CA | 00h |
| 2nd Parameter | Read | LA | 0 | 0 | 0 | IDMON | 0 | SLPOUT | NORON | 00h |
| 3rd Parameter | Read | 0 | 0 | INVON | ALLPON | ALLPOFF | DSPON | TEON | GAMM_SEL[2] | 00h |
| 4th Parameter | Read | GAMMA_SEL[1] | GAMMA_SEL[0] | TEMOD | 0 | 0 | 0 | 0 | EOD | 00h |

Description

- This command indicates the current status of the display as described in the table below :

| Bit | Description | Value |
|---|---|---|
| Booster_on | Booster status | '1' = Booster on.<br>'0' = Booster off |
| CA | Column Address Order (CA) | '1' = Decrement, (Right to Left, when MADCTL (36h) CA='1')<br>'0' = Increment, (Left to Right, when MADCTL (36h) CA='0') |
| LA | Row Address Order (LA) | '1' = Row/column exchange, (when MADCTL (36h) LA='1')<br>'0' = Normal, (when MADCTL (36h) LA='0' |
| BGR | RGB/ BGR Order | '1' = BGR Order, (When MADCTL (36h) BGR='1')<br>'0' = RGB Order, (When MADCTL (36h) BGR='0') |
| IDMON | Idle Mode On/Off | '1' = Idle Mode On,<br>'0' = Idle Mode Off |
| SLPOUT | Sleep In/Out Mode | '1' = Sleep Out Mode<br>'0' = Sleep In Mode |
| NORON | Normal mode | '1' = Normal Mode on<br>'0' = Normal Mode off |
| DSPON | Display On/Off Mode | '1' = Display On Mode<br>'0' = Display Off Mode |
| INVON | Inversion On/Off Mode | '1' = Inversion On Mode<br>'0' = Inversion Off Mode |
| ALLPON | All pixel on | '1' = All pixel on<br>'0' = Normal mode |
| ALLPOFF | All pixel off | '1' = All pixel off<br>'0' = Normal mode |
| TEON | Tearing effect line on/off | '1' = Inversion On Mode<br>'0' = Inversion Off Mode |
| GAMMA_SEL[2:0] | Gamma curve selection | '000' = Gamma curve 1(gamma 2.2)<br>'others' = reserved |
| TEMOD | Tearing effect line mode | '1' = V- blanking and H-blanking |

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| | | | |
|---|---|---|---|
| | | | ‘0’ = V-blanking only |
| | EOD | Error on DSI | ‘1’ = Error<br>‘0’ = No Error |
| Restriction | - | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.6 RDDPM (0AH): Read Display Power Mode

| 0AH | RDDPM (Read Display Power Mode) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDPM | Write | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | (0AH) |
| 1st Parameter | Read | Booster_on | IDMON | 0 | SLPOUT | NORON | DSPON | 0 | 0 | 00h |

Description

- This command indicates the current status of the display as described in the table below:

| Bit | Description | Value |
|---|---|---|
| Booster_on | Booster status | '1' = Booster on.<br>'0' = Booster off |
| IDMON | Idle Mode On/Off | '1' = Idle Mode On,<br>'0' = Idle Mode Off |
| SLPOUT | Sleep In/Out Mode | '1' = Sleep Out Mode<br>'0' = Sleep In Mode |
| NORON | Normal mode | '1' = Normal Mode on<br>'0' = Normal Mode off |
| DSPON | Display On/Off Mode | '1' = Display On Mode<br>'0' = Display Off Mode |

Restriction

-

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.7 RDDMADCTR (0BH): Read Display MADCTR

| 0BH | RDDMADCTR (Read Display MADCTR) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDMADCTR | Write | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | (0BH) |
| 1st Parameter | Read | 0 | 0 | 0 | 0 | BGR | 0 | CA | LA | 00h |

**Description**

- This command indicates the current status of the display as described in the table below:

| Bit | Description | Value |
|---|---|---|
| CA | Column Address Order (CA) | '1' = Decrement, (Right to Left, when MADCTL (36h) CA='1')<br>'0' = Increment, (Left to Right, when MADCTL (36h) CA='0') |
| LA | Row Address Order (LA) | '1' = Row/column exchange, (when MADCTL (36h) LA='1')<br>'0' = Normal, (when MADCTL (36h) LA='0' |
| BGR | RGB/ BGR Order | '1' = BGR Order, (When MADCTL (36h) BGR='1')<br>'0' = RGB Order, (When MADCTL (36h) BGR='0') |

**Restriction**

-

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.8 RDDCOLM (0CH): Read Display Color Mode

| 0DH | RDDIM (Read Display Image Mode) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDIM | Write | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | (0CH) |
| 1st Parameter | Read | 0 | 0 | 0 | 0 | 0 | DBI[2:0] | | | 00h |

**Description**

- This command indicates the current status of the display as described in the table.

| DBI[2:0] | Color Format |
|---|---|
| 0h~4h | Reserved |
| 5h | 16-bit/pixel |
| 6h | 18-bit/pixel |
| 7h | 24-bit/pixel |

Note: For DBI[2:0] definition refer to interface Pixel Format (3Ah)

**Restriction**

-

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.9 RDDIM (0DH): Read Display Image Mode

| 0DH | RDDIM (Read Display Image Mode) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDIM | Write | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | (0DH) |
| 1st Parameter | Read | 0 | 0 | INVON | ALLPON | ALLPOFF | GAMMA_SEL[2:0] | | | 00h |

**Description**

- This command indicates the current status of the display as described in the table below:

| Bit | Description | Value |
|---|---|---|
| INVON | Inversion On/Off Mode | '1' = Inversion On Mode<br>'0' = Inversion Off Mode |
| ALLPON | All pixel on | '1' = All pixel on<br>'0' = Normal display |
| ALLPOFF | All pixel off | '1' = All pixel off<br>'0' = Normal display |
| GAMMA_SEL[2:0] | Gamma curve selection | '000' = Gamma curve 1(gamma 2.2)<br>'others' = Reserved |

**Restriction**

-

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.10 RDDSM (0EH): Read Display Signal Mode

| 0EH | RDDSM (Read Display Signal Mode) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDSM | Write | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | (0EH) |
| 1st Parameter | Read | TEON | TEM | 0 | 0 | 0 | 0 | 0 | EOD | 00h |

**Description**

- This command indicates the current status of the display as described in the table below:

| Bit | Description | Value |
|---|---|---|
| TEON | Tearing effect line on/off | '1' = Inversion On Mode<br>'0' = Inversion Off Mode |
| TEM | Tearing effect line mode | '1' = V- blanking and H-blanking<br>'0' = V-blanking only |
| EOD | Error on DSI | '1' = Error<br>'0' = No Error |

**Restriction**

-

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.11 RDDSDR(0FH): Read Display Self-Diagnostic Result

| 0FH | RDDSM (Read Display Signal Mode) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDSDR | Write | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | (0FH) |
| 1st Parameter | Read | RLD | FD | 0 | 0 | 0 | 0 | 0 | CCR | 00h |

| | |
|---|---|
| Description | - This command indicates the current status of the display as described in the below:<br>-RLD: This bit is the Register Loading Detection<br>-FD: This bit is the Functionality Detection<br>-CCR: The checksum compare result.<br>0 = Checksum is the same.<br>1 = Checksum is not the same |
| Restriction | - |
| Register Availability | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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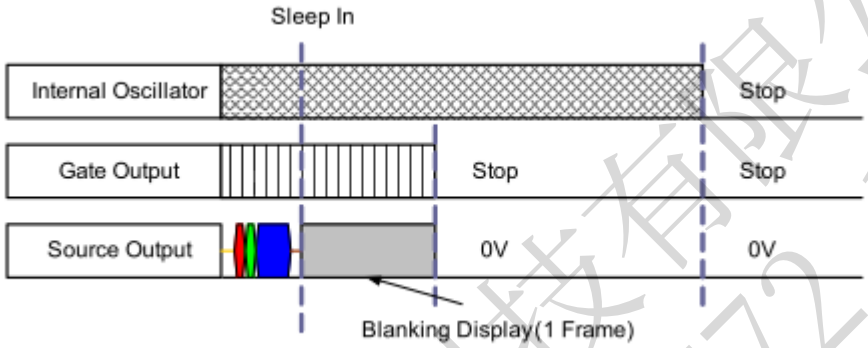
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### 6.2.12 SLPIN (10H): Sleep In

| 10H | SLPIN (Sleep In) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| SLPIN | Write | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | (10H) |
| Parameter | No Parameter | | | | | | | | | - |

**Description**

- This command causes the LCD module to enter the minimum power consumption mode.
- In this mode the DC/DC converter is stopped, Internal display oscillator is stopped, and panel scanning is stopped.




**Restriction**

- This command has no effect when module is already in sleep in mode. Sleep In Mode can only be exit by the Sleep Out Command (11H).
- It will be necessary to wait 5ms before sending next command; this is to allow time for the supply voltages and clock circuits to stabilize.
- It will be necessary to wait 120ms after sending Sleep Out command (when in Sleep In Mode) before Sleep In command can be sent.

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | No |

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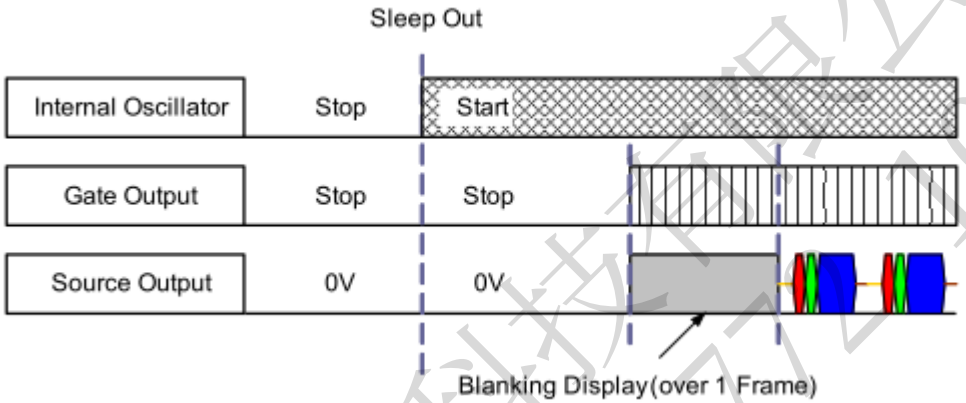
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### 6.2.13 SLPOUT (11H): Sleep Out

| 11H | SLPOUT (Sleep Out) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| SLPOUT | Write | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | (11H) |
| Parameter | No Parameter | | | | | | | | | - |

**Description**

- This command turns off sleep mode.
- In this mode the DC/DC converter is enabled, Internal display oscillator is started, and panel scanning is started.




**Restriction**

- This command has no effect when module is already in sleep out mode. Sleep Out Mode can only be exit by the Sleep In Command (10H).
- It will be necessary to wait 5ms before sending next command; this is to allow time for the supply voltages and clock circuits to stabilize.
- DRIVER loads all default values of extended and test command to the registers during this 5msec and there cannot be any abnormal visual effect on the display image if those default and register values are same when this load is done and when the DRIVER is already Sleep Out mode.
- It will be necessary to wait 120ms after sending Sleep In command (when in Sleep Out mode) before Sleep Out command can be sent

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | No |
| Sleep In | Yes |

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### 6.2.14 NORON (13H): Normal display mode On

| 13H | NORON (Normal Display Mode On ) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| INVOFF | Write | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | (13H) |
| Parameter | No Parameter | | | | | | | | | - |

| | |
|---|---|
| Description | - This command is used to return display to normal display mode.<br>-.This command can exit all pixel on/off mode to normal display mode. |
| Restriction | |
| Register Availability | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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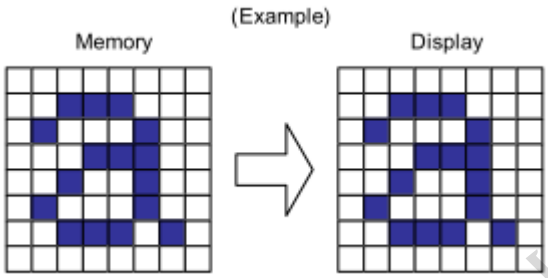
### 6.2.15 INVOFF (20H) : Display Inversion Off

| 20H | INVOFF (Display Inversion Off ) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| INVOFF | Write | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | (20H) |
| Parameter | No Parameter | | | | | | | | | - |

**Description**

- This command is used to recover from display inversion mode.
- This command makes no change of contents of frame memory.
- This command does not change any other status.

(Example)




**Restriction**

- This command has no effect when module is already in inversion off mode.

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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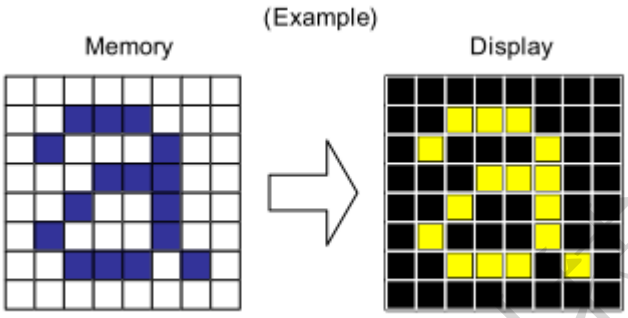
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### 6.2.16 INVON (21H) : Display Inversion On

| 21H | INVON (Display Inversion On ) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| INVON | Write | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | (21H) |
| Parameter | No Parameter | | | | | | | | | - |

**Description**

- This command is used to enter into display inversion mode.
- This command makes no change of contents of frame memory. Every bit is inverted from the frame memory to the display.
- This command does not change any other status.




**Restriction**

- This command has no effect when module is already in inversion off mode.

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.17 ALLPOFF (22H): All pixel off

| 22H | ALLPOFF (All pixel off) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| ALLPOFF | Write | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | (22H) |
| Parameter | No Parameter | | | | | | | | | - |
| Description | - This command is used to black display In sleep out mode.<br>- Exit from this command by All pixel on(23H) or Normal display mode(13H) | | | | | | | | | |
| Restriction | | | | | | | | | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | No |

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### 6.2.18 ALLPON (23H): All pixel on

| 23H | ALLPOFF (All pixel off) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| ALLPON | Write | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | (23H) |
| Parameter | No Parameter | | | | | | | | | - |
| Description | - This command is used to white display In sleep out mode.<br>- Exit from this command by All pixel off(22H) or Normal display mode(13H) | | | | | | | | | |
| Restriction | | | | | | | | | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | No |

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### 6.2.19 GAMSEL (26H): Gamma Curve Select

| 26H | (Gamma Curve Select) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| GAMSEL | Write | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | (26H) |
| Parameter | Write | 0 | 0 | 0 | 0 | GC[3:0] | | | | |

**Description**

- This command is used to select gamma curve, and only 1 curve (gamma 2.2) can be selected.

| GC[3:0] | Gamma Curve |
|---|---|
| 1 | Gamma 2.2 |
| others | Reserved |

**Restriction**

**Register Availability**

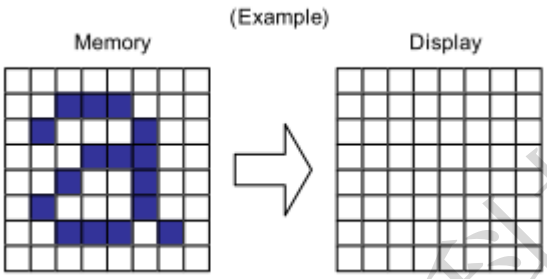
| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | No |

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### 6.2.20 DISPOFF (28H): Display Off

| 28H | DISPOFF (Display Off) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| DISPOFF | Write | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | (28H) |
| Parameter | No Parameter | | | | | | | | | - |
| Description | - This command is used to enter into DISPLAY OFF mode. In this mode.<br>- This command makes no change of contents of frame memory.<br>- This command does not change any other status.<br>- There will be no abnormal visible effect on the display.<br>- Exit from this command by Display On (29H)<br>  | | | | | | | | | |
| Restriction | -This command has no effect when module is already in Display Off mode. | | | | | | | | | |
| Register Availability | Status: Sleep Out, Display On — Availability: Yes<br>Status: Sleep In — Availability: No | | | | | | | | | |

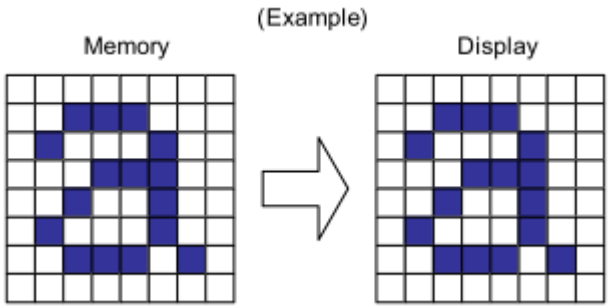
### 6.2.21 DISPON (29H): Display On

| 29H | DISPON (Display On) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| DISPON | Write | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | (29H) |
| Parameter | No Parameter | | | | | | | | | - |
| Description | - This command is used to recover from DISPLAY OFF mode.<br>- This command makes no change of contents of frame memory.<br>- This command does not change any other status. | | | | | | | | | |

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<table>
<tr><td></td><td> 
</td></tr>
<tr><td>Restriction</td><td>-This command has no effect when module is already in Display Off mode.</td></tr>
<tr><td>Register Availability</td><td>
<table>
<tr><th>Status</th><th>Availability</th></tr>
<tr><td>Sleep Out, Display On</td><td>No</td></tr>
<tr><td>Sleep In</td><td>Yes</td></tr>
</table>
</td></tr>
</table>

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### 6.2.22 TEOFF (34H): Tearing Effect Line OFF

| 34H | TEOFF (Tearing Effect Line OFF) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| TEOFF | Write | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | (34H) |
| Parameter | No Parameter | | | | | | | | | - |
| Description | - This command is used to turn OFF (Active Low) the Tearing Effect output signal from the TE signal line. | | | | | | | | | |
| Restriction | - This command has no effect when Tearing Effect output is already OFF. | | | | | | | | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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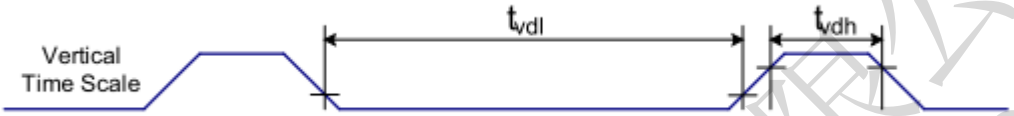
### 6.2.23 TEON (35H): Tearing Effect Line ON

| 35H | TEON (Tearing Effect Line ON) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| TEON | Write | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | (35H) |
| 1st Parameter | Write | 0 | 0 | 0 | 0 | 0 | 0 | 0 | TEM | 00h |

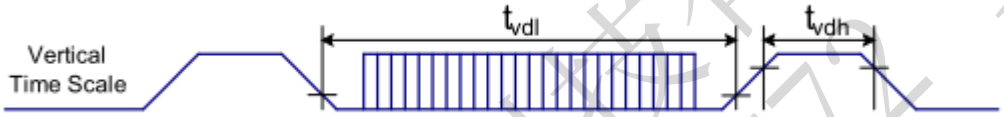
**Description**

-Tearing Effect Mode ON

- Mode 1: When TEM = '0', The tearing effect output line consists of V-blanking information only.




- Mode 2: When TEM = '1': The tearing effect output line consists of both V-blanking and H-blanking information.




*Note: During Sleep In Mode with Tearing Effect Line On, Tearing Effect Output pin will be active Low.*

**Restriction**

- This command has no effect when Tearing Effect output is already ON.

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.24 MADCTR (36H): Memory Data Access Control

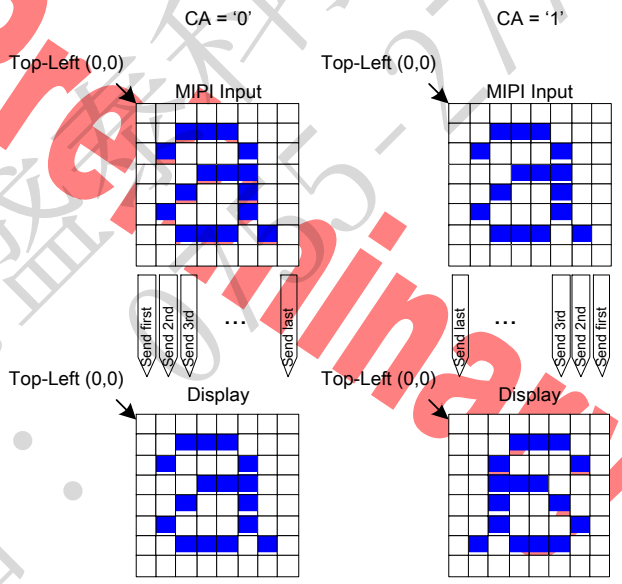
| 36H | MADCTR (Memory Data Access Control) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| MADCTR | Write | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | (36H) |
| 1st Parameter | Read | 0 | 0 | 0 | 0 | BGR | 0 | CA | LA | 00h |

Description

- Bit Assignment

| Bit | Description | Value |
|---|---|---|
| CA | Column Address Order (CA) | '1' = Decrement, (Right to Left, when MADCTL (36h) CA='1')<br>'0' = Increment, (Left to Right, when MADCTL (36h) CA='0') |
| LA | Row Address Order (LA) | '1' = Row/column exchange, (when MADCTL (36h) LA='1')<br>'0' = Normal, (when MADCTL (36h) LA='0' |
| BGR | RGB/ BGR Order | '1' = BGR Order, (When MADCTL (36h) BGR='1')<br>'0' = RGB Order, (When MADCTL (36h) BGR='0') |

-CA :

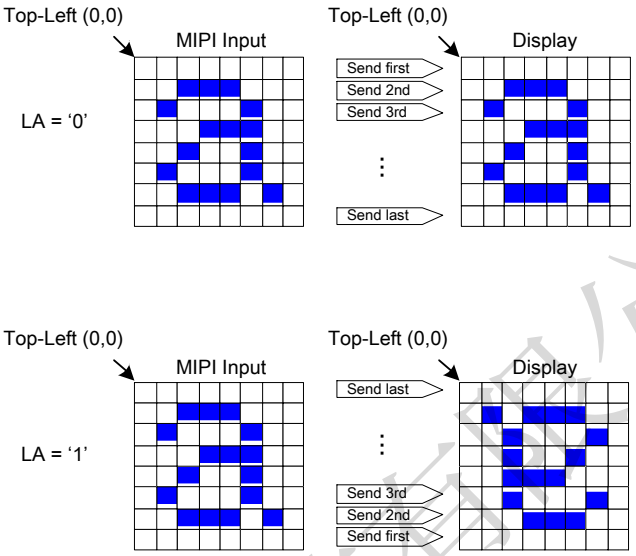



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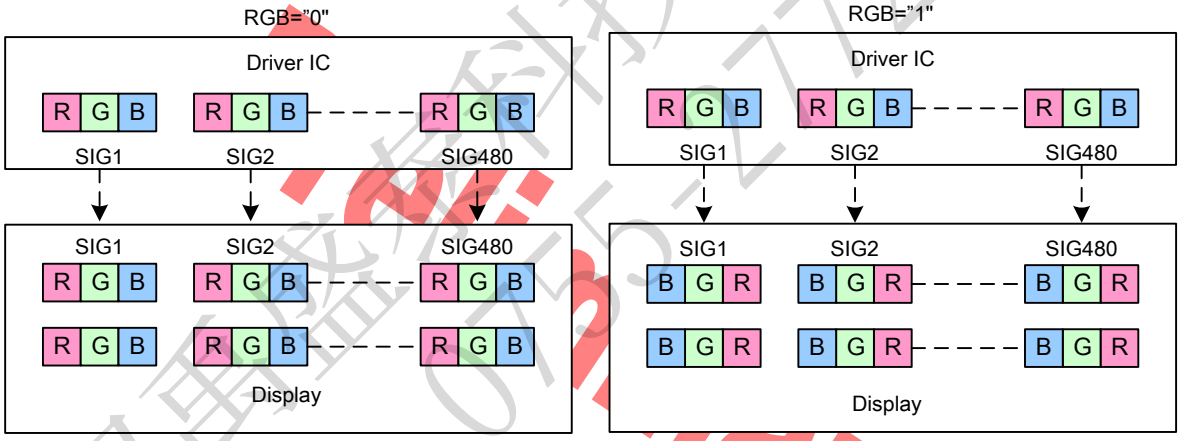


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LA :




BGR :




Restriction -

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.25 IDMOFF (38H): Idle Mode Off

| 38H | IDMOFF (Idle Mode Off) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| IDMOFF | Write | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | (38H) |
| Parameter | No Parameter | No Parameter | | | | | | | | - |
| Description | -This command is used to recover from Idle mode on. | | | | | | | | | |
| Restriction | -This command has no effect when module is already in idle off mode | | | | | | | | | |
| Register Availability | | | | | | | | | | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.26 IDMON (39H): Idle Mode On

| 38H | IDMOFF (Idle Mode Off) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| IDMON | Write | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | (39H) |
| Parameter | No Parameter | No Parameter | | | | | | | | - |

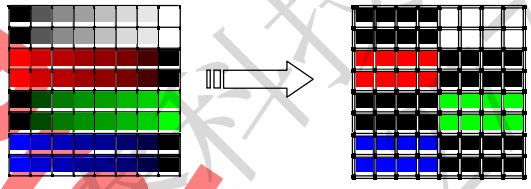
**Description**

-This command is used to enter into Idle mode on.

-There will be no abnormal visible effect on the display mode change transition.

-In the idle on mode,

1. Color expression is reduced. The primary and the secondary colors using MSB of each R,G and B, 8 color depth data is displayed.
2. 8-Color mode frame frequency is applied.
3. Exit from IDMON by Idle Mode Off (38h) command



| Color | R[7:0] | G[7:0] | B[7:0] |
|---|---|---|---|
| Blank | 0xxxxxxx | 0xxxxxxx | 0xxxxxxx |
| Blue | 0xxxxxxx | 0xxxxxxx | 1xxxxxxx |
| Red | 1xxxxxxx | 0xxxxxxx | 0xxxxxxx |
| Magenta | 1xxxxxxx | 0xxxxxxx | 1xxxxxxx |
| Green | 0xxxxxxx | 1xxxxxxx | 0xxxxxxx |
| Cyan | 0xxxxxxx | 1xxxxxxx | 1xxxxxxx |
| Yellow | 1xxxxxxx | 1xxxxxxx | 0xxxxxxx |
| White | 1xxxxxxx | 1xxxxxxx | 1xxxxxxx |

**Restriction**

-This command has no effect when module is already in idle on mode

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.27 TESLWR (44H): Write TE Scan Line

| 44H | WRTESCN (Write TE Scan Line) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| TESLWR | Write | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | (44H) |
| 1st Parameter | Write | - | - | - | - | TESN[11:8] | | | | 00h |
| 2nd Parameter | Write | TESN[7:0] | | | | | | | | 00h |
| Description | - This command turns on the display module's TE signal when the display module reaches line TESN. | | | | | | | | | |
| Restriction | - The command takes affect with the end of one frame. | | | | | | | | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.28 RDSCNL (45H): Read Scan Line

| 45H | RDSCNL (Read Scan Line) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDSCNL | Write | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | (45H) |
| 1st Parameter | Read | - | - | - | - | TESN[11:8] | | | | 00h |
| 2nd Parameter | Read | TESN[7:0] | | | | | | | | 00h |

| | |
|---|---|
| Description | - This read byte returns the current scan line. |
| Restriction | - The command takes affect with the end of one frame. |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.29 DSTB (4FH): Deep Standby Mode ON

| 4FH | DSTB (Deep Standby Mode ON) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| DSTB | Write | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | (4FH) |
| Parameter | Write | 0 | 0 | 0 | 0 | 0 | 0 | 0 | DSTB | 00H |

| | |
|---|---|
| Description | - When DSTB = 1 :Deep Standby Mode ON<br>Exit deep standby mode by pulling low "RESX" pin at least 1ms |
| Restriction | - Only effect at Sleep In mode. |
| Register Availability | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | No |
| Sleep In | Yes |

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### 6.2.30 WRDISBV (51H) Write Display Brightness

| 51H | WRDISBV (Write Display Brightness) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDSCNL | Write | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | (51h) |
| 1st Parameter | Write | DBV7 | DBV6 | DBV5 | DBV4 | DBV3 | DBV2 | DBV1 | DBV0 | 00h |

| | |
|---|---|
| Description | - This command is used to adjust the brightness value of the display.<br>- It should be checked what is the relationship between this written value and output brightness of the display. This relationship is defined on the display module specification.<br>- In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness. |
| Restriction | - |
| Register Availability | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.31 RDDISBV (52H) Read Display Brightness Value

| 52H | RDDISBV (Read Display Brightness Value) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDISBV | Write | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | (52h) |
| 1st Parameter | Read | DBV7 | DBV6 | DBV5 | DBV4 | DBV3 | DBV2 | DBV1 | DBV0 | 00h |

**Description**

- This command returns the brightness value of the display.
- It should be checked what the relationship between this returned value and output brightness of the display. This relationship is defined on the display module specification.
- In principle the relationship is that 00h value means the lowest brightness and FFh value means the highest brightness. - See command “Write Display Brightness (51H)”.
- This command can be used to read the brightness value of the display also when Display brightness control is in automatic mode.
- DBV [7:0] is reset when display is in sleep-in mode.
- DBV [7:0] is ‘0’ when bit BCTRL of “Write CTRL Display (53H)” command is ‘0’.
- DBV[7:0] is manual set brightness specified with “Write CTRL Display (53H)” command when bit BCTRL is ‘1’

**Restriction**

-

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.32 WRCTRLD (53H) Write CTRL Display

| 53H | WRCTRLD (Write CTRL Display) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| WRCTRLD | Write | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | (53h) |
| 1st Parameter | Write | 0 | 0 | BCTRL | 0 | DD | BL | 0 | 0 | 00h |

**Description**

- This command is used to control ambient light, brightness and gamma settings.

-BCTRL：Brightness Control Block On/Off. This bit is always used to switch brightness for display and keyboard.

'0' = Off (Brightness registers are 00h)

'1' = On (Brightness registers are active, according to the other parameters.)

-DD：Display Dimming

'0' = Display Dimming is off

'1' = Display Dimming is on

-BL：Backlight On/Off

'0' = Off (Completely turn off backlight circuit. Control lines must be low. )

'1' = On

-Dimming function is adapted to the brightness registers for display and keyboard when bit BCTRL is changed at dimming-on (DD=1).

-When BL bit changed from 'on' to 'off', backlight is turned off without gradual dimming, even if dimming-on (DD=1) are selected.

**Restriction**

-

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.33 RDCTRLD (54H) Read CTRL Display

| 54H | RDCTRLD (Read CTRL Display) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDCTRLD | Write | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | (54h) |
| 1st Parameter | Read | 0 | 0 | BCTRL | 0 | DD | BL | 0 | 0 | 00h |

| | |
|---|---|
| Description | -This command returns ambient light and brightness control values.<br>-BCTRL: Brightness Control Block On/Off, This bit is always used to switch brightness for display.<br>'0' = Off<br>'1' = On<br><br>-DD: Display Dimming On/Off (Only for manual brightness setting)<br>'0' = Off<br>'1' = On<br><br>-BL: Backlight Control On/Off<br>'0' = Off<br>'1' = On |
| Restriction | - |
| Register Availability | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.34 WRCABC (55H) Write Content Adaptive Brightness Control

| 55H | WRCABC (Write Content Adaptive Brightness Control) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| WRCABD | Write | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | (55h) |
| 1st Parameter | Write | CEON | SREON | EN_LEVEL[1:0] | | 0 | 0 | CABC_MODE[1:0] | | 00h |

Description

- CEON: Color and Skin Enhancement ON/OFF.

  '0' = OFF

  '1' = ON

- SREON: Sunlight Readability Enhancement ON/OFF

  '0' = OFF

  '1' = ON

-EN_LEVEL: Enhancement mode Selection

| CEON | SREON | EN_LEVEL[1:0] | | Function |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | Enhancement OFF |
| 1 | 0 | 0 | 0 | CE LOW |
| 1 | 0 | 0 | 1 | CE MIDDLE |
| 1 | 0 | 1 | 1 | CE HIGH |
| 0 | 1 | 0 | 0 | SRE LOW |
| 0 | 1 | 0 | 1 | SRE MIDDLE |
| 0 | 1 | 1 | 0 | SRE HIGH |
| 0 | 1 | 1 | 1 | SRE USER DEFINE |
| 1 | 1 | 0 | 0 | CE / SRE LOW |
| 1 | 1 | 0 | 1 | CE / SRE MIDDLE |
| 1 | 1 | 1 | X | CE / SRE HIGH |

Note: "X" is don't care

-CABC_MODE: CABC mode selection

| CABC_MODE | Function |
|---|---|
| 0 | Off |
| 1 | User Interface Mode |
| 2 | Still Picture |
| 3 | Moving Image |

Restriction -

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Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.35 RDCABC (56H) Read Content Adaptive Brightness Control

| 56H | WRCABC (Write Content Adaptive Brightness Control) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDCABD | Write | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | (56h) |
| 1st Parameter | Read | CEON | SREON | EN_LEVEL[1:0] | | 0 | 0 | CABC_MODE[1:0] | | 00h |

Description

- CEON: Color and Skin Enhancement ON/OFF.

  '0' = OFF

  '1' = ON

- SREON: Sunlight Readability Enhancement ON/OFF

  '0' = OFF

  '1' = ON

-EN_LEVEL: Enhancement mode Selection

| CEON | SREON | EN_LEVEL[1:0] | | Function |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | Enhancement OFF |
| 1 | 0 | 0 | 0 | CE LOW |
| 1 | 0 | 0 | 1 | CE MIDDLE |
| 1 | 0 | 1 | 1 | CE HIGH |
| 0 | 1 | 0 | 0 | SRE LOW |
| 0 | 1 | 0 | 1 | SRE MIDDLE |
| 0 | 1 | 1 | 0 | SRE HIGH |
| 0 | 1 | 1 | 1 | SRE USER DEFINE |

Note: "X" is don't care.

-CABC_MODE: CABC mode selection

| CABC_MODE[1:0] | Function |
|---|---|
| 00b | Off |
| 01b | User Interface Mode |
| 10b | Still Picture |
| 11b | Moving Image |

Restriction -

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.36 WRCABCMB (5EH) Write CABC Minimum Brightness

| 5EH | WRCABCMB (Write CABC Minimum Brightness) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| WRCABCMB | Write | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | (5Eh) |
| 1st Parameter | Write | CMB7 | CMB6 | CMB5 | CMB4 | CMB3 | CMB2 | CMB1 | CMB0 | 00h |
| Description | - This command is used to set the minimum brightness value of the display for CABC function.<br>- In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the highest brightness for CABC. | | | | | | | | | |
| Restriction | - | | | | | | | | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.37 RDCABCMB (5FH) Read CABC Minimum Brightness

| 5FH | RDCABCMB (Read CABC Minimum Brightness) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDCABCMB | Write | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | (5Fh) |
| 1st Parameter | Read | CMB7 | CMB6 | CMB5 | CMB4 | CMB3 | CMB2 | CMB1 | CMB0 | 00h |

| | |
|---|---|
| Description | - This command returns the minimum brightness value of CABC function.<br>- In principle the relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.<br>- See command “Write CABC Minimum Brightness (5EH)”. |
| Restriction | - |
| Register Availability | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.38 RDDID1 (DAH) Read Display Identification Information

| DAH | RDDID1 (Read Display Identification Information 1) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDID1 | Write | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | (DAH) |
| 1st Parameter | Read | ID1[7:0] | | | | | | | | FFh |
| Description | - Read Display Identification Information<br>ID1:LCD module's manufacturer ID (FFh: not programmed) | | | | | | | | | |
| Restriction | - | | | | | | | | | |
| Register Availability | | | | | | | | | | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.39 RDDID2 (DBH) Read Display Identification Information

| DBH | RDDID2 (Read Display Identification Information 2) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDID2 | Write | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | (DBH) |
| 1st Parameter | Read | ID2[7:0] | | | | | | | | FFh |
| Description | - Read Display Identification Information<br>ID2:LCD module/driver version ID (FFh: not programmed) | | | | | | | | | |
| Restriction | - | | | | | | | | | |

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.40 RDDID3 (DCH) Read Display Identification Information

| DCH | RDDID 3 (Read Display Identification Information 3) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDID3 | Write | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | (DCH) |
| 1st Parameter | Read | ID3[7:0] | | | | | | | | FFh |
| Description | - Read Display Identification Information<br>ID3:LCD module/driver ID (FFh: not programmed) | | | | | | | | | |
| Restriction | - | | | | | | | | | |
| Register Availability | | | | | | | | | | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.41 RDDDBS (A1H) : Read DDB Start

| A1H | RDDDBS (Read DDB Start) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDDBS | Write | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | (A1H) |
| 1st Parameter | Read | SID[7:0] | | | | | | | | FFh |
| 2nd Parameter | Read | SID[15:8] | | | | | | | | FFh |
| 3rd Parameter | Read | MID[7:0] | | | | | | | | FFh |
| 4th Parameter | Read | MID[15:8] | | | | | | | | FFh |
| 5th parameter | Read | RID[7:0] | | | | | | | | FFh |
| 6th Parameter | Read | RID[15:8] | | | | | | | | FFh |
| 7th Parameter | Read | DDB_ID_1[7:0] | | | | | | | | FFh |
| 8th Parameter | Read | DDB_ID_2[7:0] | | | | | | | | FFh |
| 9th Parameter | Read | DDB_ID_3[7:0] | | | | | | | | FFh |
| 10th Parameter | Read | DDB_ID_4[7:0] | | | | | | | | FFh |

**Description**

- This command returns supplier identification and display module / revision information.

Note:

1. This information is not the same what DAh/DBh/DCh commands are returning.
2. Parameter 8th is an “Exit code”, this means that there is no more data in the DDB block.

This read sequence can be interrupted by any command and it can be continued by “Read DDB Continue(A8h)” command when the first parameter, what has been transferred, is the parameter, which has not been sent e.g. RDDDBS => 1st parameter has been sent => 2nd parameter has been sent => interrupt => RDDDBC => 3rd parameter of the RDDDBS has been sent.

**Restriction**

-

**Register Availability**

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.42 RDDDBC (A8H) : Read DDB Continue

| A8H | RDDDBC (Read DDB Continue) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDDDBC | Write | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | (A8H) |
| 1st Parameter | Read | DDB1[7:0] | | | | | | | | FFh |
| 2nd Parameter | Read | DDB2[7:0] | | | | | | | | FFh |
| 3rd Parameter | Read | DDB3[7:0] | | | | | | | | FFh |
| 4th Parameter | Read | DDB4[7:0] | | | | | | | | FFh |
| 5th parameter | Read | DDB5[7:0] | | | | | | | | FFh |
| 6th Parameter | Read | DDB6[7:0] | | | | | | | | FFh |
| 7th Parameter | Read | DDB7[7:0] | | | | | | | | FFh |

| | |
|---|---|
| Description | - This command returns supplier's identification and display module model/revision information from the point where RDDDBS command was interrupt by an other command e.g. RDDDBS was interrupt after 3rd parameter (DDB3). The first parameter (DDB1), what RDDDBC is returning, is DDB4[7:0].<br>See also section "6.2.36 Read DDB Start (A1h)". |
| Restriction | - |
| Register Availability | Status / Availability<br>Sleep Out, Display On / Yes<br>Sleep In / Yes |

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### 6.2.43 RDFCS (AAH) : Read First Checksum

| AAH | RDFCS (Read First Checksum) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDFCS | Write | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | (AAH) |
| 1st Parameter | Read | FCS[7:0] | | | | | | | | 00h |

| | |
|---|---|
| Description | - Read the first checksum that has been calculated from user commands after write access to these commands has been done. |
| Restriction | -only in sleep out mode |
| Register Availability | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.44 RDCCS (AFH) : Read Continue Checksum

| AFH | RDCCS (Read Continue Checksum) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDCCS | Write | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | (AFH) |
| 1st Parameter | Read | CCS[7:0] | | | | | | | | 00h |

| | |
|---|---|
| Description | -Read the continue checksum that has been calculated continuously after the first checksum has calculated from user commands after write access to these commands has been done. |
| Restriction | - |
| Register Availability | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.45 RDICID (F4H) : Read Sitronix IC ID

| F4H | RDICID (Read Sitronix IC ID Code) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| RDICID | Write | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | (F4H) |
| 1st Parameter | Read | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 71h |
| 2nd Parameter | Read | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 23h |

| | |
|---|---|
| Description | - Read Sitronix IC ID Code |
| Restriction | - |
| Register Availability | |

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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### 6.2.46 MIPIEXTFMAT (F9H) : MIPI Extension Format

| F9H | MIPIEXTFMAT (MIPI Extension Format) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Inst / Para | Write/Read | D7 | D6 | D5 | D4 | D3 | D3 | D1 | D0 | (Default) |
| MIPIEXTFMAT | Write | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | (F9H) |
| 1st Parameter | Write | 0 | 0 | 0 | 0 | 0 | 0 | PIXEL_EXTEN[1:0] | | 00h |

Description

-The PIXEL_EXTEN is used for pixel extension format.

| PIXEL_EXTEN[1:0] | 5-6-5 Format | 6-6-6 Format | 8-8-8 Format |
|---|---|---|---|
| 00 | R[7:0] = {R[4:0] , 000b }<br>G[7:0] = {G[5:0] , 00b }<br>B[7:0] = {B[4:0] , 000b } | R[7:0] = {R[5:0] , 00b }<br>G[7:0] = {G[5:0] , 00b }<br>B[7:0] = {B[5:0] , 00b } | R[7:0] = R[7:0]<br>G[7:0] = G[7:0]<br>B[7:0] = B[7:0] |
| 01 | R[7:0] = {R[4:0] , 111b}<br>G[7:0] = {G[5:0] , 11b}<br>B[7:0] = {B[4:0] , 111b} | R[7:0] = {R[5:0] , 11b}<br>G[7:0] = {G[5:0] , 11b}<br>B[7:0] = {B[5:0] , 11b} | R[7:0] = R[7:0]<br>G[7:0] = G[7:0]<br>B[7:0] = B[7:0] |
| 10 | R[7:0] = {R[4:0] ,R[4:2]}<br>G[7:0] = {G[5:0] , G[5:4]}<br>B[7:0] = {B[4:0] , B[4:2]} | R[7:0] = {R[5:0] ,R[5:4]}<br>G[7:0] = {G[5:0] , G[5:4]}<br>B[7:0] = {B[5:0] , B[5:4]} | R[7:0] = R[7:0]<br>G[7:0] = G[7:0]<br>B[7:0] = B[7:0] |
| 11 | R[7:0] = {R[4:0] , G[5:3]}<br>G[7:0] = {G[5:0] , G[5:4]}<br>B[7:0] = {B[4:0] , G[5:3]} | R[7:0] = {R[5:0] , G[5:4]}<br>G[7:0] = {G[5:0] , G[5:4]}<br>B[7:0] = {B[5:0] , G[5:4]} | R[7:0] = R[7:0]<br>G[7:0] = G[7:0]<br>B[7:0] = B[7:0] |

Restriction

-

Register Availability

| Status | Availability |
|---|---|
| Sleep Out, Display On | Yes |
| Sleep In | Yes |

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# 7 ELECTRICAL CHARACTERISTICS

## 7.1 Absolute Maximum Ratings

| Item | Symbol | Range | Unit |
|---|---|---|---|
| Supply Voltage (Analog) | IOVCC | - 0.3 ~ +2.1 | V |
| Supply Voltage (I/O) | DPHYVCC | - 0.3 ~ +2.1 | V |
| Driver Supply Voltage | AVDD-VSS | -0.3 ~ +6.6 | V |
| Driver Supply Voltage | AVEE-VSS | - 6.6 ~ +0.3 | V |
| Driver Supply Voltage | VGH-VGL | -0.3 ~ +30.0 | V |
| Logic Input Voltage Range | VIN | -0.3 ~ DPHYVCC + 0.3 | V |
| Logic Output Voltage Range | VO | -0.3 ~ DPHYVCC + 0.3 | V |
| Operating Temperature Range | TOPR | -30 ~ +75 | ℃ |
| Storage Temperature Range | TSTG | -40 ~ +110 | ℃ |

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### 7.2 DC Characteristics

#### 7.2.1 Basic Characteristics

| Parameter | Symbol | Condition | Specification | | | Unit | Note |
|---|---|---|---|---|---|---|---|
| | | | MIN. | TYP. | MAX. | | |
| Power & Operation Voltage | | | | | | | |
| Power supply Voltage | IOVCC | | 1.65 | 1.8 | 1.95 | V | |
| | DPHYVCC | | 1.65 | 1.8 | 1.95 | V | |
| | AVDD | | 4.5 | 5.5 | 6.3 | V | |
| | AVEE | | 4.5 | 5.5 | 6.3 | V | |
| Gate Driver High Voltage | VGH | | 7 | | 18 | V | Note 2 |
| Gate Driver Low Voltage | VGL | | -7 | | -12 | V | |
| Input / Output | | | | | | | |
| Logic-High Input Voltage | VIH | | 0.7 IOVCC | | IOVCC | V | Note 1 |
| Logic-Low Input Voltage | VIL | | VSS | | 0.3IOVCC | V | Note 1 |
| Logic-High Output Voltage | VOH | IOH = -1.0mA | 0.8 IOVCC | | IOVCC | V | Note 1 |
| Logic-Low Output Voltage | VOL | IOL = +1.0mA | VSS | | 0.2 IOVCC | V | Note 1 |
| Input Leakage Current | IIL | VIN=IOVCC/VSS | -1 | | +1 | uA | Note 1 |
| VCOM Voltage | | | | | | | |
| VCOM amplitude | VCOM | | -2.75 | | -0.2 | V | |
| Source Driver | | | | | | | |
| Gamma Amplitude (Positive) | VGPAMP | | 3 | | 6 | V | |
| Gamma Amplitude (Negative) | VGNAMP | | -6 | | -3 | V | |

*Notes:*

*1. TA= -30 to 70℃ (to +85℃ no damage).*

*2. When evaluating the maximum and minimum of VGH/VGL. IOVCC/AVDD/AVEE is typical value.*

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### 7.2.2 Current Consumption

(Ta = 25℃)

| Parameter | Symbol | Test Condition | Specification | | | Unit |
|---|---|---|---|---|---|---|
| | | | MIN. | TYP. | MAX. | |
| Power supply voltage | IOVCC = 1.8 | Deep standby mode<br>No load on panel | - | TBD | | uA |
| | AVDD = 5.9 | | - | TBD | | uA |
| | AVEE = -5.9 | | - | TBD | | uA |
| | IOVCC = 1.8 | Sleep in mode<br>No load on panel | - | TBD | | mA |
| | AVDD = 5.9 | | - | TBD | | mA |
| | AVEE = -5.9 | | - | TBD | | mA |

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### 7.2.3 MIPI DC Characteristic

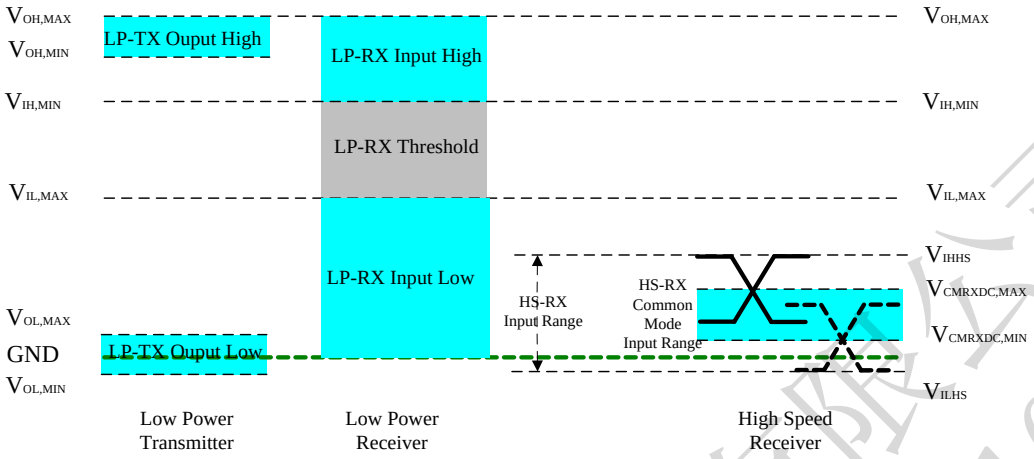



**Figure 76 MIPI Signaling Voltage Levels**

| Parameter | Symbol | Specification | | | Unit |
|---|---|---|---|---|---|
| | | MIN | TYP | MAX | |
| **Operation Voltage for MIPI Receiver** | | | | | |
| Low power mode operating voltage | VLPH | - | TBD | - | V |
| **MIPI Characteristics for High Speed Receiver** | | | | | |
| Single-ended input low voltage | VILHS | TBD | - | - | mV |
| Single-ended input high voltage | VIHHS | - | - | TBD | mV |
| Common-mode voltage | VCMRXDC | TBD | - | TBD | mV |
| Differential input impedance | ZID | TBD | TBD | TBD | ohm |
| **MIPI Characteristics for Low Power Mode** | | | | | |
| Pad signal voltage range | VI | TBD | - | TBD | mV |
| Logic 0 input threshold | VIL | TBD | - | TBD | mV |
| Logic 1 input threshold | VIH | TBD | - | - | mV |
| Output low level | VOL | TBD | - | TBD | mV |
| Output high level | VOH | TBD | TBD | TBD | V |

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## 7.3 AC Characteristics

### 7.3.1 MIPI Timing

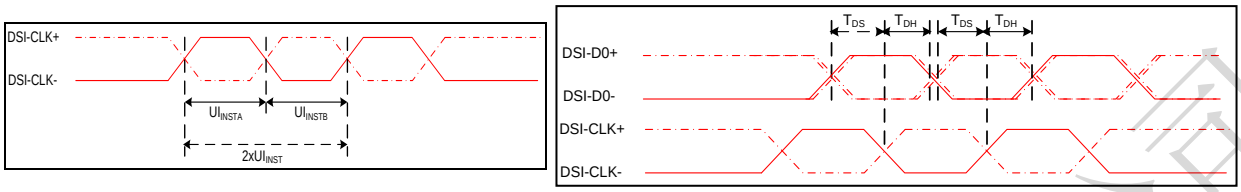



**Figure 77 High Speed Mode – Clock Channel Timing**

| Signal | Symbol | Parameter | Specification | | | Unit | Description |
|---|---|---|---|---|---|---|---|
| | | | MIN | TYP | MAX | | |
| DSI-CLK+/- | $2xUI_{INST}$ | Double UI instantaneous | TBD | - | TBD | ns | |
| DSI-CLK+/- | $UI_{INSTA}$ ,$UI_{INSTB}$ | UI instantaneous half | TBD | - | TBD | ns | |
| fDSI-CLK+/- | $F_{DSICLK}$ | DSI-CLK+/- frequency | TBD | - | TBD | MHz | |
| DSI-Dn+/- | $T_{DS}$ | Data to clock setup time | TBD | - | - | UI | |
| DSI-Dn+/- | $T_{DH}$ | Data to clock hold time | TBD | - | - | UI | |

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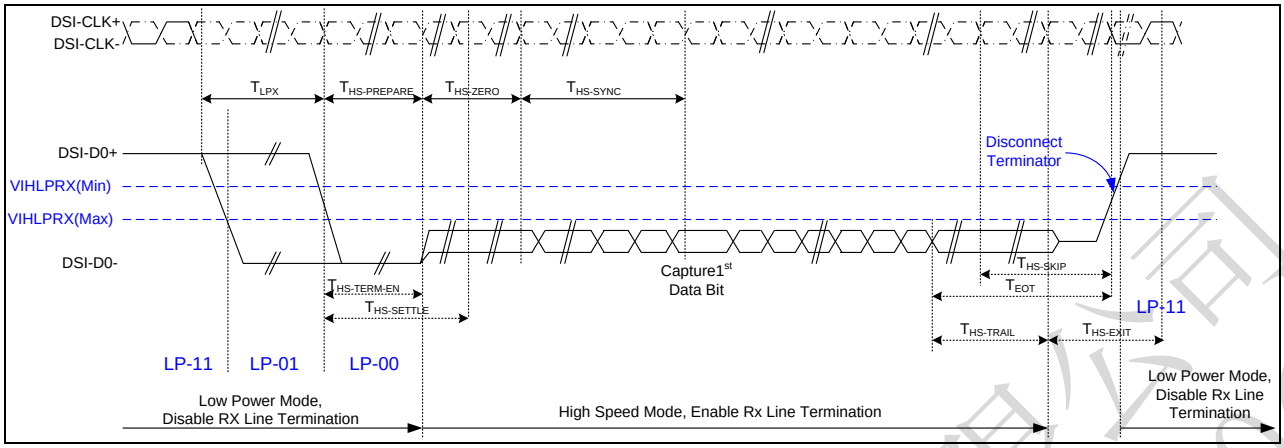



**Figure 78 High-Speed Data Transmission**

| Parameter | Symbol | Specification | | | Unit |
|---|---|---|---|---|---|
| | | MIN | TYP | MAX | |
| Time to drive LP-00 to prepare for HS transmission | $T_{HS-PREPARE}$ | TBD | - | TBD | ns |
| Time from start of t HS-TRAIL or t CLK-TRAIL period to start of LP-11 state | $T_{EOT}$ | - | - | TBD | ns |
| Time to enable data receiver line termination measured from when Dn crosses VILMAX | $T_{HS-TERM-EN}$ | - | - | TBD | ns |
| Time to drive flipped differential state after last payload data bit of a HS transmission | $T_{HS-TRAIL}$ | TBD | - | - | ns |
| Time-out at RX to ignore transition period of EoT | $T_{HS-SKIP}$ | TBD | - | TBD | ns |
| Time to drive LP-11 after HS burst | $T_{HS-EXIT}$ | TBD | - | - | ns |
| Length of any Low-Power state period | $T_{LPX}$ | TBD | - | - | ns |
| Sync sequence period | $T_{HS-SYNC}$ | - | TBD | - | ns |
| Minimum lead HS-0 drive period before the Sync sequence | $T_{HS-ZERO}$ | TBD | - | - | ns |
| Time interval during which the HS receiver should ignore any Clock Lane HS transitions, starting from the beginning of $T_{CLK-PREPAR}$ | $T_{CLK-SETTLE}$ | TBD | - | TBD | ns |
| Time interval during which the HS receiver shall ignore any Data Lane HS transitions, starting from the beginning of $T_{HS-PREPAR}$. The HS receiver shall ignore any Data Lane transitions before the minimum value, and the HS receiver shall respond to any Data Lane transitions after the maximum value. | $T_{HS-SETTLE}$ | TBD | - | TBD | ns |

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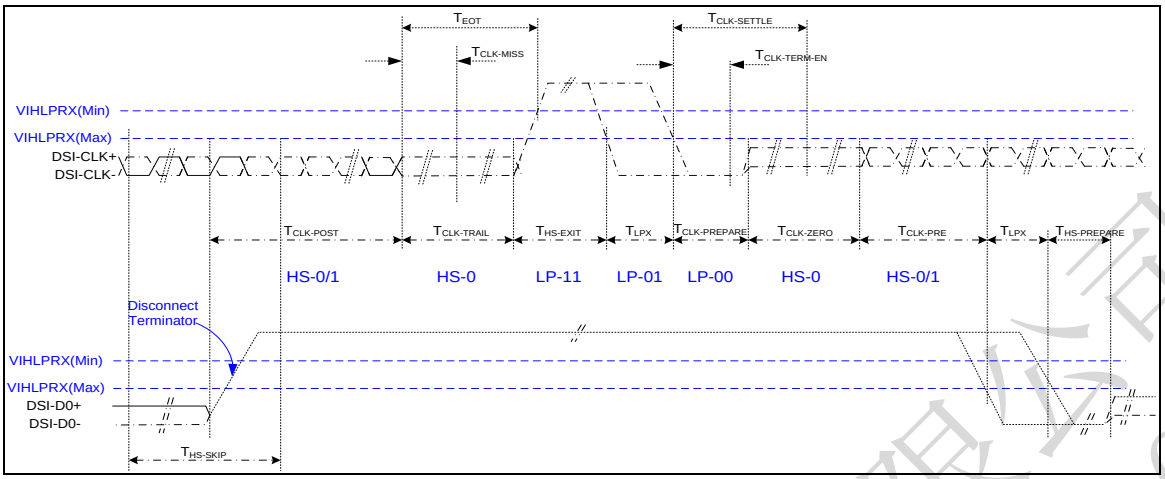



**Figure 79 Switching the Clock Lane between Clock Transmission and Low-Power Mode**

| Parameter | Symbol | Specification | | | Unit |
|---|---|---|---|---|---|
| | | MIN | TYP | MAX | |
| Time that the transmitter shall continue sending HS clock after the last associated Data Lane has transitioned to LP mode | $T_{CLK-POST}$ | TBD | - | - | ns |
| Detection time that the clock has stopped toggling | $T_{CLK-MISS}$ | - | - | TBD | ns |
| Time to drive LP-00 to prepare for HS clock transmission | $T_{CLK-PREPARE}$ | TBD | - | TBD | ns |
| Minimum lead HS-0 drive period before starting Clock | $T_{CLK-PREPARE}+T_{CLK-ZERO}$ | TBD | - | - | ns |
| Time to enable Clock Lane receiver line termination measured from when Dn cross VIL,MAX | $T_{CLK-TERM-EN}$ | - | - | TBD | ns |
| Minimum time that the HS clock must be set prior to any associated date lane beginning the transmission from LP to HS mode | $T_{CLK-PRE}$ | TBD | - | - | UI |
| Time to drive HS differential state after last payload clock bit of a HS transmission burst | $T_{CLK-TRAIL}$ | TBD | - | - | ns |

*Note: 3-Lane = (1000 / $f_{mipi}$ * 8 * 52 -$T_{HS_TRAIL}$- 60) / (1000/$f_{mipi}$)*

*4-Lane = (1000 / $f_{mipi}$ * 8 * 39 -$T_{HS_TRAIL}$- 60) / (1000/$f_{mipi}$)*

*Example: 3-Lane, 600Mbps : n = (1000 / 600 * 8 * 52 - $T_{HS_TRAIL}$- 60) / (1000/600) = 340*

*4-Lane, 600Mbps : n = (1000 / 600 * 8 * 39 - $T_{HS_TRAIL}$- 60) / (1000/600) = 236*

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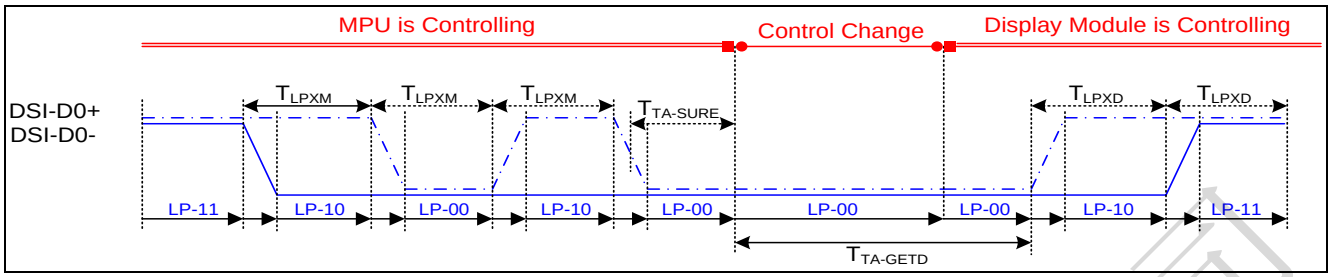



**Figure 80 Bus Turn-around Procedure**

| Parameter | Symbol | Specification | | | Unit |
|---|---|---|---|---|---|
| | | MIN | TYP | MAX | |
| Length of any Low-Power state period : Master side | $T_{LPX}$ | TBD | - | TBD | ns |
| Length of any Low-Power state period : Slave side | $T_{LPX}$ | TBD | - | TBD | ns |
| Ratio of $T_{LPX}$ (MASTER)/ $T_{LPX}$ (SLAVE) between Master and Slave side | Ratio $T_{LPX}$ | TBD | - | TBD | |
| Time-out before new TX side start driving | $T_{TA-SURE}$ | TBD | - | TBD | ns |
| Time to drive LP-00 by new TX | $T_{TA-GET}$ | - | TBD | - | ns |
| Time to drive LP-00 after Turnaround Request | $T_{TA-GO}$ | - | TBD | - | ns |

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### 7.3.2 MIPI Interface Timing

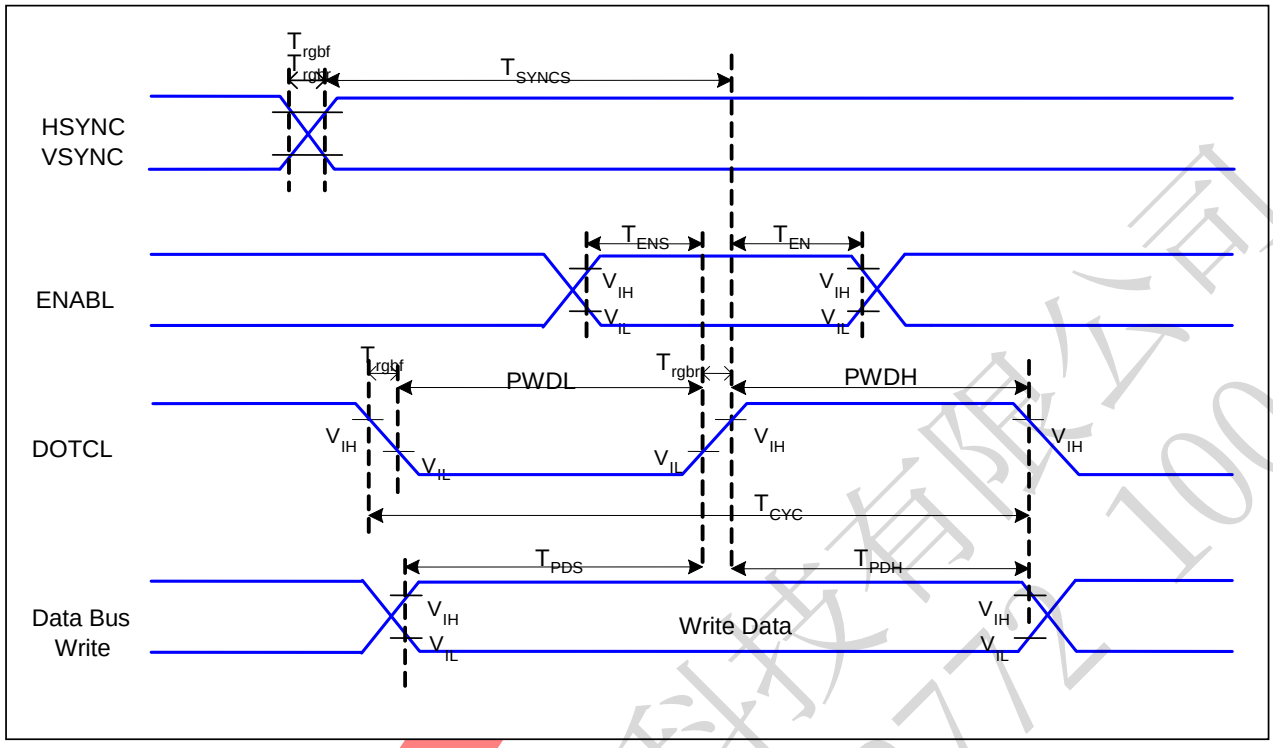
| Parameter | Symbol | Min. | Typ.. | Max. | Unit |
|---|---|---|---|---|---|
| Horizontal Sync. Width | HPW | TBD | - | - | Byte Clock |
| Horizontal Sync. Back Porch | HBP | TBD | - | - | Byte Clock |
| Horizontal Sync. Front Porch | HFP | TBD | - | - | Byte Clock |
| Vertical Sync. Width | VSW | TBD | - | - | Line |
| Vertical Sync. Back Porch | VBP | TBD | - | - | Line |
| Vertical Sync. Front Porch | VFP | TBD | - | - | Line |
| Vertical Frequency | | - | TBD | - | Hz |

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### 7.3.3 RGB timing




*IOVCC=1.8V, Ta=25℃*

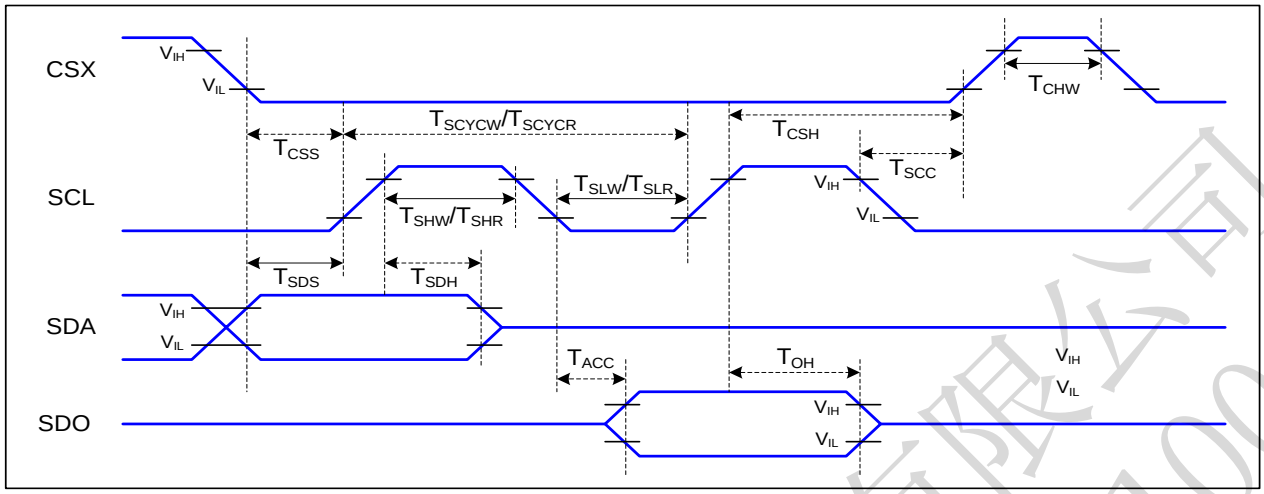
| Signal | Symbol | Parameter | MIN | MAX | Unit | Description |
|---|---|---|---|---|---|---|
| HSYNC, VSYNC | $T_{SYNCS}$ | VSYNC, HSYNC Setup Time | TBD | - | ns | |
| ENABLE | $T_{ENS}$ | Enable Setup Time | TBD | - | ns | |
| | $T_{ENH}$ | Enable Hold Time | TBD | - | ns | |
| DOTCLK | PWDH | DOTCLK High-level Pulse Width | TBD | - | ns | |
| | PWDL | DOTCLK Low-level Pulse Width | TBD | - | ns | |
| | $T_{CYCD}$ | DOTCLK Cycle Time | TBD | - | ns | |
| DB | $T_{PDS}$ | PD Data Setup Time | TBD | - | ns | |
| | $T_{PDH}$ | PD Data Hold Time | TBD | - | ns | |

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### 7.3.4 SPI9 & SPI16 Timing




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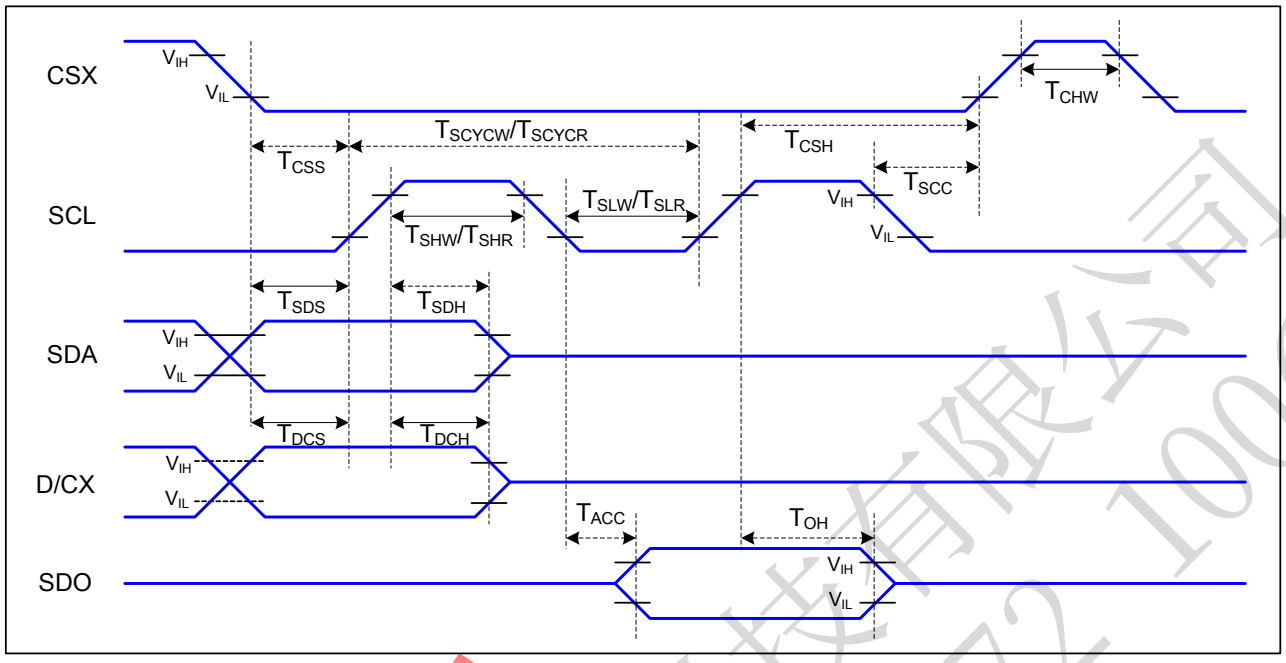
| Signal | Symbol | Parameter | Min | Max | Unit | Description |
|---|---|---|---|---|---|---|
| CSX | $T_{CSS}$ | Chip select setup time (write) | TBD | | ns | |
| | $T_{CSH}$ | Chip select hold time (write) | TBD | | ns | |
| | $T_{CSS}$ | Chip select setup time (read) | TBD | | ns | |
| | $T_{SCC}$ | Chip select hold time (read) | TBD | | ns | |
| | $T_{CHW}$ | Chip select “H” pulse width | TBD | | ns | |
| SCL | $T_{SCYCW}$ | Serial clock cycle (Write) | TBD | | ns | |
| | $T_{SHW}$ | SCL “H” pulse width (Write) | TBD | | ns | |
| | $T_{SLW}$ | SCL “L” pulse width (Write) | TBD | | ns | |
| | $T_{SCYCR}$ | Serial clock cycle (Read) | TBD | | ns | |
| | $T_{SHR}$ | SCL “H” pulse width (Read) | TBD | | ns | |
| | $T_{SLR}$ | SCL “L” pulse width (Read) | TBD | | ns | |
| SDA (DIN) | $T_{SDS}$ | Data setup time | TBD | | ns | |
| | $T_{SDH}$ | Data hold time | TBD | | ns | |
| DOUT | $T_{ACC}$ | Access time | TBD | TBD | ns | For maximum CL=30pF |
| | $T_{OH}$ | Output disable time | TBD | TBD | ns | For minimum CL=8pF |

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### 7.3.5 SPI8 (4 line) Timing




*IOVCC=1.8V, Ta=25℃*

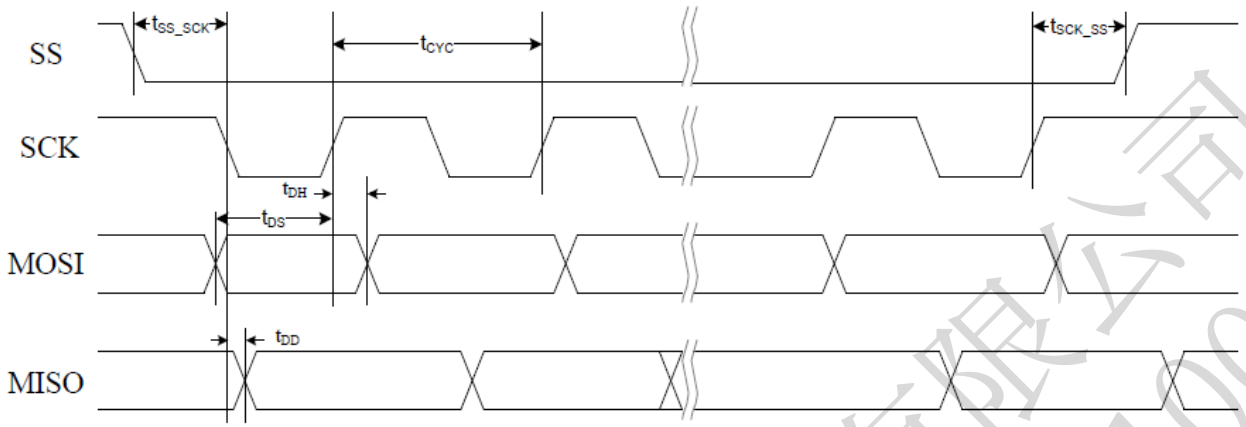
| Signal | Symbol | Parameter | MIN | MAX | Unit | Description |
|---|---|---|---|---|---|---|
| CSX | $T_{CSS}$ | Chip select setup time (write) | TBD | | ns | |
| | $T_{CSH}$ | Chip select hold time (write) | TBD | | ns | |
| | $T_{CSS}$ | Chip select setup time (read) | TBD | | ns | |
| | $T_{SCC}$ | Chip select hold time (read) | TBD | | ns | |
| | $T_{CHW}$ | Chip select “H” pulse width | TBD | | ns | |
| SCL | $T_{SCYCW}$ | Serial clock cycle (Write) | TBD | | ns | -write command & data ram |
| | $T_{SHW}$ | SCL “H” pulse width (Write) | TBD | | ns | |
| | $T_{SLW}$ | SCL “L” pulse width (Write) | TBD | | ns | |
| | $T_{SCYCR}$ | Serial clock cycle (Read) | TBD | | ns | -read command & data ram |
| | $T_{SHR}$ | SCL “H” pulse width (Read) | TBD | | ns | |
| | $T_{SLR}$ | SCL “L” pulse width (Read) | TBD | | ns | |
| D/CX | $T_{DCS}$ | D/CX setup time | TBD | | ns | |
| | $T_{DCH}$ | D/CX hold time | TBD | | ns | |
| SDA (DIN) | $T_{SDS}$ | Data setup time | TBD | | ns | |
| | $T_{SDH}$ | Data hold time | TBD | | ns | |
| DOUT | $T_{ACC}$ | Access time | TBD | 50 | ns | For maximum CL=30pF |
| | $T_{OH}$ | Output disable time | TBD | 50 | ns | For minimum CL=8pF |

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### 7.3.6 Touch SPI Timing




*IOVCC=1.8V, Ta=25℃*

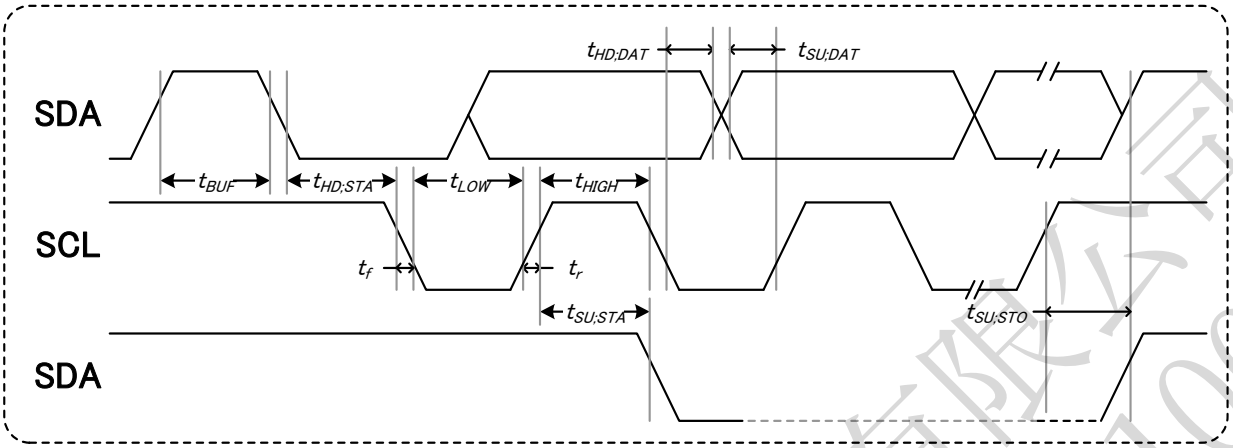
| Signal | Symbol | Parameter | MIN | MAX | Unit | Description |
|---|---|---|---|---|---|---|
| SCK | $f_{SCK}$ | SCK frequency | - | TBD | Mhz | |
| SCK | $t_{CYC}$ | SCK cycle time | TBD | - | ns | |
| MOSI | $t_{DS}$ | Data setup time prior SCK rising | TBD | - | ns | |
| | $t_{DH}$ | Data hold time after SCK rising | TBD | - | ns | |
| MISO | $t_{DD}$ | MISO data output delay from SCK falling | - | TBD | ns | |
| SS | $t_{SS}$_SCK | SS falling to 1st SCK falling | TBD | - | ns | |
| | $t_{SCK}$_SS | SCK rising to SS rising | TBD | - | ns | |
| | $t_R$ | CS recovery time | TBD | - | us | |

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### 7.3.7 Touch I2C timing




*IOVCC=1.8V, Ta=25℃*

| Signal | Symbol | Parameter | MIN | MAX | Unit | Description |
|---|---|---|---|---|---|---|
| SCL | $f_{SCL}$ | SCL clock frequency | - | TBD | kHz | |
| | $t_{LOW}$ | SCL clock low period | TBD | - | ns | |
| | $t_{HIGH}$ | SCL clock high period | TBD | - | ns | |
| SDA | $t_{SU}$;Data | Data set-up time | TBD | - | ns | |
| | $t_{HD}$;Data | Data hold time | TBD | - | ns | |
| SDA | $t_{SU}$;STA | Setup time for a repeated START condition | TBD | - | ns | |
| | $t_{HD}$;STA | Start condition hold time | TBD | - | ns | |
| | $t_{SU}$;STO | Setup time for STOP condition | TBD | - | ns | |
| | $t_{BUF}$ | Bus free time between a STOP and START | TBD | - | us | |

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### 7.3.8 Reset Timing

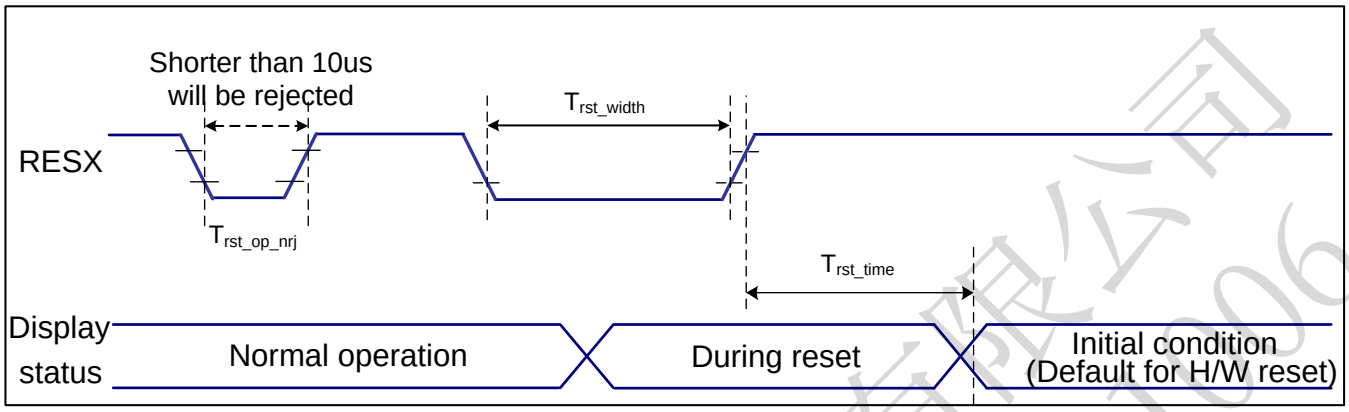



**Figure 81 Reset Operation**

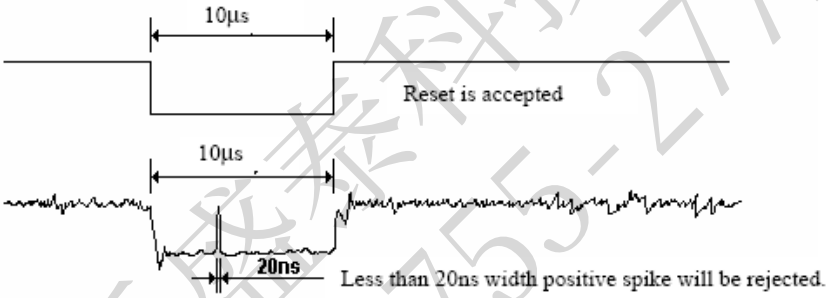



**Figure 82 Reset Noise Rejected Diagram**

**Reset Timing Characteristics IOVCC=1.8v Ta=25℃**

| Parameter | Symbol | Specification | | | Unit |
|---|---|---|---|---|---|
| | | Min. | Typ. | Max. | |
| Reset low width | Trst_width | TBD | - | - | ms |
| Reset time | Trst_time | TBD | - | - | ms |
| OP noise reject | Trst_op_nrj | - | - | TBD | us |

- During the reset period, the display will be blanked, then return to default condition for IC initialization



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### 7.3.9 Abnormal Timing:

ST7102 provides abnormal power drop detection function, but there has a premise that external power must drop at low ramp, or the discharge function will not work. The following diagram is show how much time that external power drop from 90% to 10% is suitable.

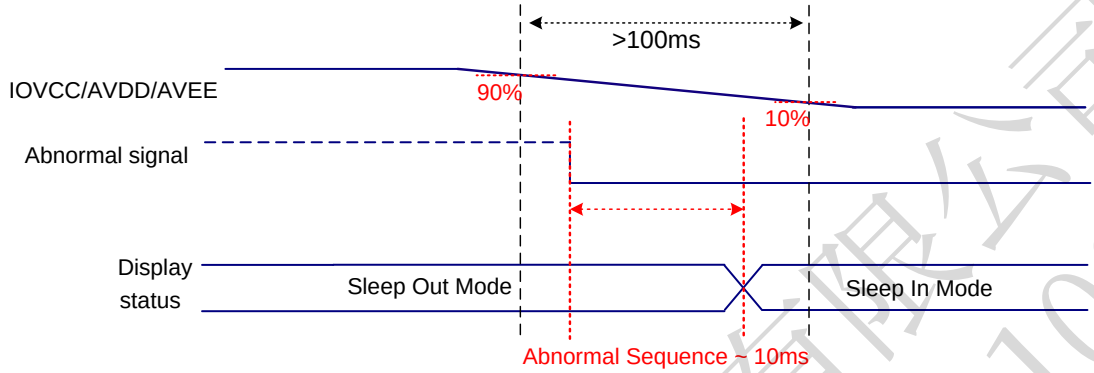



**Figure 83 Reset Noise Rejected Diagram**

**Reset Timing Characteristics IOVCC=1.8v, AVDD=5.8v, AVEE=-5.8v Ta=25℃**

| External power | Power drop time from 90% to 10% | | | Unit |
|---|---|---|---|---|
| | Min. | Typ. | Max. | |
| IOVCC | 100 | | | ms |
| AVDD | 100 | | | ms |
| AVEE | 100 | | | ms |

*Note: Due to IOVCC is critical for whole chip, keep IOVCC as stable as possible*

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# 8 REVISION HISTORY

| Version | Date | Description |
|---|---|---|
| V0.0 | 2023/10 | **First issue** |

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